STPM Physics · Semester 3

Oscillations, Waves & Modern Physics

FULL NOTES

Cikgu Suhaili

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PhysiQ · ellyelias.github.io/physiq · Worked examples, exercises with answers and the simulations from the app.

Chapter 20 · Semester 3

OSCILLATIONS

Notes by Cikgu Suhaili

15 worked examples 4 simulations Exercises with answers

Oscillations: introduction

Oscillation (periodic motion)

is **the back-and-forth motion of an object about a fixed point**.

- a mass on the end of a spring (horizontal or vertical)
- atoms vibrating in a crystal
- a simple pendulum
- a child on a swing

The simplest oscillation is a mass on a spring sliding on a smooth horizontal surface (Figure 20.1). We assume the spring is light (its mass is ignored) and there is no friction.

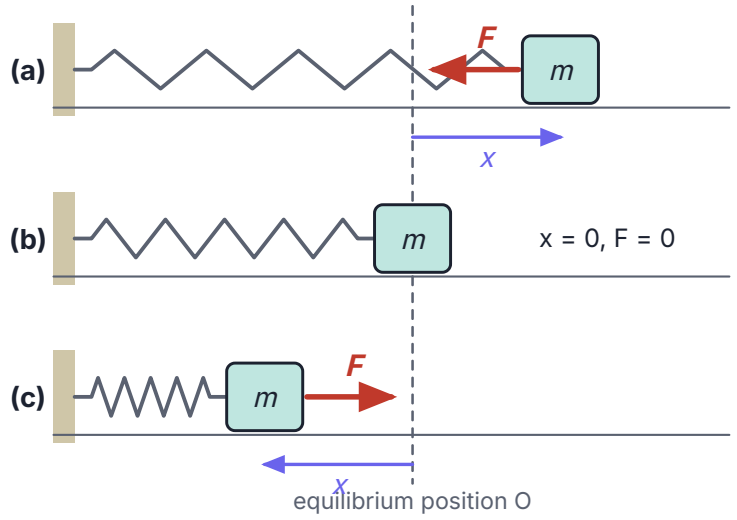


Figure 20.1: (a) spring stretched, the force acts to the left; (b) at the equilibrium position, F = 0; (c) spring compressed, the force acts to the right.

- The spring has a natural length at which it exerts no force on the mass. The position of the mass at this point is the **equilibrium position**.
- If the mass is pulled to the right (spring stretched) or pushed to the left (spring compressed), the spring exerts a force that acts to return it to equilibrium. This is the **restoring force**.
- At $x = 0$ the restoring force is zero, but the mass keeps moving past O because of its **inertia**.
- The restoring force is always directed towards the equilibrium position.

20.1 Characteristics of simple harmonic motion

Equation of SHM

By Hooke's law, the restoring force varies with the displacement x as

$$F = -kx$$

k = spring constant (N m^{-1}), a measure of the stiffness of the spring. The minus sign shows that F is always opposite to the displacement x.

With no friction this is the net force, so by Newton's second law, $F = ma$:

$$ma = -kx$$

$$a = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{k}{m}x$$

Since k/m is a positive constant, write it as ω^2:

$$a = -\omega^2 x$$

ω = angular frequency (rad s^{-1}). Acceleration $\propto$ displacement, and is opposite in direction to the displacement.

Simple harmonic motion (SHM)

is the periodic motion of an object in which **the acceleration (and the restoring force) is directly proportional to the displacement from a fixed point (the equilibrium position), and is always directed towards that point**.

Teacher's note

Whenever a question gives an equation of motion such as $\mathrm{d}^2x/\mathrm{d}t^2 + 100x = 0$, rewrite it as $a = -\omega^2 x$ first. Here $a = -100x$, so $\omega^2 = 100$ and $\omega = 10$ rad s^{-1}. Every other SHM quantity follows from ω.

20.2 Kinematics of simple harmonic motion

The reference circle

> **Reference circle**
> is the circle in which a particle moves at constant angular velocity so that **its projection onto a diameter of the circle performs SHM**.

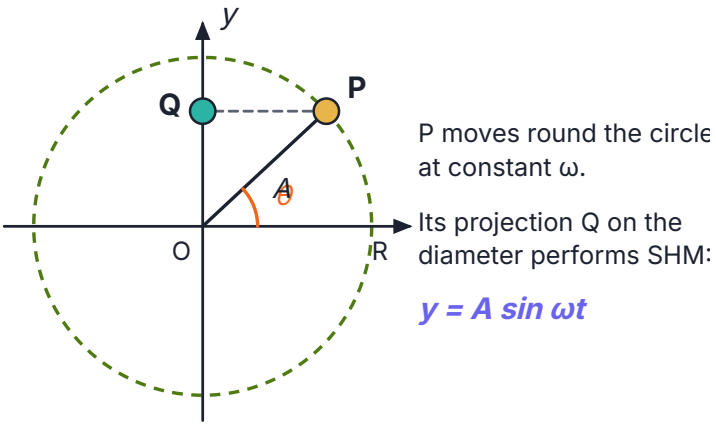


Figure 20.2: P moves anticlockwise round a circle of radius A at constant ω. Its projection Q moves up and down the diameter in SHM.

If OQ = y and θ = ωt, then $y = A \sin \theta$, so

$$y = A \sin \omega t \quad \text{or} \quad x = A \sin \omega t$$

This is why ω is called the **angular frequency**, even though the oscillating mass moves only in a straight line.

Terms used in SHM

Quantity	Meaning
Displacement, x (m)	the distance of the particle from the equilibrium position (with direction)
Amplitude, A (m)	the maximum magnitude of the displacement
Period, T (s)	the time for one complete oscillation
Frequency, f (Hz)	the number of complete oscillations per second; $f = \frac{1}{T}$
Angular frequency, ω (rad s^{-1})	$\omega = 2\pi f = \frac{2\pi}{T}$

Solutions of $a = -\omega^2 x$

Both of these satisfy $a = -\omega^2 x$. The only difference is where the clock starts ($t = 0$).

$x = A \sin \omega t$	$x = A \cos \omega t$
starts at the equilibrium position ($x = 0$), moving at maximum speed	starts at the maximum displacement ($x = A$), momentarily at rest
check: $v = \mathrm{d}x/\mathrm{d}t = A\omega \cos \omega t$; $a = \mathrm{d}v/\mathrm{d}t = -A\omega^2 \sin \omega t = -\omega^2 x$ ✓	check: $v = -A\omega \sin \omega t$; $a = -A\omega^2 \cos \omega t = -\omega^2 x$ ✓

A mass pulled to $x = +A$ and released from rest follows $x = A \cos \omega t$. A mass given a push through the equilibrium position at $t = 0$ follows $x = A \sin \omega t$.

Velocity and acceleration

Velocity (for $x = A \sin \omega t$)

$$v = \frac{\mathrm{d}x}{\mathrm{d}t} = A\omega \cos \omega t$$

Maximum when $\cos \omega t = \pm 1$:

$$v_{\max} = A\omega$$

Acceleration

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = -A\omega^2 \sin \omega t = -\omega^2 x$$

Maximum magnitude when $\sin \omega t = \pm 1$:

$$a_{\max} = A\omega^2$$

Velocity in terms of displacement

$$x = A \sin \omega t \Rightarrow \sin \omega t = \frac{x}{A}$$

$$\sin^2 \omega t + \cos^2 \omega t = 1 \Rightarrow \cos \omega t = \pm\sqrt{1 - \frac{x^2}{A^2}}$$

$$v = A\omega \cos \omega t = \pm A\omega\sqrt{1 - \frac{x^2}{A^2}} = \pm\omega\sqrt{A^2 - x^2}$$

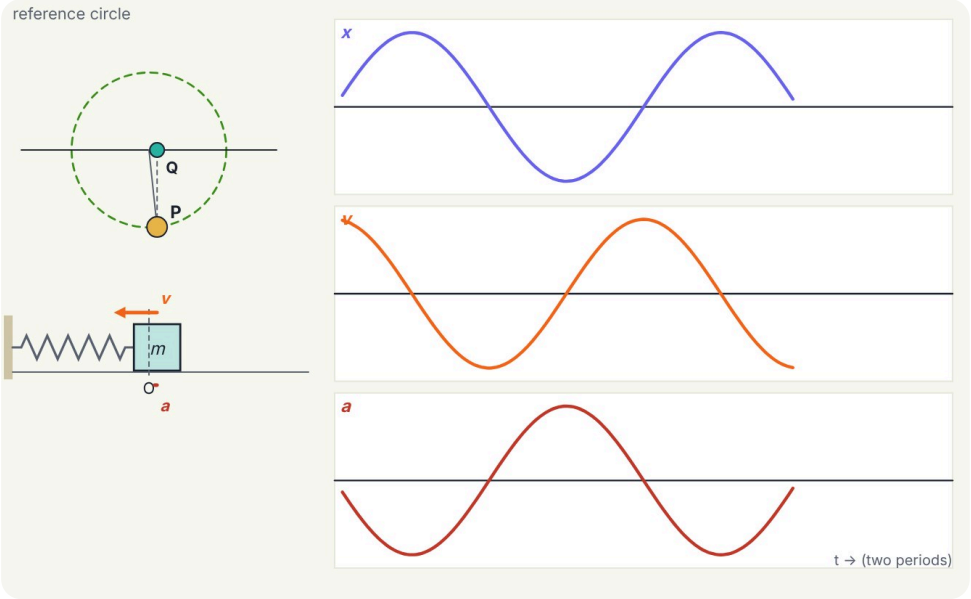
$$v = \pm\omega\sqrt{A^2 - x^2}$$

$\pm$: the particle can pass through the same point x in either direction. At $x = 0$, $v = \pm\omega A$ (maximum speed); at $x = \pm A$, $v = 0$.

Simulation

SHM and the reference circle

P goes round the circle at constant ω; its shadow Q (and the mass on the spring) moves in SHM. Watch the x–t, v–t and a–t graphs being drawn together. Change the amplitude and the period, and choose whether the clock starts at the equilibrium position or at the maximum displacement.



t	ω = 2π/T	x	v	a = −ω²x
2.97 s	3.14 rad/s	1.1 cm	−0.312 m/s	−0.104 m/s²

K (m = 0.5 kg)	U	E = K + U
24.4 mJ	0.3 mJ	24.7 mJ

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

Example 20.2.1

A particle vibrates according to the equation $y = 4 \sin 3t$, where y is in metres and t in seconds. Calculate (a) the amplitude, (b) the angular frequency, (c) the frequency, (d) the period, (e) the maximum speed, (f) the maximum magnitude of the acceleration.

Solution

Compare with $y = A \sin \omega t$.

(a) A = **4 m** (b) ω = **3 rad s^{-1}**

(c) $f = \dfrac{\omega}{2\pi} = \dfrac{3}{2\pi}$ = **0.48 Hz** (d) $T = \dfrac{1}{f}$ = **2.09 s**

(e) $v_{max} = A\omega = (4)(3)$ = **12 m s^{-1}**

(f) $a_{max} = A\omega^2 = (4)(3)^2$ = **36 m s^{-2}**

Example 20.2.2

A 300 g mass at the end of a spring oscillates with an amplitude of 7.0 cm and a frequency of 1.80 Hz. When it is 3.0 cm from its equilibrium position, calculate (a) its speed, (b) the magnitude of its acceleration.

Solution

$\omega = 2\pi f = 2\pi(1.80) = 11.3$ rad s^{-1}

(a) $v = \omega\sqrt{A^2 - x^2} = 11.3\sqrt{0.070^2 - 0.030^2}$ = **0.71 m s^{-1}**

(b) $a = \omega^2 x = (11.3)^2(0.030)$ = **3.8 m s^{-2}**

Example 20.2.3

The equation of motion of a mass at the end of a spring is $y = 0.30 \cos 0.50t$ (in metres). Find the displacement, velocity and acceleration of the mass at (a) $t = 0$, (b) $t = 3.0$ s.

Solution

$v = \dfrac{dy}{dt} = -(0.30)(0.50) \sin 0.50t = -0.15 \sin 0.50t$; $a = \dfrac{dv}{dt} = -(0.30)(0.50)^2 \cos 0.50t = -0.075 \cos 0.50t$

(a) $t = 0$: $y =$ **0.30 m**, $v =$ **0**, $a =$ **$-0.075\ \text{m s}^{-2}$**

(b) $t = 3.0$ s (calculator in radians): $y = 0.30 \cos 1.5 =$ **0.021 m**; $v = -0.15 \sin 1.5 =$ **$-0.15\ \text{m s}^{-1}$**; $a = -0.075 \cos 1.5 =$ **$-5.3 \times 10^{-3}\ \text{m s}^{-2}$**

STPM 2023 · Section A · Q2

A particle undergoes simple harmonic motion with a period of 2.0 s and amplitude 2.4 cm. What is the velocity of the particle when the displacement is 1.2 cm?

A. $6.5 \times 10^{-2}\ \text{m s}^{-1}$

B. $3.8 \times 10^{-2}\ \text{m s}^{-1}$

C. $2.1 \times 10^{-2}\ \text{m s}^{-1}$

D. $1.3 \times 10^{-2}\ \text{m s}^{-1}$

Answer

Answer: A

$\omega = 2\pi/T = \pi$ rad s^{-1}; $v = \omega\sqrt{A^2 - x^2} = \pi\sqrt{0.024^2 - 0.012^2} = 6.5 \times 10^{-2}$ m s^{-1}.

Common mistake: using T as ω, or forgetting to change cm to m.

Graphs of SHM against time

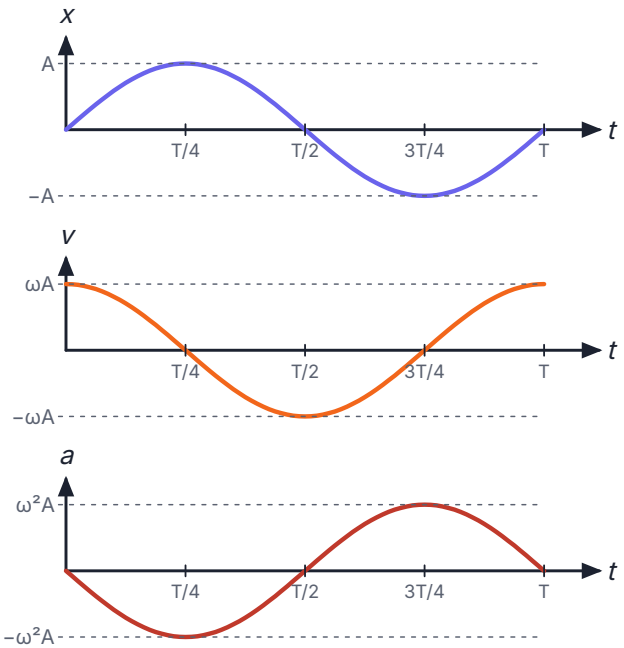
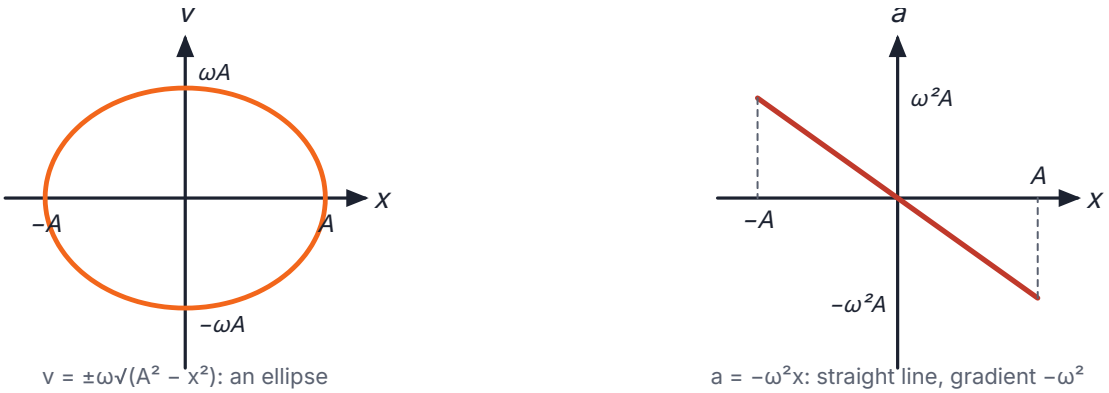


Figure 20.3: for x = A sin ωt, the velocity leads the displacement by a quarter period (T/4), and the acceleration is always opposite to the displacement.

t	x	v	a
0	0	$+\omega A$	0
$T/4$	$+A$	0	$-\omega^2 A$
$T/2$	0	$-\omega A$	0
$3T/4$	$-A$	0	$+\omega^2 A$
T	0	$+\omega A$	0

- When x is at a maximum or a minimum, $v = 0$ and $\lvert a \rvert$ is maximum.
- When $x = 0$ (passing through equilibrium), $\lvert v \rvert$ is maximum and $a = 0$.

Graphs of velocity and acceleration against displacement



v = ±ω√(A² − x²): an ellipse

a = −ω²x: straight line, gradient −ω²

Figure 20.4: v against x is an ellipse; a against x is a straight line through the origin with gradient −ω².

Since $F = ma = -m\omega^2 x$, the graph of restoring force against displacement is also a straight line through the origin. Its gradient has magnitude $k = m\omega^2$.

Example 20.2.4

The equation of motion of a particle in SHM is $x = 5 \sin 3t$, where x is in cm. (a) What is its amplitude? (b) Find its period. (c) Find its displacement at $t = 0.2$ s. (d) Sketch the graph of displacement against time.

Solution

(a) A = **5 cm**

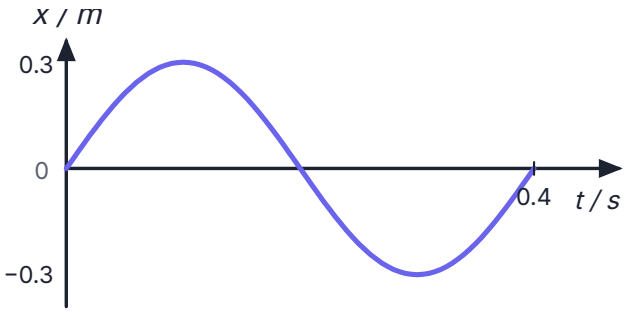
(b) $\omega = 3 = 2\pi/T \Rightarrow T$ = **2.09 s**

(c) $x = 5 \sin 3(0.2) = 5 \sin 0.6$ = **2.82 cm**

(d) a sine curve starting at 0, peak 5 cm at $t = 0.52$ s, back to 0 at 1.05 s, −5 cm at 1.57 s, 0 at 2.09 s.

Example 20.2.5

The displacement–time graph of an oscillating object is shown. Determine (a) the amplitude, the period and the frequency, (b) the angular frequency, (c) the equation of displacement as a function of time, (d) the equations of velocity and acceleration as functions of time.



Solution

(a) A = **0.3 m**, T = **0.4 s**, $f = 1/T$ = **2.5 Hz**

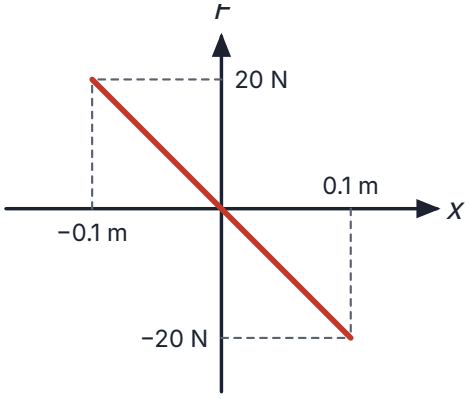
(b) $\omega = 2\pi f$ = **5π rad s^{-1}** (15.7 rad s^{-1})

(c) at $t = 0$, $x = 0$ and increasing, so x **= 0.3 sin 5πt**

(d) $v = \mathrm{d}x/\mathrm{d}t$ = **1.5π cos 5πt** m s^{-1}; $a = \mathrm{d}v/\mathrm{d}t$ = **−7.5π^2 sin 5πt** m s^{-2}

Example 20.2.6

The graph shows the force F acting on a particle of mass 2 kg against its displacement x. What type of motion is the particle following? Give a reason. What is its amplitude? Find (i) its angular frequency, (ii) its period, (iii) its maximum speed.



Solution

SHM, because the force is directly proportional to the displacement and opposite in direction to it (a straight line through the origin with negative gradient).

Amplitude = **0.1 m**

(i) $|\text{gradient}| = m\omega^2$: $\dfrac{20}{0.1} = 2\omega^2 \Rightarrow \omega =$ **10 rad s^{-1}**

(ii) $T = 2\pi/\omega =$ **0.63 s** (iii) $v_{max} = A\omega = (0.1)(10) =$ **1.0 m s^{-1}**

20.3 Energy in simple harmonic motion

Kinetic energy

$K = \dfrac{1}{2}mv^2$ and $v^2 = \omega^2(A^2 - x^2)$

$$K = \frac{1}{2}m\omega^2(A^2 - x^2)$$

maximum at $x = 0$: $K_{max} = \frac{1}{2}m\omega^2A^2$; zero at $x = \pm A$. In terms of time ($x = A \sin \omega t$): $K = \frac{1}{2}m\omega^2A^2 \cos^2 \omega t$

Potential energy

Work done against the restoring force: $U = \int_0^x kx\,\mathrm{d}x = \frac{1}{2}kx^2$

$k = m\omega^2$

$$U = \frac{1}{2}m\omega^2 x^2$$

zero at $x = 0$; maximum at $x = \pm A$: $U_{max} = \tfrac{1}{2}m\omega^2 A^2$. In terms of time: $U = \tfrac{1}{2}m\omega^2 A^2 \sin^2 \omega t$

Total energy is conserved

$E = K + U = \frac{1}{2}m\omega^2(A^2 - x^2) + \frac{1}{2}m\omega^2 x^2$

$$E = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}kA^2 = \text{constant}$$

the total mechanical energy of SHM is conserved if no external force acts on it. $E \propto A^2$: doubling the amplitude makes the energy four times larger.

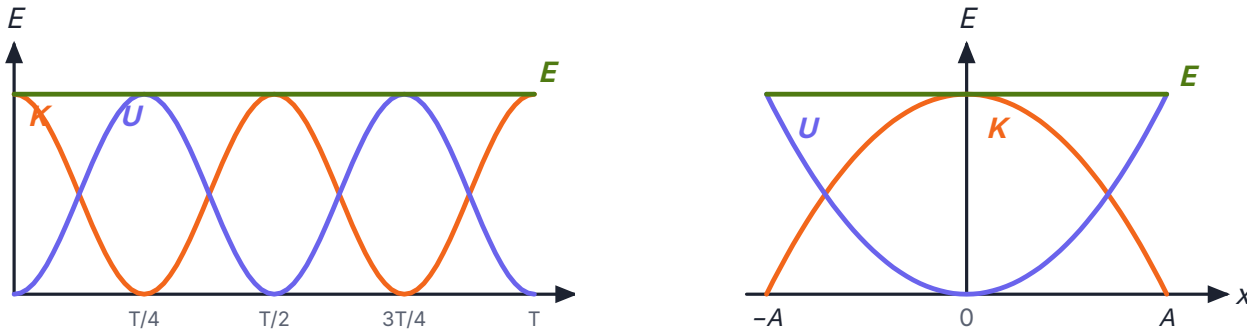


Figure 20.5: left, K and U against time (they vary at twice the frequency of x); right, K and U against displacement (two parabolas). E is constant in both.

t	x	K	U
0	A	0	$\frac{1}{2}kA^2$
$T/4$	0	$\frac{1}{2}kA^2$	0
$T/2$	$-A$	0	$\frac{1}{2}kA^2$
$3T/4$	0	$\frac{1}{2}kA^2$	0
T	A	0	$\frac{1}{2}kA^2$

(This table starts from $x = A$, i.e. $x = A\cos\omega t$.)

Misconception	Correct idea
Total energy changes as the object oscillates.	$E = \frac{1}{2}m\omega^2A^2$ is constant. Only K and U change.
Kinetic energy is greatest at the extremes.	K is greatest at $x = 0$ and zero at $x = \pm A$.
K–t and U–t graphs look like the x–t graph.	They vary at twice the frequency of x and are never negative.

Example 20.3.1

A 1.0 kg mass vibrates according to $y = 0.42\cos 7.4t$ (y in m, t in s). Determine (a) the amplitude, (b) the frequency, (c) the total energy, (d) the kinetic energy and the potential energy when $y = 0.16$ m.

Solution

(a) A = **0.42 m** (b) $f = \omega/2\pi = 7.4/2\pi$ = **1.2 Hz**

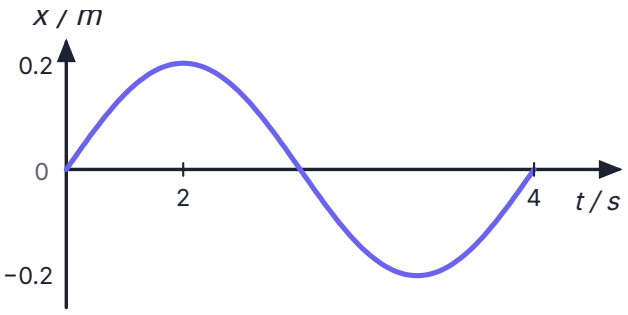
(c) $E = \frac{1}{2}m\omega^2A^2 = \frac{1}{2}(1.0)(7.4)^2(0.42)^2$ = **4.8 J**

(d) $v = \omega\sqrt{A^2 - y^2} = 7.4\sqrt{0.42^2 - 0.16^2}$ = 2.87 m s^{-1}; $K = \frac{1}{2}(1.0)(2.87)^2$ = **4.1 J**

$U = E - K = 4.8 - 4.1$ = **0.7 J**

Example 20.3.2

The displacement–time graph of an oscillating object of mass 0.3 kg is shown. Calculate the total energy (kinetic + potential) of the object.



Solution

A = 0.2 m, T = 4 s ⇒ ω = 2π/T = π/2 rad s^{-1}

$E = \frac{1}{2}m\omega^2A^2 = \frac{1}{2}(0.3)(\pi/2)^2(0.2)^2$ = **0.015 J**

STPM 2022 · Section A · Q1

An object of mass 0.10 kg undergoes SHM along the x-axis according to the equation $d^2x/dt^2 + 100x = 0$. If the total energy of the system is 0.0020 J, what is the amplitude of the motion?

A. 0.040 cm

B. 0.20 cm

C. 0.40 cm

D. 2.0 cm

Answer

Answer: D

$d^2x/dt^2 = -100x \Rightarrow \omega^2 = 100$.

$A^2 = 2E/(m\omega^2) = 2(0.0020)/(0.10 \times 100) = 4.0 \times 10^{-4}$ m^2 ⇒ A = 0.02 m = 2.0 cm.

STPM 2023 · Section A · Q1

Which statement is true about simple harmonic motion?

A. The acceleration is uniform.

B. The amplitude decreases with time.

C. The restoring force dissipates naturally.

D. The kinetic and potential energies interchange continuously.

Answer

Answer: D

A: the acceleration varies ($a = -\omega^2 x$). B and C describe **damped** oscillation, not ideal SHM. D: the total energy is constant while K and U trade places.

Tip: "true about SHM" questions often hide a damped-oscillation distractor.

Exercises: kinematics and energy of SHM

1. A particle moves in SHM according to $y = 10 \sin \omega t$ m. If the period is 30 s, find (a) the amplitude, (b) the maximum velocity, (c) the maximum acceleration, (d) the displacement, velocity and acceleration when $t = 15$ s.

   Answer

   (a) 10 m (b) 2.1 m s^{-1} (c) 0.44 m s^{-2} (d) at $t = T/2$: $y = 0$, $v = -2.1$ m s^{-1}, $a = 0$

2. An object moving with SHM has an amplitude of 2 cm and a frequency of 20 Hz. Calculate (a) the period, (b) the acceleration at the middle and at the end of an oscillation, (c) the velocities at the same two points.

   Answer

   (a) 0.05 s (b) middle: 0; end: 316 m s^{-2} (c) middle: 2.5 m s^{-1}; end: 0

3. A block of mass 0.25 kg on a frictionless surface is attached to a light spring of spring constant 180 N m^{-1}. It is displaced 15 cm from equilibrium and released. Find (a) the total energy of the system, (b) the speed of the block when it is 10 cm from equilibrium.

   Answer

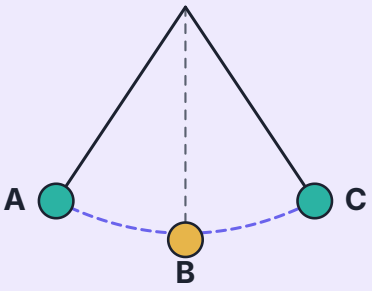
   (a) 2.0 J (b) 3.0 m s^{-1}

4. A 50 g mass vibrates in SHM at the end of a spring. The amplitude is 12 cm and the period is 1.70 s. Calculate (a) the frequency, (b) the spring constant, (c) the maximum speed, (d) the maximum acceleration, (e) the speed when the displacement is 6.0 cm, (f) the acceleration when $x = 6.0$ cm.

   Answer

   (a) 0.588 Hz (b) 0.68 N m^{-1} (c) 0.44 m s^{-1} (d) 1.6 m s^{-2} (e) 0.38 m s^{-1} (f) −0.82 m s^{-2}

5. The diagram shows a pendulum oscillating between A and C; B is the lowest point. State all the positions where (a) the acceleration is maximum, (b) the acceleration is zero, (c) the speed is maximum, (d) the velocity is zero, (e) the kinetic energy is maximum, (f) the potential energy is maximum.



Answer

(a) A and C (b) B (c) B (d) A and C (e) B (f) A and C

20.4 Systems in simple harmonic motion

Spring–mass system

Hooke's law and Newton's second law: $ma = -kx \Rightarrow a = -\frac{k}{m}x$

Compare with $a = -\omega^2 x$: $\omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}}$

$T = \frac{2\pi}{\omega}$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

a larger mass gives a longer period; a stiffer spring (larger k) gives a shorter period. T does not depend on the amplitude.

For a **vertical** spring, gravity only shifts the equilibrium position down by mg/k. The mass oscillates about this new equilibrium with the same period, $T = 2\pi\sqrt{(m/k)}$. g does not appear in the formula.

Example 20.4.1

A mass of 200 g is attached to a horizontal spring on a frictionless surface. The mass is pulled 2.00 cm to the right of its equilibrium position and released. If the spring constant is 100 N m^{-1}, find the frequency of oscillation.

Solution

$\omega^2 = k/m$

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{100}{0.200}} = \mathbf{3.56\ Hz}$$ (the 2.00 cm amplitude is not needed)

Example 20.4.2

A 0.50 kg mass is attached to a spring of spring constant 20 N m^{-1} and set oscillating. Find the period. What happens to the period if the spring is replaced by one four times as stiff?

Solution

$T = 2\pi\sqrt{0.50/20} = 2\pi(0.158) = \mathbf{0.99\ s}$

$T \propto 1/\sqrt{k}$, so with k = 80 N m^{-1} the period **halves** (0.50 s).

Simple pendulum

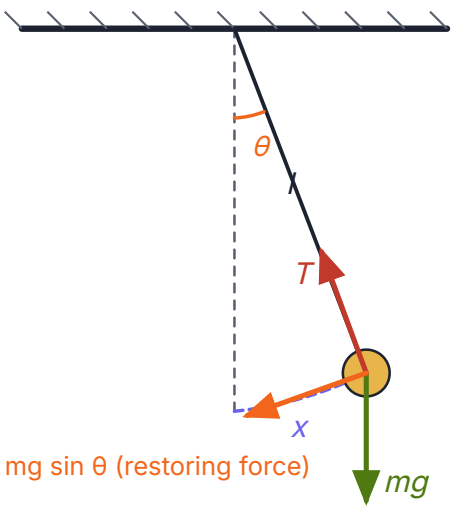


Figure 20.6: the component mg sin θ along the arc pulls the bob back towards the equilibrium position.

Restoring force along the arc: $F = -mg \sin\theta$

For small angles ($\theta < 10°$), $\sin\theta \approx \theta = \frac{x}{l}$ (θ in radians)

$$ma = -mg\frac{x}{l} \Rightarrow a = -\frac{g}{l}x$$

Compare with $a = -\omega^2 x$: $\omega = \sqrt{\frac{g}{l}}$

$$T = 2\pi\sqrt{\frac{l}{g}}$$

depends only on the length l and on g, not on the mass of the bob or (for small angles) on the amplitude

	Spring–mass system	Simple pendulum
Period	$T = 2\pi\sqrt{(m/k)}$	$T = 2\pi\sqrt{(l/g)}$
Depends on	mass m, spring constant k	length l, gravitational acceleration g
Does not depend on	amplitude A (and g)	mass of the bob, amplitude (small angles)

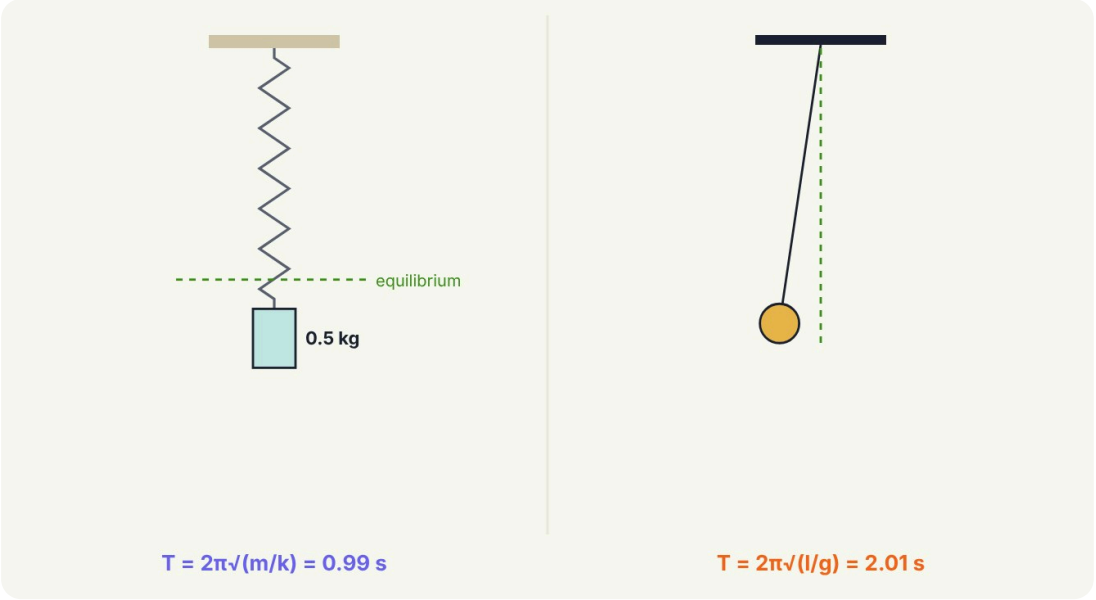
Teacher's note

A heavier phone on a spring bounces more slowly, but a heavier pendulum bob swings at exactly the same rate. Always check which system the question describes before you choose the formula. On the Moon (smaller g) a pendulum clock runs slow; a spring–mass system keeps the same period.

Simulation

Spring–mass system and simple pendulum

Change the mass, spring constant, length and g, and compare the two periods. Change the amplitude too: the period does not change. Try the Moon button.



spring–mass period	spring–mass frequency	pendulum period	pendulum frequency
0.99 s	**1.01 Hz**	**2.01 s**	**0.50 Hz**

the pendulum does not depend on **mass, amplitude**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

Example 20.4.3

A simple pendulum has a length of 1.00 m. Taking $g = 9.81\ \text{m s}^{-2}$, find its period. What length gives a period of 1.0 s?

Solution

$T = 2\pi\sqrt{1.00/9.81} = \mathbf{2.01\ s}$

$l = g(T/2\pi)^2 = 9.81(1.0/2\pi)^2 = \mathbf{0.25\ m}$

Example 20.4.4

A student uses a simple pendulum of length 80.0 cm to determine g. There are 20 oscillations in 35.9 s. Find (a) the value of g, (b) the period if the experiment is done on the Moon, where the gravitational field strength is 1/6 of that on Earth.

Solution

T = 35.9/20 = 1.795 s

(a) $T^2 = 4\pi^2 \frac{l}{g} \Rightarrow g = \frac{4\pi^2(0.800)}{(1.795)^2}$ = **9.80 m s^{-2}**

(b) $T^2 g$ = constant: $T^2(g/6) = (1.795)^2 g \Rightarrow T = 1.795\sqrt{6}$ = **4.40 s**

20.5 Damped oscillation

Damping

is **the decrease in amplitude of an oscillation caused by dissipative forces** such as friction and air resistance.

- Part of the energy of the system is used to overcome these resistive forces, and is lost as heat.
- So the amplitude does not stay constant: it decreases gradually (exponentially) until the oscillation stops. Since $E \propto A^2$, the energy decreases too.
- A damped oscillation is therefore not perfect SHM.

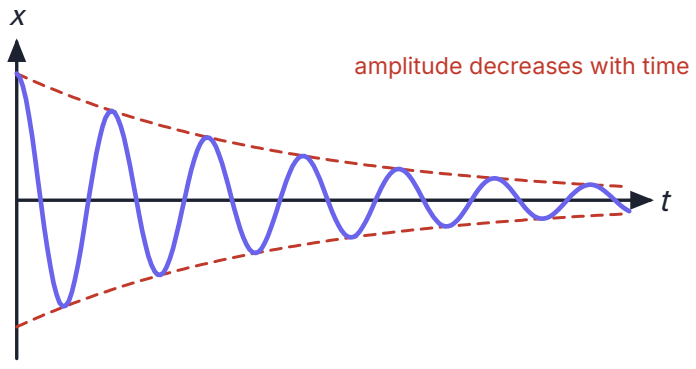


Figure 20.7: under damping: the oscillation continues but the amplitude decreases with time.

Three types of damping

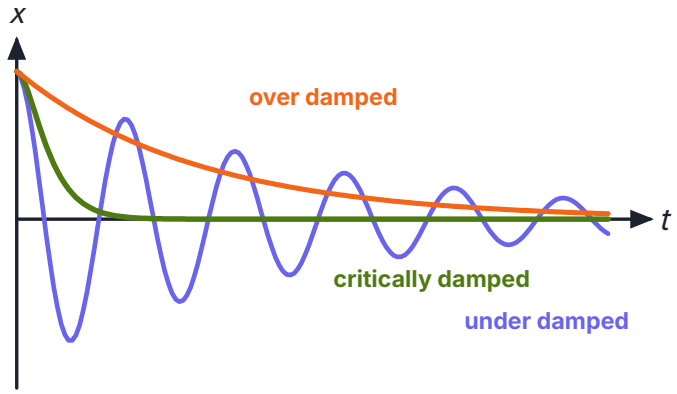


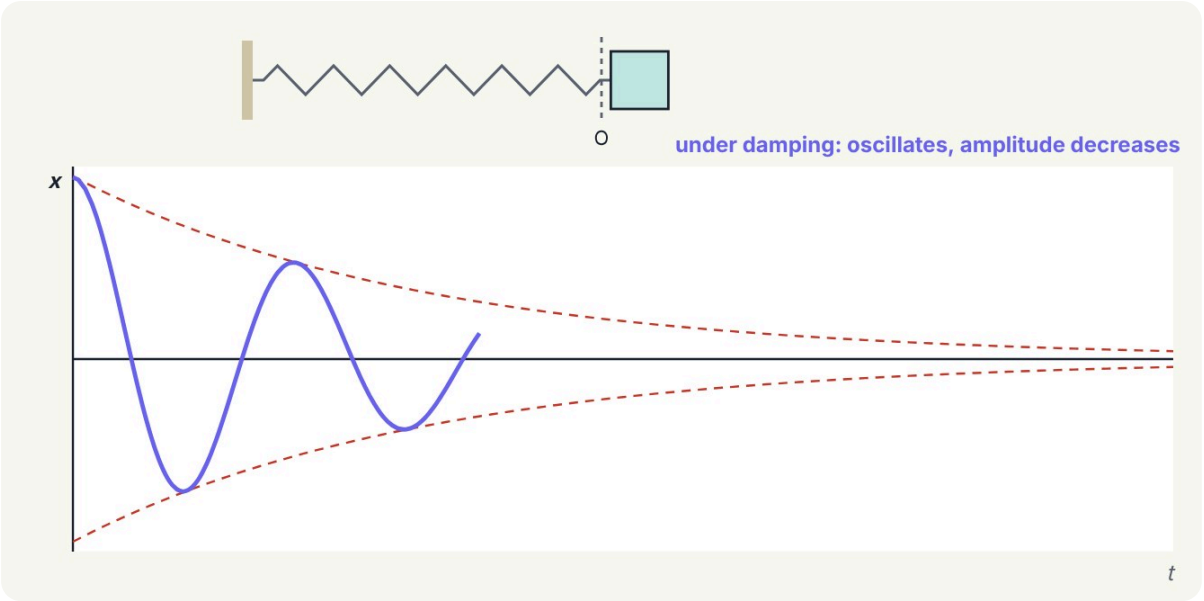
Figure 20.8: displacement against time for the three types of damping, each released from the same displacement.

Type	What happens	Example
Under damping (light damping)	The system oscillates, with the amplitude decreasing with time until it becomes zero. It takes a long time to die away.	a mass oscillating in air; a plucked guitar string
Critical damping	The system does not oscillate. It returns to the equilibrium position in the **shortest possible time**.	the shock absorbers of a car
Over damping (heavy damping)	The system does not oscillate, but it takes a **long time** to return to equilibrium.	a car fuel gauge; bathroom scales; a heavy door closer

Simulation

Damped oscillation

Increase the damping from none to under, critical and over damping. Watch the mass, the displacement–time graph and the energy. Critical damping brings the mass back to O fastest without overshooting.



time	displacement	energy left
3.0 s	0.14 × A_0	9 %

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

20.6 Forced oscillation and resonance

Free oscillation	Forced oscillation
The system is displaced and released, then left alone: no driving force.	A periodic external force (the driving force) acts on the system continuously.
It oscillates at its own **natural frequency** f_0.	After a short time, it oscillates at the **driving (forcing) frequency** f, not at f_0.
Its amplitude depends only on the energy given at the start (it is damped gradually if there is resistance).	Its amplitude depends on how close f is to f_0.

Natural frequency

is **the frequency at which a system oscillates when there is no driving force and no damping**. For a simple pendulum, $f_0 = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$.

Barton's pendulums

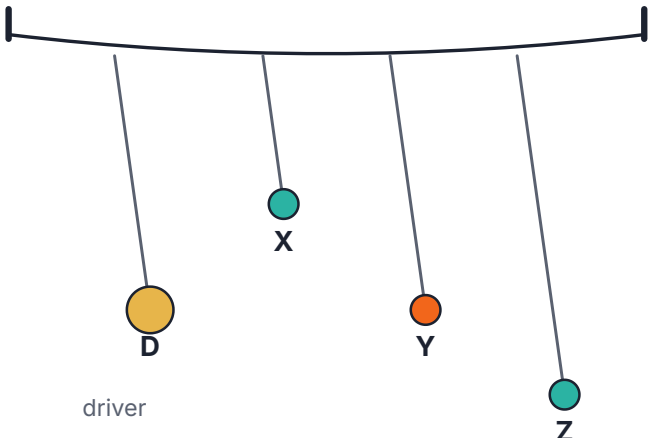


Figure 20.9: the driver pendulum D and pendulums X, Y, Z hang from the same string. Only Y has the same length as D.

- X, Y and Z are at rest at first. D is set swinging.
- Energy is passed along the string, so X, Y and Z start to oscillate (forced oscillation), all at the frequency of D.
- Y, with the same length (and so the same natural frequency) as D, oscillates with the **largest amplitude**. Y is in **resonance**.

Resonance

occurs when **the frequency of the driving force is equal to the natural frequency of the system**. The system then absorbs energy from the driving force most efficiently, and the amplitude of oscillation is maximum.

resonance: $f = f_0$

the condition for resonance. This frequency is the resonance frequency.

Resonance and damping

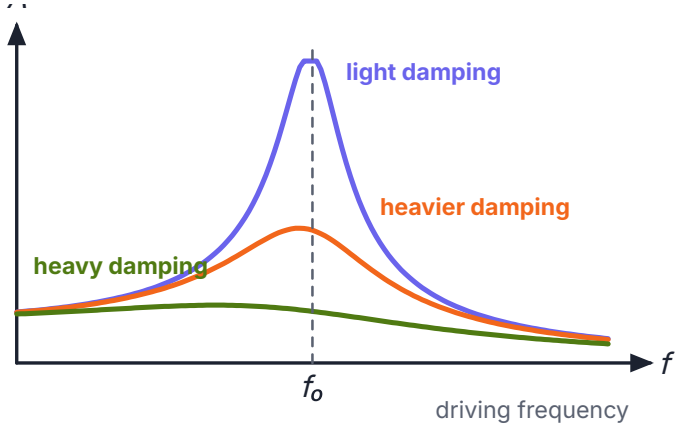
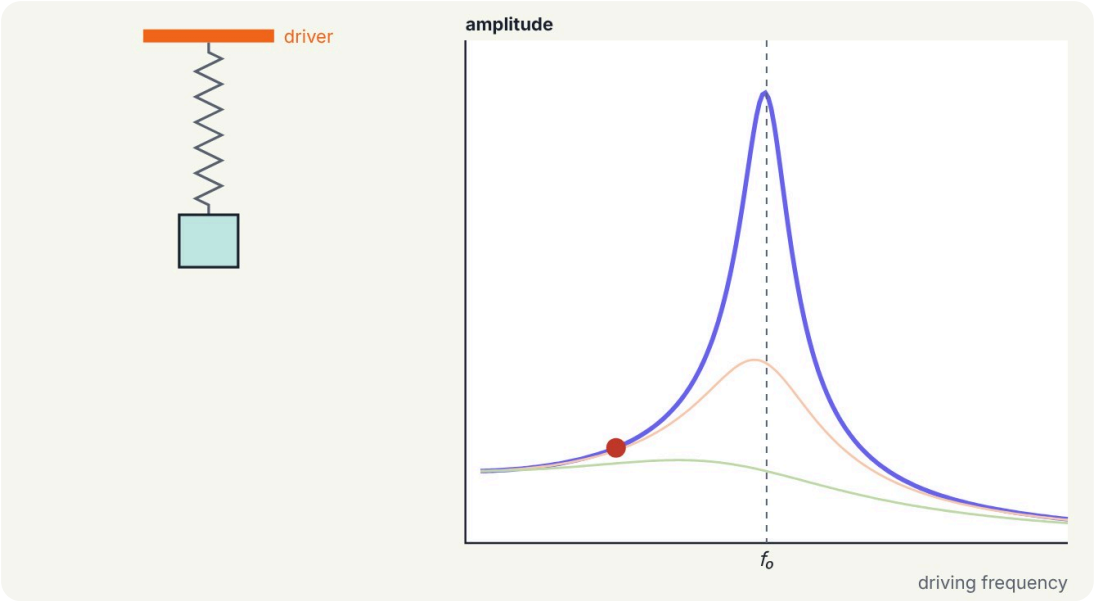


Figure 20.10: amplitude of the forced oscillation against driving frequency for different amounts of damping.

- Damping absorbs energy and reduces the effect of resonance.
- As the damping increases, the amplitude at resonance decreases and the resonance peak becomes broader (flatter).
- With heavier damping, the peak also moves slightly below f_0.

Simulation Forced oscillation and resonance

Slowly change the driving frequency and watch the amplitude of the driven mass. The amplitude is largest when the driving frequency equals the natural frequency. Then increase the damping and see the peak fall and become broader.



natural frequency f_o	driving frequency f	steady amplitude	the mass oscillates at
1.00 Hz	**0.50 Hz**	**1.3 × driver**	**0.50 Hz (= f)**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

Examples of resonance

Destructive

- 1940, Washington State, USA: the Tacoma Narrows suspension bridge collapsed about four months after opening. A strong wind drove the bridge into large oscillations.
- 1850, Angers, France: more than 200 soldiers died when a bridge they were marching over collapsed. This is why soldiers break step when they march across a bridge.
- A singer can shatter a wine glass with a note at its natural frequency.

Useful

- The tuning circuit of a radio or TV resonates at the frequency of the chosen station.
- Wind instruments produce sound by making the air column inside resonate.
- Microwave ovens heat food because water molecules absorb energy from the microwaves.

Teacher's note

Questions often ask for the condition for resonance. Write it as a full sentence: "the driving frequency is equal to the natural frequency of the system".

Summary

Subtopic	Key results
20.1	$F = -kx$; $a = -\omega^2 x$
20.2	$x = A \sin \omega t$ or $A \cos \omega t$; $v = \pm\omega\sqrt{(A^2 - x^2)}$; $v_{max} = \omega A$; $a_{max} = \omega^2 A$
20.3	$K = \frac{1}{2}m\omega^2(A^2 - x^2)$; $U = \frac{1}{2}m\omega^2 x^2$; $E = \frac{1}{2}m\omega^2 A^2$ (constant)
20.4	$T = 2\pi\sqrt{(m/k)}$ (spring–mass); $T = 2\pi\sqrt{(l/g)}$ (simple pendulum)
20.5	amplitude and energy decrease; under, critical and over damping
20.6	resonance when the driving frequency = natural frequency

- Always match the given equation to $a = -\omega^2 x$ first to find ω.
- Keep every quantity in SI units (m, s) before substituting.
- Use radians on your calculator for $\sin \omega t$ and $\cos \omega t$.
- T is independent of amplitude in SHM; it is damping that makes the amplitude decrease.

Chapter 21 · Semester 3

WAVE MOTION

Notes by Cikgu Suhaili

10 worked examples 5 simulations Exercises with answers

Waves: introduction

A wave transfers **energy** from one place to another without transferring matter. The particles of the medium only vibrate about their equilibrium positions.

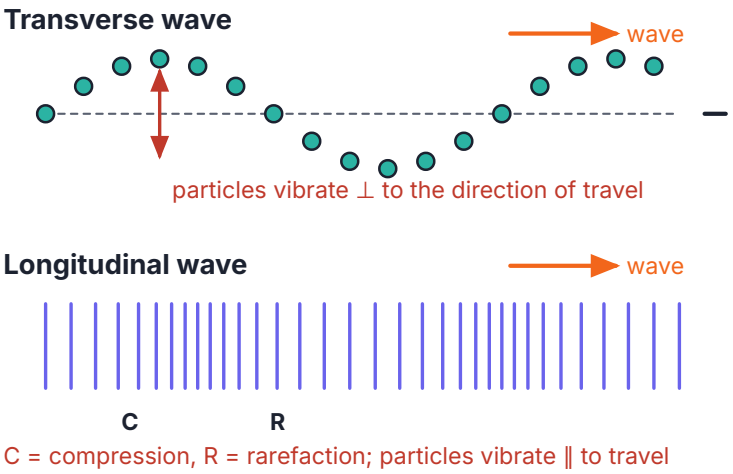


Figure 21.1: transverse and longitudinal waves.

Transverse wave	Longitudinal wave
The particles vibrate **perpendicular** to the direction of propagation.	The particles vibrate **parallel** to the direction of propagation.
crests and troughs	compressions and rarefactions
e.g. waves on a string, water waves, electromagnetic waves	e.g. sound waves, waves along a slinky spring pushed back and forth

21.1 Progressive waves

Progressive wave

is **a wave in which the wave profile moves through the medium, transferring energy from one point to another**.

Equation of a progressive wave

If the particle at $x = 0$ moves in SHM, $y = A \sin \omega t$, the disturbance reaches a particle at distance x a time x/v later. So the particle at x does the same motion, but later:

$$y = A \sin \omega\left(t - \frac{x}{v}\right) = A \sin \left(\omega t - \frac{\omega}{v} x\right)$$

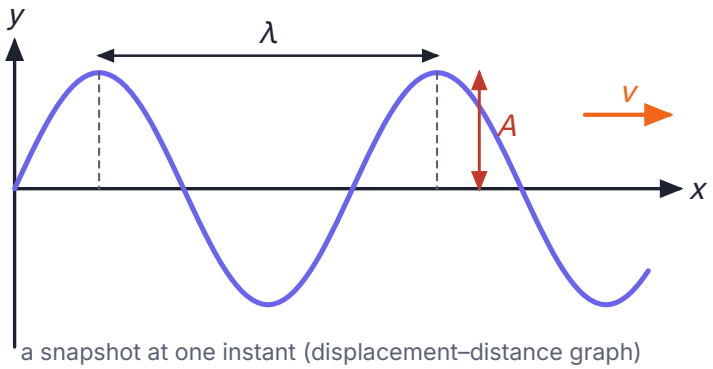
With $k = \frac{\omega}{v} = \frac{2\pi}{\lambda}$ (the wave number):

$$y = A \sin (\omega t - kx) \quad \text{or} \quad y = A \cos (\omega t - kx)$$

$(\omega t - kx)$: wave travelling in the $+x$ direction. $(\omega t + kx)$: wave travelling in the $-x$ direction. A = amplitude, $\omega = 2\pi f$, $k = 2\pi/\lambda$ (rad m^{-1}).

Quantity	Meaning
Wavelength, λ (m)	the distance between two successive points that vibrate in phase (e.g. crest to crest)
Period, T (s)	the time for one complete oscillation of a particle
Frequency, f (Hz)	the number of oscillations per second; $f = 1/T$
Wave number, k (rad m^{-1})	$k = 2\pi/\lambda$
Wave speed, v ($m\ s^{-1}$)	the distance moved by the wave profile per second

Displacement–distance and displacement–time graphs



a snapshot at one instant (displacement–distance graph)

Figure 21.2: the displacement–distance (y–x) graph is a snapshot of all the particles at one instant. The distance for one cycle is λ.

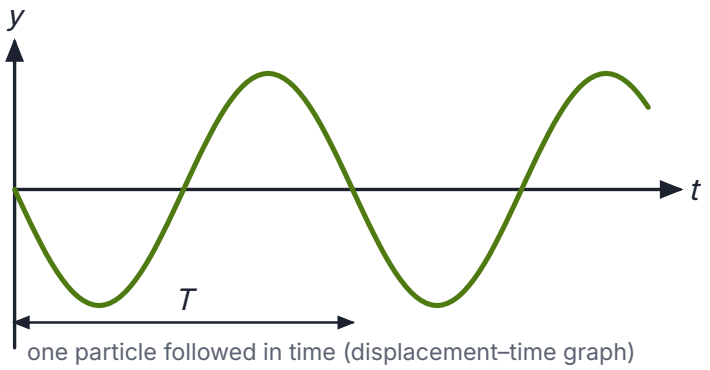


Figure 21.3: the displacement–time (y–t) graph follows one particle. The time for one cycle is T.

Teacher's note

Read the axis first. On a y–x graph, one cycle is the wavelength λ. On a y–t graph, one cycle is the period T. Many students mix these up.

Phase difference

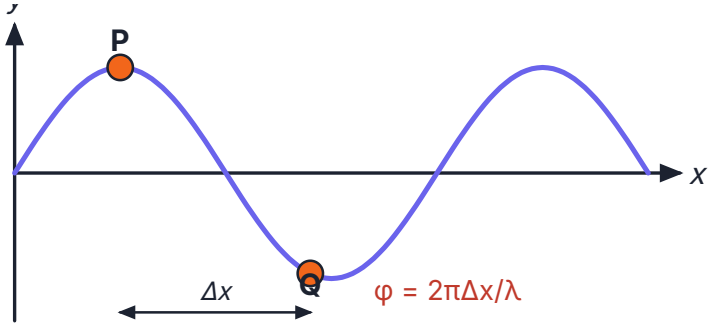


Figure 21.4: two points P and Q a distance Δx apart on the same wave.

At P (x_1): phase = $\omega t - kx_1$; at Q (x_2): phase = $\omega t - kx_2$

phase difference $\varphi = k(x_2 - x_1) = k\Delta x$

$$\varphi = kx = \frac{2\pi x}{\lambda}$$

the phase difference between two points a distance x apart. Points λ apart are in phase ($\varphi = 2\pi$); points λ/2 apart are in antiphase ($\varphi = \pi$).

Wave speed

In one period T, the wave profile moves forward one wavelength λ.

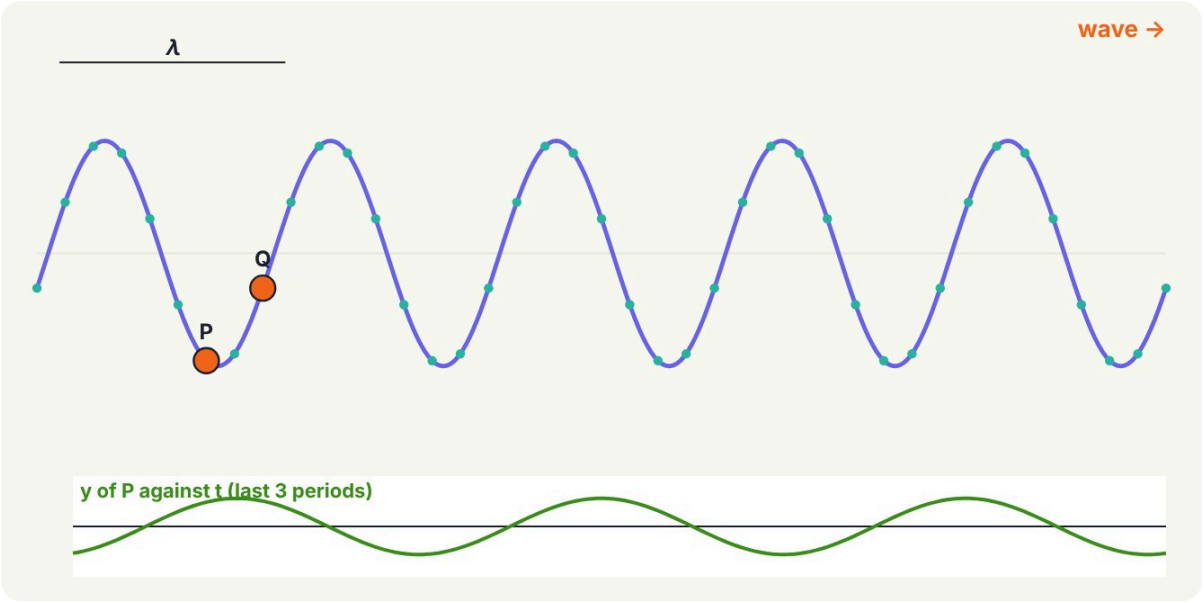
$v = \frac{\lambda}{T} = f\lambda$ (also $v = \frac{\omega}{k}$)

$$v = f\lambda$$

The **wave speed** $v = f\lambda$ is constant in a given medium. The **particle velocity** $\partial y/\partial t = A\omega \cos(\omega t - kx)$ changes all the time, with maximum $A\omega$. Do not mix them up.

Simulation **Progressive wave**

Change the amplitude, wavelength, frequency and direction. Follow the marked particle: it only moves up and down (or back and forth for a longitudinal wave) while the wave profile moves along. The two orange points show the phase difference.



v = fλ	k = 2π/λ	ω = 2πf	T = 1/f
1.00 m/s	**3.14 rad/m**	**3.14 rad/s**	**2.00 s**

phase difference P–Q (2πΔx/λ)	P and Q
1.57 rad = 0.25 cycle	**out of phase**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 21 to use the simulation.

Example 21.1.1

A progressive wave is represented by $y = 0.05 \sin(8\pi t - 0.4\pi x)$, where x and y are in metres and t in seconds. Find (a) the amplitude, (b) the frequency, (c) the wavelength, (d) the wave speed and its direction, (e) the maximum speed of a particle.

Solution

Compare with $y = A \sin(\omega t - kx)$.

(a) A = **0.05 m** (b) $\omega = 8\pi \Rightarrow f$ = **4 Hz**

(c) $k = 0.4\pi = 2\pi/\lambda \Rightarrow \lambda$ = **5 m**

(d) $v = f\lambda$ = **20 m s^{-1}**, in the **+x** direction (minus sign)

(e) $v_{max} = A\omega = 0.05 \times 8\pi$ = **1.26 m s^{-1}**

Example 21.1.2

A wave of frequency 500 Hz travels at 350 m s^{-1}. (a) What is the distance between two points that are $\pi/3$ rad out of phase? (b) What is the phase difference between two displacements at one point 1.0 ms apart?

Solution

$\lambda = v/f = 350/500 = 0.70$ m

(a) $\varphi = 2\pi x/\lambda \Rightarrow x = (\pi/3)(0.70)/(2\pi)$ = **0.117 m**

(b) $\varphi = \omega t = 2\pi(500)(1.0 \times 10^{-3})$ = **π rad**

21.2 Wave intensity

Intensity of a wave

is **the rate of energy transferred (power) per unit area perpendicular to the direction of propagation**.

$$I = \frac{P}{A}$$

unit: W m^{-2}

Intensity and amplitude

The energy of a particle in SHM is $\frac{1}{2}m\omega^2A^2$, so for waves of the same frequency the energy carried is proportional to the square of the amplitude:

$$I \propto A^2$$

doubling the amplitude makes the intensity four times larger

Intensity from a point source

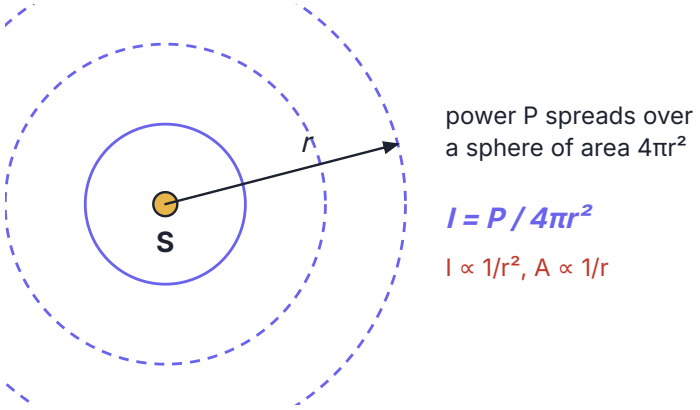


Figure 21.5: a point source radiates power P equally in all directions.

- The energy spreads out over the surface of a sphere of area $4\pi r^2$, so $I = \frac{P}{4\pi r^2}$.
- The intensity therefore decreases with distance: $I \propto \frac{1}{r^2}$ (the inverse square law).
- Since $I \propto A^2$, the amplitude decreases as $A \propto \frac{1}{r}$.

Example 21.2.1

A point source emits sound of power 2.0 W uniformly in all directions. Find (a) the intensity 5.0 m from the source, (b) the ratio of the amplitudes at 5.0 m and at 10.0 m.

Solution

(a) $I = \frac{P}{4\pi r^2} = \frac{2.0}{4\pi(5.0)^2} = \mathbf{6.4 \times 10^{-3}\ W\ m^{-2}}$

(b) $A \propto 1/r \Rightarrow A_5 : A_{10} = 10 : 5 = \mathbf{2 : 1}$

Example 21.2.2

The amplitude of a wave is reduced to 60% of its original value. What percentage of the original intensity remains?

Solution

$I \propto A^2 \Rightarrow I'/I = (0.60)^2 = 0.36$, i.e. **36%**

21.3 Principle of superposition

Principle of superposition

states that **when two or more waves meet at a point, the resultant displacement at that point is the vector sum of the displacements due to each individual wave**.

$$y = y_1 + y_2$$

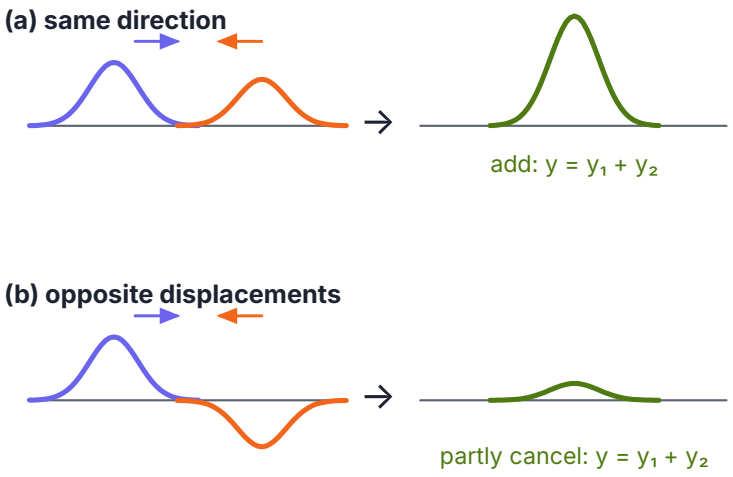


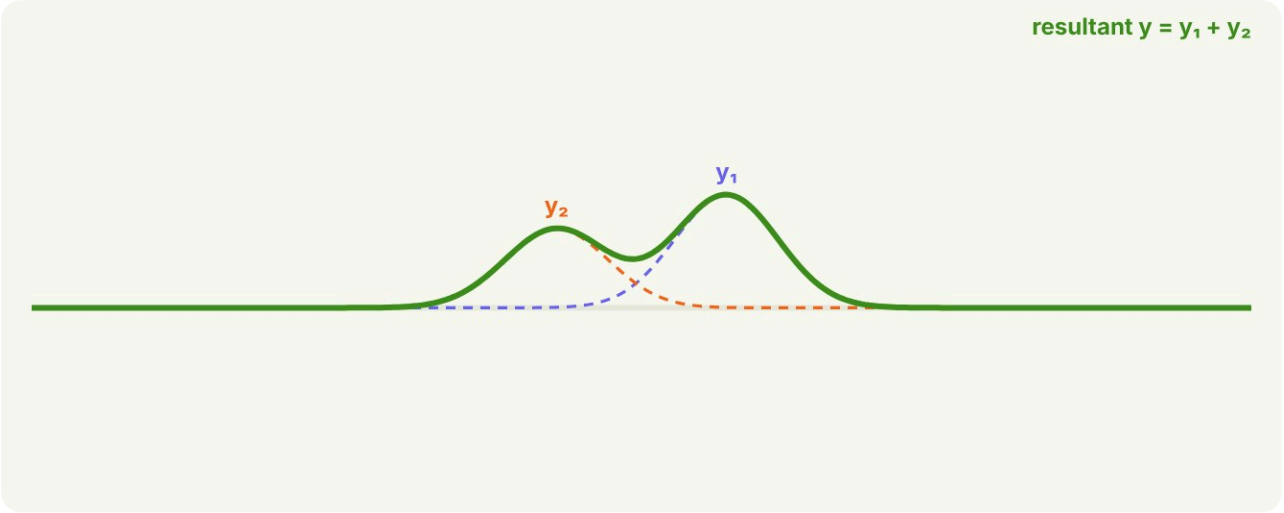
Figure 21.6: two pulses meeting. (a) Displacements in the same direction add. (b) Displacements in opposite directions partly cancel. After meeting, each pulse continues unchanged.

- The waves **interfere** with each other only while they overlap.
- **Constructive interference**: the displacements are in the same direction, giving a larger resultant.
- **Destructive interference**: the displacements are in opposite directions, giving a smaller resultant (zero if they are equal and opposite).

Simulation

Superposition of two pulses

Send two pulses towards each other, on the same side or on opposite sides. Pause while they overlap: the resultant (green) is the sum of the two displacements at every point. After they pass, each pulse is unchanged.



This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 21 to use the simulation.

Example 21.3.1

Two waves $y_1 = 3 \sin(\omega t - kx)$ cm and $y_2 = 4 \sin(\omega t - kx + \pi)$ cm travel along the same string. Find the amplitude of the resultant wave.

Solution

> $\sin(\theta + \pi) = -\sin\theta$, so $y_2 = -4 \sin(\omega t - kx)$
>
> $y = y_1 + y_2 = -1 \sin(\omega t - kx) \Rightarrow$ amplitude **1 cm** (destructive interference)

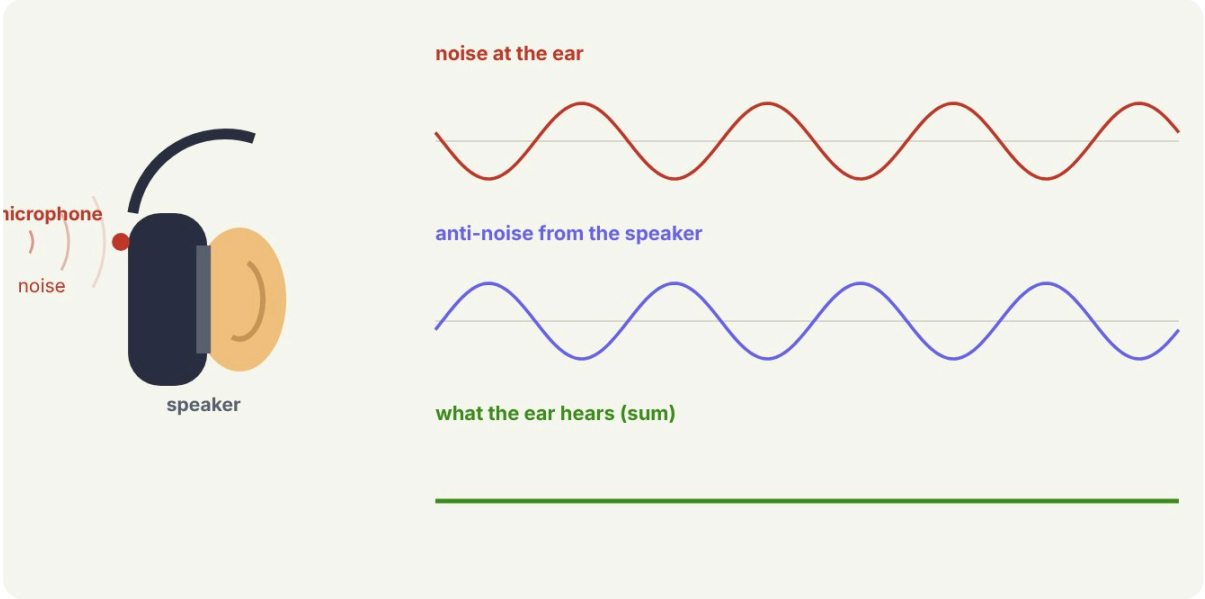
A real use: noise-cancelling headphones

- A small microphone on the outside of each ear cup picks up the noise around you (an aeroplane engine, a bus, an air-conditioner).
- The electronics produce an "anti-noise" wave with the **same amplitude** but **opposite phase** (a phase difference of π), and the speaker plays it into your ear.
- At your ear the two waves superpose: $y = y_{\text{noise}} + y_{\text{anti}} \approx 0$. This is **destructive interference**.
- It works best for steady, low-frequency sounds. For high frequencies even a tiny delay in the electronics is a large fraction of a period, so the anti-noise is no longer in antiphase.

Simulation

Noise-cancelling headphones

Switch the noise cancelling on and off and compare the sound reaching the ear (green). Then add a delay in the electronics or change the noise frequency: the cancellation gets worse once the phase difference moves away from π. Switch on the sound to hear it (use earphones for the low hum).



phase difference between noise and anti-noise
180°

amplitude reaching the ear
0% of the noise

delay as a fraction of one period
0.00 T

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 21 to use the simulation.

21.4 Standing waves

> **Standing (stationary) wave**
> is formed by **the superposition of two progressive waves of the same amplitude and frequency, travelling at the same speed in opposite directions**.

Formation by reflection

A wave travelling along a string is reflected at the end. The incident and reflected waves travel in opposite directions and superpose to form a standing wave.



Figure 21.7: a pulse is inverted when it is reflected at a fixed end, and is not inverted at a free end.

- At some points the two waves are always equal and opposite, so they always cancel: these points do not move. They are **nodes** (N).
- Midway between the nodes, the two waves always add, giving the largest oscillation. These points are **antinodes** (A).

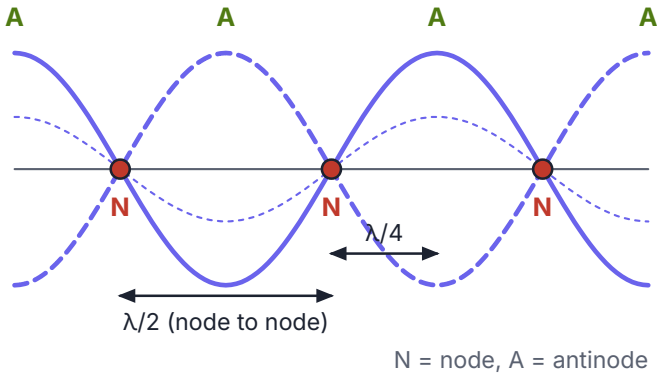


Figure 21.8: a standing wave. The profile does not move along; each point oscillates with its own amplitude.

The standing wave equation

$y_1 = a \sin(\omega t - \frac{2\pi x}{\lambda})$ (to the right), $y_2 = a \sin(\omega t + \frac{2\pi x}{\lambda})$ (to the left)

$y = y_1 + y_2$, using $\sin P + \sin Q = 2 \sin \frac{P+Q}{2} \cos \frac{P-Q}{2}$

$y = 2a \sin \omega t \cos(-\frac{2\pi x}{\lambda}) = 2a \cos \frac{2\pi x}{\lambda} \sin \omega t$

$$y = 2a \cos \frac{2\pi x}{\lambda} \sin \omega t = (2a \cos kx) \sin \omega t$$

every point oscillates in SHM ($\sin \omega t$) with an amplitude $A = 2a \cos kx$ that depends on its position x

Other books write $y = 2A \sin kx \cos \omega t$. It is the same standing wave with the origin at a node instead of an antinode.

Nodes: amplitude = 0

$\cos \frac{2\pi x}{\lambda} = 0 \Rightarrow \frac{2\pi x}{\lambda} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$

$x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \ldots = \frac{n\lambda}{4}, n = 1, 3, 5, \ldots$

distance between successive nodes $= \frac{3\lambda}{4} - \frac{\lambda}{4} = \boldsymbol{\frac{\lambda}{2}}$

Antinodes: amplitude = $2a$ (maximum)

$\cos \frac{2\pi x}{\lambda} = \pm 1 \Rightarrow \frac{2\pi x}{\lambda} = 0, \pi, 2\pi, \ldots$

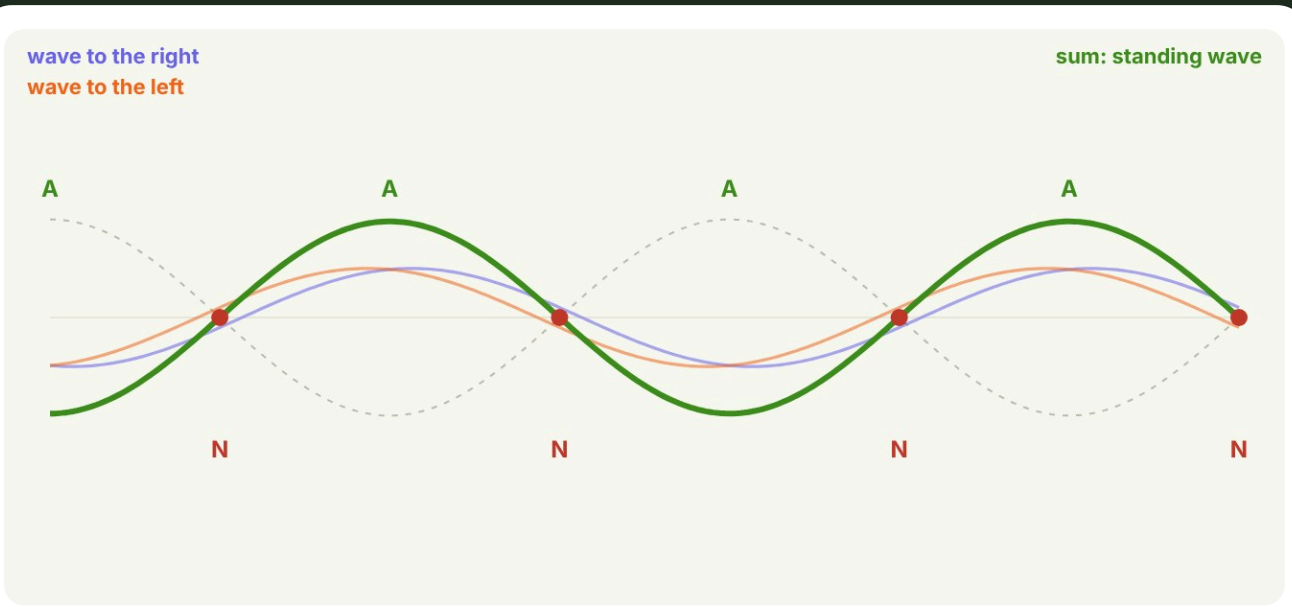
$x = 0, \frac{\lambda}{2}, \lambda, \ldots = \frac{n\lambda}{4}, n = 0, 2, 4, \ldots$

distance between successive antinodes $= \boldsymbol{\frac{\lambda}{2}}$;

node to next antinode $= \boldsymbol{\frac{\lambda}{4}}$

Simulation Forming a standing wave

Two identical waves travel in opposite directions (blue and orange). Their sum (green) is a standing wave: the nodes never move and the antinodes have the largest amplitude. Switch to the string to see the harmonics of a string fixed at both ends.



node to node	node to antinode	amplitude at an antinode	amplitude at a node
λ/2	**λ/4**	**2a**	**0**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 21 to use the simulation.

Standing waves on a stretched string

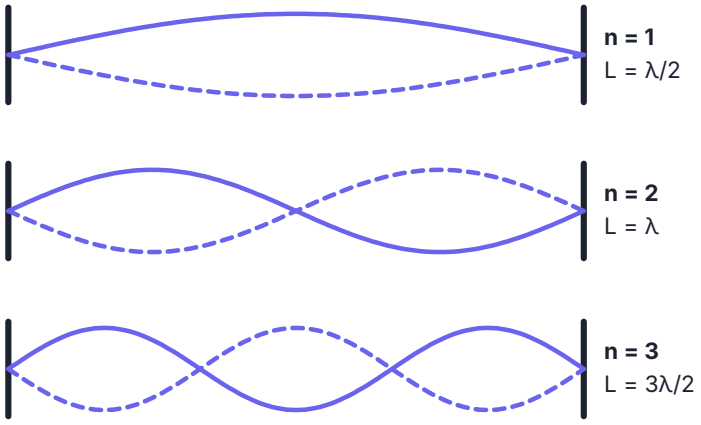


Figure 21.9: a string fixed at both ends has a node at each end. Only waves that fit a whole number of half-wavelengths can form.

- The lowest frequency (the **fundamental frequency** or **first harmonic**): $L = \lambda/2$, so $\lambda = 2L$ and $f_1 = v/2L$.
- Next: $L = \lambda$, the second harmonic (first overtone), $f_2 = 2f_1$.
- In general $L = n\lambda/2$ and $f_n = nv/2L$. More about strings and pipes in Chapter 22 Sound waves.

Standing waves and progressive waves

Standing wave	Progressive wave
The wave profile does not move; energy is not transferred from one particle to the next.	The wave profile moves in the direction of propagation and energy is transferred.
There are points (nodes) that are not displaced at all.	No nodes.
Midway between two successive nodes is an antinode, where the amplitude is maximum.	No antinodes.
Points between a node and an antinode have different amplitudes of oscillation.	All points have the same amplitude of oscillation.
Points between two successive nodes oscillate in phase, with the same frequency.	Neighbouring points are not in phase, but have the same frequency.

Example 21.4.1

A standing wave is represented by $y = 2\cos 4x \sin 6t$ (SI units). Calculate (a) the frequency, (b) the wavelength, (c) the distance between successive nodes, (d) the amplitude at $x = 0.50$ m.

Solution

Compare with $y = 2a\cos\dfrac{2\pi x}{\lambda}\sin\omega t$.

(a) $\omega = 6 = 2\pi f \Rightarrow f =$ **0.955 Hz**

(b) $\dfrac{2\pi}{\lambda} = 4 \Rightarrow \lambda =$ **1.57 m**

(c) $\dfrac{\lambda}{2} = \dfrac{1.57}{2} =$ **0.785 m**

(d) $A = 2\cos 4(0.50) = 2\cos 2.0 = -0.832$; amplitude = **0.832 m** (calculator in radians; the amplitude is the magnitude)

Example 21.4.2

A standing-wave pattern is observed in a thin wire of length 3.00 m. The equation of the wave is y = (0.002 m) sin (πx) cos (100πt), with x in metres and t in seconds. (a) How many loops does the pattern have? (b) What is the frequency? (c) What is the distance between adjacent nodes?



Solution

Compare with $y = 2A \sin kx \cos \omega t$.

$k = 2\pi/\lambda = \pi \Rightarrow \lambda = 2.00$ m; each loop is $\lambda/2 = 1.00$ m long

(a) number of loops = 3.00/1.00 = **3**

(b) $\omega = 2\pi f = 100\pi \Rightarrow f =$ **50.0 Hz**

(c) $\lambda/2 =$ **1.00 m**

Example 21.4.3

The speed of waves in a guitar string is 425 m s^{-1}. Find the fundamental frequency (first harmonic) of the string if its length is 76.5 cm.

Solution

$\lambda = 2L = 2(0.765) = 1.53$ m

$f = v/\lambda = 425/1.53 =$ **278 Hz**

Example 21.4.4

Find the length of guitar string needed to produce a fundamental frequency of 256 Hz, if the speed of waves in the string is 405 $\mathrm{m\,s^{-1}}$.

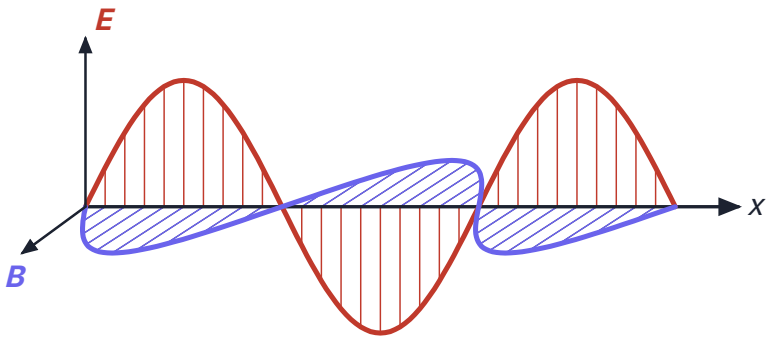
Solution

$\lambda = v/f = 405/256 = 1.58$ m

$L = \lambda/2 =$ **0.791 m**

21.5 Electromagnetic waves

An electromagnetic (EM) wave is made up of an **electric field** and a **magnetic field** that vibrate perpendicular to each other and to the direction of propagation, in phase.



E ⊥ B ⊥ direction of travel; E and B in phase

Figure 21.10: an electromagnetic wave travelling in the +x direction.

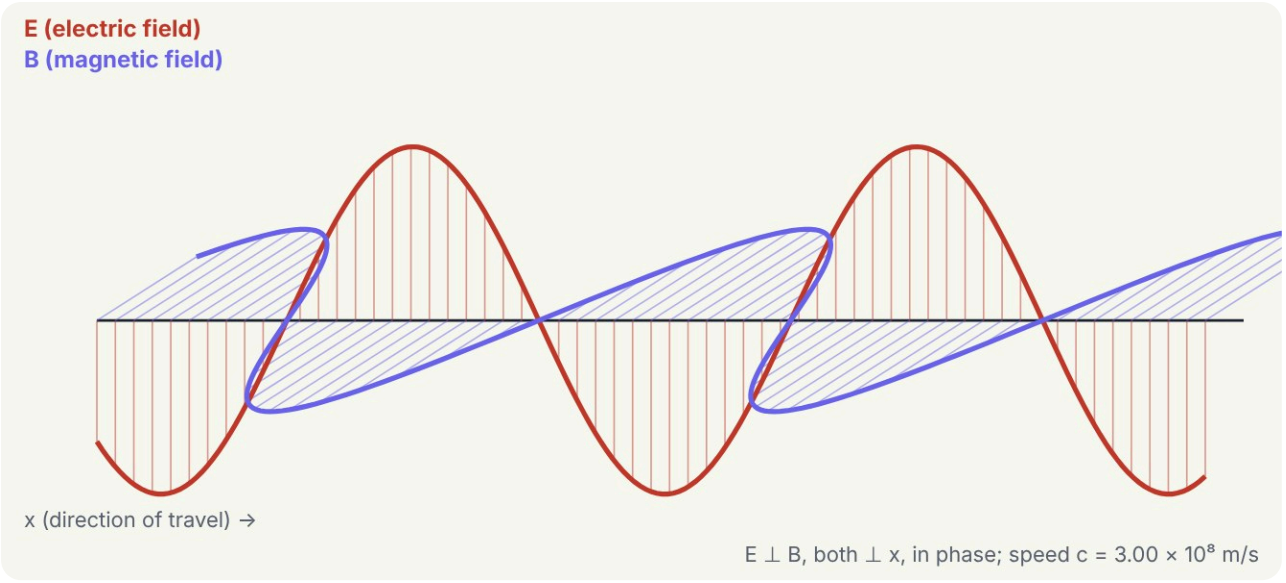
$$E = E_0 \sin(\omega t - kx) \quad B = B_0 \sin(\omega t - kx)$$

E_0, B_0 = amplitudes of the electric and magnetic fields; both have the same ω and k, so they are in phase

Simulation

Electromagnetic wave

The electric field (red) and the magnetic field (violet) vibrate at right angles to each other and to the direction of travel, and reach their maxima together.



This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 21 to use the simulation.

Characteristics of electromagnetic waves

- They are transverse waves.
- They do not need a medium: they can travel through a vacuum.
- They all travel at the same speed in a vacuum, $c = 3.00 \times 10^8\ \text{m s}^{-1}$.
- They carry no charge, so they are not deflected by electric or magnetic fields.
- They show reflection, refraction, interference, diffraction and polarisation.
- They obey $c = f\lambda$.

Electromagnetic waves and mechanical waves

Electromagnetic waves	Mechanical waves
vibrations of electric and magnetic fields	vibrations of the particles of a medium
do not need a medium; can travel in a vacuum	need a medium; cannot travel in a vacuum
always transverse	transverse or longitudinal
speed in a vacuum = 3.00×10^8 m s^{-1}	much slower (e.g. sound in air ≈ 340 m s^{-1})
e.g. light, radio waves, X-rays	e.g. sound, water waves, waves on a string

The speed of light

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1} (permeability of free space); $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} (permittivity of free space)

Significance: the speed of EM waves can be calculated from two constants of electricity and magnetism: $1/\sqrt{(4\pi \times 10^{-7} \times 8.85 \times 10^{-12})} = 3.00 \times 10^8$ m s^{-1}, which is the measured speed of light. This showed that **light is an electromagnetic wave**.

The electromagnetic spectrum

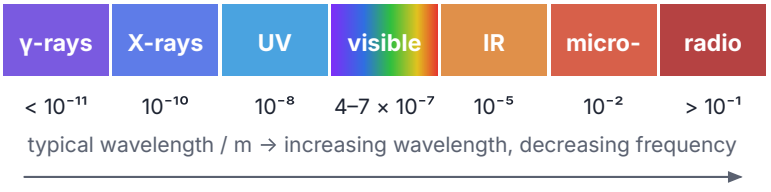


Figure 21.11: the electromagnetic spectrum.

Radiation	Typical wavelength / m	Typical frequency / Hz
γ-rays	$< 10^{-11}$	$> 10^{19}$
X-rays	$10^{-11} – 10^{-8}$	$10^{16} – 10^{19}$
ultraviolet	$10^{-8} – 4 \times 10^{-7}$	$10^{15} – 10^{16}$
visible light	$4 \times 10^{-7} – 7 \times 10^{-7}$	$\approx 10^{14} – 10^{15}$
infrared	$7 \times 10^{-7} – 10^{-3}$	$10^{11} – 10^{14}$
microwaves	$10^{-3} – 10^{-1}$	$10^{9} – 10^{11}$
radio waves	$> 10^{-1}$	$< 10^{9}$

Teacher's note

For the exam, remember the **order** of the spectrum and the order of magnitude of each region. The boundaries are not sharp, so books give slightly different values.

Example 21.5.1

The electric field of an EM wave in a vacuum is $E = 150 \sin (6.0 \times 10^{15} t - kx)$ V m^{-1}. Find (a) the frequency, (b) the wavelength, (c) the region of the spectrum.

Solution

(a) $f = \omega/2\pi = 6.0 \times 10^{15}/2\pi =$ **9.5×10^{14} Hz**

(b) $\lambda = c/f = 3.00 \times 10^{8}/9.5 \times 10^{14} =$ **3.1×10^{-7} m**

(c) **ultraviolet**

Summary and exercises

Subtopic	Key results
21.1	$y = A\sin(\omega t \pm kx)$; $k = 2\pi/\lambda$; $\varphi = 2\pi x/\lambda$; $v = f\lambda$
21.2	$I = P/A$; $I \propto A^2$; point source $I = P/4\pi r^2$
21.3	$y = y_1 + y_2$
21.4	$y = 2a\cos kx \sin \omega t$; node to node = $\lambda/2$; node to antinode = $\lambda/4$
21.5	$E = E_0 \sin(\omega t - kx)$, $B = B_0 \sin(\omega t - kx)$; $c = 1/\sqrt{(\mu_0\varepsilon_0)}$

Exercises

1. A wave on a string is described by $y = 0.02\sin(30t + 4x)$ (SI units). Find the wavelength, the frequency, and the speed and direction of the wave.

   Answer

   $\lambda = 1.57$ m, $f = 4.77$ Hz, $v = 7.5\ \text{m s}^{-1}$ in the $-x$ direction

2. Two points on a progressive wave of wavelength 1.2 m are 0.30 m apart. What is the phase difference between them?

   Answer

   $\pi/2$ rad

3. The intensity of a wave 2.0 m from a point source is 0.80 W m^{-2}. What is the intensity 8.0 m from the source?

   Answer

   0.050 W m^{-2}

4. A standing wave on a string has nodes 0.25 m apart. The wave speed is 100 m s^{-1}. Find the frequency.

   Answer

   $\lambda = 0.50$ m, $f = 200$ Hz

5. State two differences between an electromagnetic wave and a sound wave.

Answer

e.g. EM waves can travel through a vacuum but sound needs a medium; EM waves are transverse but sound is longitudinal; EM waves travel at 3.00 × 10^8 m s^{-1} in a vacuum, much faster than sound.

Real-World Physics: Japan's tsunami sea wall

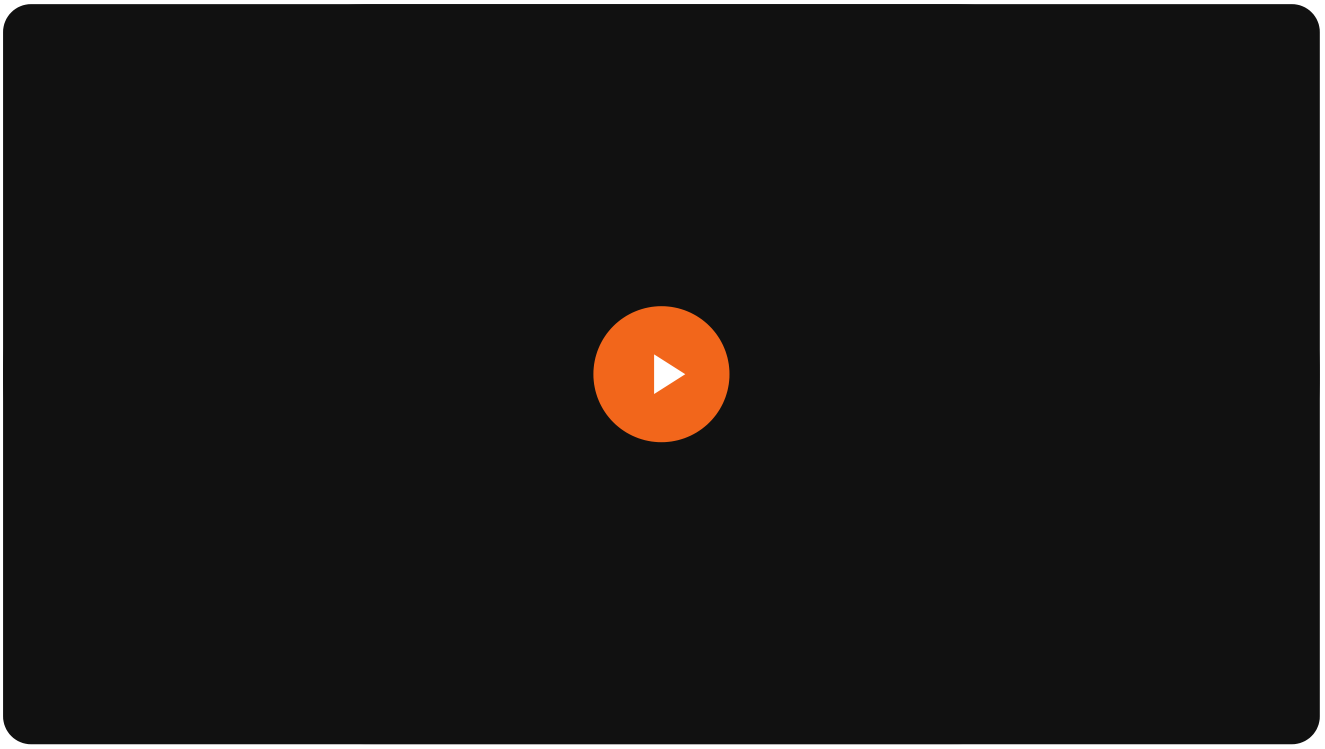
REAL-WORLD PHYSICS

Tōhoku coast, Japan · built after 11 March 2011

Why does a tsunami grow taller as it reaches the shore?

On 11 March 2011 a magnitude 9 earthquake off north-east Japan sent a tsunami onto the Tōhoku coast. The old sea walls were designed for waves of about 8 m, but the waves reached 12–15 m, poured over the walls and destroyed many of them. About 20 000 people died.

Japan then built nearly **400 km** of new concrete sea walls along the coast, up to **14.7 m** high, with deep foundations and wider bases so they do not collapse if a wave goes over the top.



Japan's giant tsunami sea wall · YouTube documentary

Open in YouTube ↗

While you watch: compare the height of the tsunami in the open ocean with its height when it reaches the coast.

The physics

- A tsunami is a **progressive wave** with a very long wavelength (hundreds of kilometres). Its speed depends on the depth of the water: $v = \sqrt{(gd)}$ (given).
- In the deep ocean it travels at about 700–900 km h^{-1} but is often less than 1 m high. Near the coast the water becomes shallow, so the wave **slows down**.
- The **frequency stays the same** (it is set by the source), so from $v = f\lambda$ the **wavelength becomes shorter**. The same energy is squeezed into a shorter, slower wave, so the **amplitude grows**. Since $I \propto A^2$, the wave arriving at the wall is extremely powerful.

Quick check 1

The speed of a tsunami is given by $v = \sqrt{(gd)}$, where d is the depth of the water. What is its speed in the open ocean where d = 4000 m?

A 63 m s^{-1}

B 198 m s^{-1}

C 3.9×10^4 m s^{-1}

D 713 m s^{-1}

Quick check 2

The tsunami in Quick check 1 has a period of 20 minutes. What is its wavelength in the open ocean?

A 9.9 m

B 2.4×10^3 m

C 2.4×10^5 m

D 1.4×10^7 m

Quick check 3

As a tsunami moves from deep water into shallow water near the coast, which statement is correct?

A Its frequency decreases and its wavelength stays the same.

B Its speed increases and its wavelength increases.

C Its frequency stays the same, its speed decreases and its wavelength decreases.

D Its speed stays the same and its amplitude decreases.

Quick check 4

Near the coast, the amplitude of a tsunami increases from 1.0 m to 4.0 m. Assuming intensity $\propto$ (amplitude)^2, by what factor does the intensity increase?

A 2

B 4

C 8

D 16

Sources: The B1M; NOAA U.S. Tsunami Warning System. The depth and period in the calculations are typical values for practice. Preview the clip before class. youtube.com/watch?v=tJ6G2lTo6xA

Chapter 22 · Semester 3

SOUND WAVES

Notes by Cikgu Suhaili

22 worked examples 5 simulations Exercises with answers

22.1 Propagation of sound waves

- A sound wave is a **longitudinal wave** that needs a medium (solid, liquid or gas) to travel.
- The source of a sound wave is a vibrating object.
- The sound wave travels as a series of **compressions and rarefactions** of the medium.
- Sound travels slowest in gases, faster in liquids and fastest in solids. In a gas the speed depends on temperature: in air, $v \approx (331 + 0.606\theta)$ m s^{-1}, θ in °C.

Material (20 °C)	Speed of sound / m s^{-1}
Air	343
Air (0 °C)	331
Helium	1005
Water	1440
Sea water	1560
Glass	≈ 4500
Iron and steel	≈ 5000
Aluminium	≈ 5100

How a tuning fork produces sound

- When the prongs move outwards, they push the air in front of them together. The pressure rises slightly: a **compression** (condensation).
- When the prongs move inwards, the air spreads out. The pressure falls slightly below normal: a **rarefaction**.
- As the fork keeps vibrating, compressions and rarefactions are produced one after another and spread away from it.

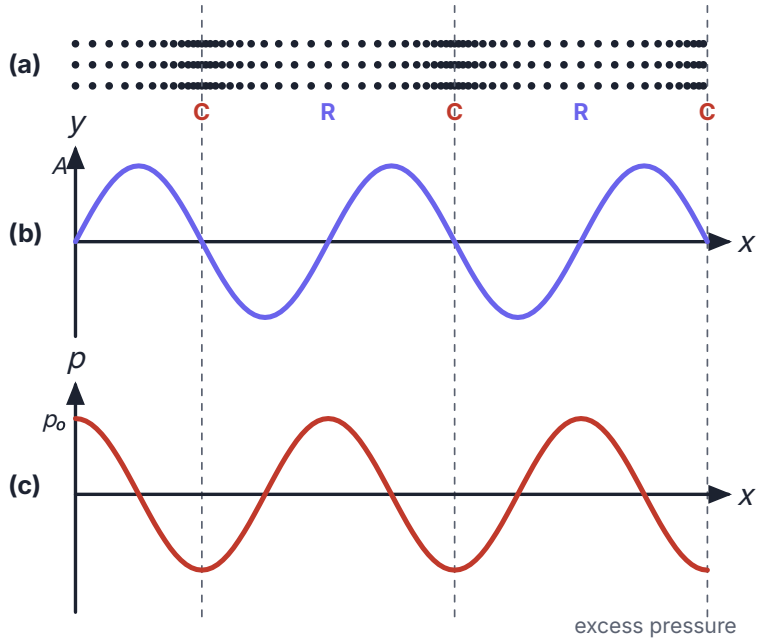


Figure 22.1: (a) the air layers at one instant (C = compression, R = rarefaction); (b) displacement y of the air particles against distance x; (c) excess pressure p against distance x.

Displacement and pressure

Displacement of the air particles

$$y = y_0 \sin(\omega t - kx)$$

Pressure variation (excess pressure)

$$p = p_0 \sin\left(\omega t - kx + \frac{\pi}{2}\right)$$

- At a compression or a rarefaction, the particles have **zero displacement** (they pass through their equilibrium position at maximum speed). Here the pressure is **maximum** (compression, above p_{atm}) or **minimum** (rarefaction, below p_{atm}).
- Where the displacement is maximum ($y = \pm y_0$), the particles are momentarily at rest, their spacing is normal, and the excess pressure is zero.
- So the pressure wave and the displacement wave are **π/2 rad (a quarter cycle) out of phase**: the pressure wave leads the displacement wave by π/2.

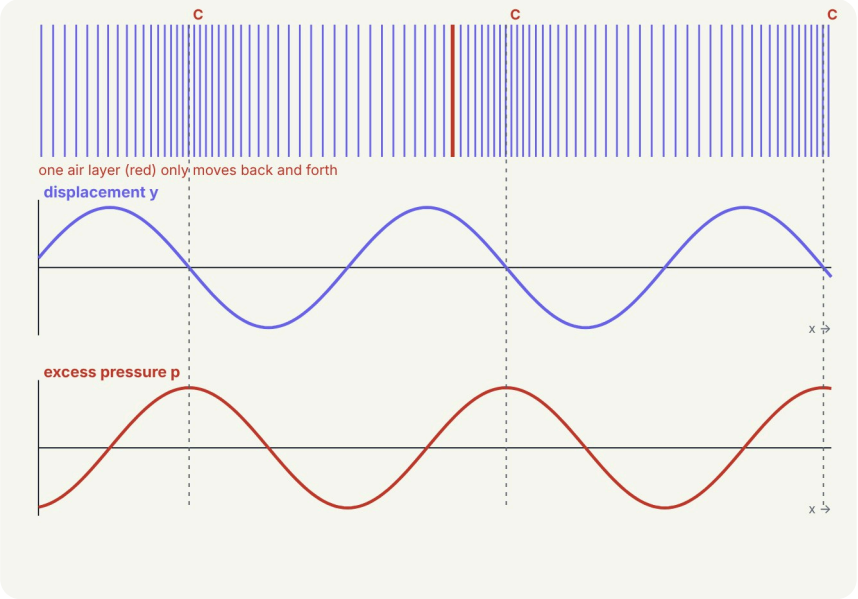
phase difference between p and y: $\Delta\varphi = \frac{\pi}{2}$ rad

p_0 = maximum excess pressure = $p_{max} - p_{atm}$

Simulation

Sound wave: displacement and pressure

Watch the air layers. Each layer only vibrates back and forth about its own position. The compressions (dark) and rarefactions (light) move along. Compare the displacement graph with the pressure graph: they are a quarter cycle apart.



This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 22 to use the simulation.

Example 22.1.1

A sound wave in air is represented by $y = 4.0 \times 10^{-8} \sin(1200\pi t - 3.5\pi x)$ m. Find (a) the amplitude, (b) the wavelength, (c) the frequency, (d) the speed of sound in air.

Solution

Compare with $y = y_0 \sin(\omega t - kx)$.

(a) $y_0 = \mathbf{4.0 \times 10^{-8}\ m}$

(b) $k = 3.5\pi \Rightarrow \lambda = 2\pi/k = \mathbf{0.571\ m}$

(c) $\omega = 1200\pi \Rightarrow f = \omega/2\pi = \mathbf{600\ Hz}$

(d) $v = f\lambda = 600 \times 0.571 = \mathbf{343\ m\,s^{-1}}$

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Describe the propagation of sound waves in air. [2]

Solution

- Sound is a series of compressions and rarefactions (pressure variations) that travel through the air.
- Sound waves travel in air as longitudinal waves: the air particles vibrate parallel to the direction of propagation.

22.2 Sources of sound

The source of any sound is a vibrating object, e.g. musical instruments (guitar, violin, flute, trumpet). The vibrations are standing waves on strings or in air columns.

Stationary waves on a stretched string

- When a stretched string is plucked, a transverse progressive wave travels to the fixed ends and is reflected.
- The incident and reflected waves (same frequency and amplitude) superpose to form a **stationary transverse wave**.
- Both fixed ends are **nodes**. The vibrating string makes the air vibrate at the same frequency, producing sound.

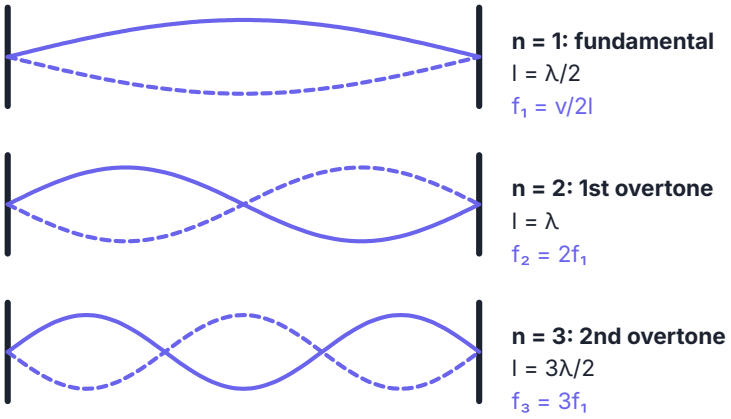


Figure 22.2: the first three modes of vibration of a string fixed at both ends.

$$f_n = \frac{nv}{2l} = nf_1, \quad n = 1, 2, 3, \ldots \qquad v = \sqrt{\frac{T}{\mu}}$$

n = harmonic number (number of loops); l = length of string; T = tension; μ = mass per unit length (kg m^{-1}); all harmonics are present

Term	Meaning
Fundamental frequency	the lowest natural frequency of a vibrating system
Harmonics	frequencies that are whole-number multiples of the fundamental frequency
Overtones	the frequencies above the fundamental that the system can produce: the first overtone is the next one after the fundamental

Example 22.2.1

A string vibrates in five segments at a frequency of 460 Hz. (a) What is its fundamental frequency? (b) What frequency makes it vibrate in three segments?

Solution

(a) five segments: $n = 5$; $f_5 = 5f_1 \Rightarrow f_1 = 460/5 =$ **92 Hz**

(b) $f_3 = 3(92) =$ **276 Hz**

Example 22.2.2

A steel piano wire 0.400 m long has a mass of 3.00 g and is under a tension of 800 N. Find (a) the frequency of its fundamental mode, (b) the highest harmonic that could be heard by a person who can hear frequencies up to 10 000 Hz.

Solution

$\mu = m/l = 3.00 \times 10^{-3}/0.400 = 7.5 \times 10^{-3}$ kg m^{-1}

(a) $f_1 = \frac{1}{2l}\sqrt{\frac{T}{\mu}} = \frac{1}{2(0.400)}\sqrt{\frac{800}{7.5 \times 10^{-3}}} =$ **408 Hz**

(b) $n = 10\,000/408 = 24.5$, so the highest harmonic heard is the **24th**.

STPM 2021 · Section A

A string is fixed at both ends and plucked in the middle. The length, mass and tension of the string are 2.0 m, 8.0 g and 1.2 kN respectively. What is the fundamental frequency of the string?

A. 4.3 Hz

B. 68 Hz

C. 140 Hz

D. 270 Hz

Answer

Answer: C

$\mu = 0.0080/2.0 = 4.0 \times 10^{-3}$ kg m^{-1}; $v = \sqrt{(1200/4.0 \times 10^{-3})} = 548$ m s^{-1}; $f_1 = v/2L = 548/4.0 = 137$ Hz ≈ 140 Hz.

STPM 2022

A string of length L is stretched between two fixed points with tension T. Sketch the shape of the string for the second overtone, and find the distance between two consecutive nodes in terms of the fundamental wavelength. [4]

Solution

Second overtone = third harmonic ($n = 3$): the string forms **3 loops**.

$L = 3(\lambda_3/2) \Rightarrow$ distance between nodes $= \lambda_3/2 = L/3$

fundamental: $\lambda_1 = 2L \Rightarrow L = \lambda_1/2$, so the distance = $\boldsymbol{\lambda_1/6}$

Stationary waves in air columns

- A sound wave sent into a pipe is reflected at the end. The incident and reflected waves superpose to form a **stationary longitudinal wave**.
- **Resonance** occurs, and a loud sound is heard, when the frequency of the source equals a natural frequency of the air column.
- A **closed end** is always a **displacement node** (the air cannot move). An **open end** is a **displacement antinode**.

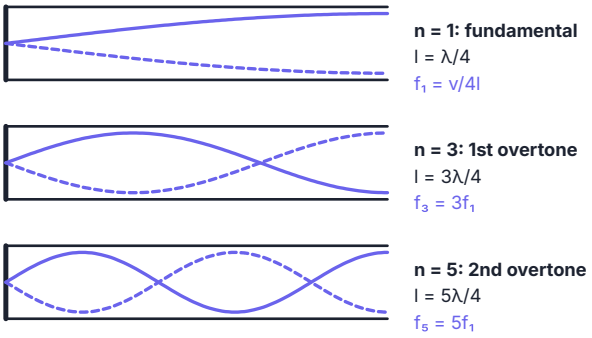


Figure 22.3: closed pipe (one end closed). Only odd harmonics.

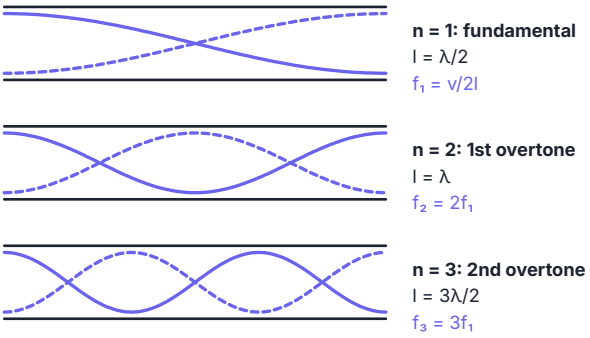


Figure 22.4: open pipe (both ends open). All harmonics.

	Closed pipe	Open pipe and string
Fundamental	$f_1 = v/4l$	$f_1 = v/2l$
General	$f_n = nv/4l$, n = 1, 3, 5, ... (odd harmonics only)	$f_n = nv/2l$, n = 1, 2, 3, ...
1st overtone	$3f_1$	$2f_1$
2nd overtone	$5f_1$	$3f_1$

End correction

End correction, c

is **the small distance between the open end of a pipe and the antinode just outside it**. The air at the open end is free to move, so its vibration extends slightly beyond the end of the pipe.



Figure 22.5: the antinode forms a distance c beyond each open end.

Closed pipe	Open pipe
$l + c = \lambda/4 \Rightarrow f_1 = \frac{v}{4(l + c)}$	$l + 2c = \lambda/2 \Rightarrow f_1 = \frac{v}{2(l + 2c)}$
$f_n = \frac{nv}{4(l + c)}$, n = 1, 3, 5, ...	$f_n = \frac{nv}{2(l + 2c)}$, n = 1, 2, 3, ...

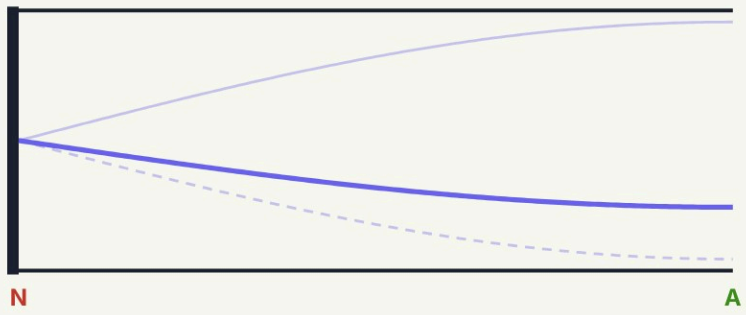
Teacher's note

To find the end correction experimentally, use a resonance tube: find the first two resonance lengths l_1 and l_2 for the same tuning fork. Then $l_2 - l_1 = \lambda/2$ (the end correction cancels), and $c = \lambda/4 - l_1$.

Simulation

Harmonics in strings and pipes

Choose a string, an open pipe or a closed pipe, then step through the harmonics. Change the length and switch on the end correction. A closed pipe only has odd harmonics.



displacement of the air (drawn sideways)

harmonic	overtone	λ	frequency (v = 340 m/s)
n = 1 (fundamental)	**—**	**2.000 m**	**170 Hz**

formula

f = nv/4(l + c)

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 22 to use the simulation.

Example 22.2.3

A pipe is 45.0 cm long. Find the fundamental frequency and the first three overtones (a) if the pipe is open at both ends, (b) if it is closed at one end. (c) For each case, find the highest harmonic that can be heard by a person who hears 20 Hz to 20 000 Hz. (v = 344 m s^{-1})

Solution

(a) open: $f_1 = v/2l$ = 344/0.90 = **382 Hz**; overtones f_2, f_3, f_4 = **764, 1147, 1529 Hz**

(b) closed: $f_1 = v/4l$ = 344/1.80 = **191 Hz**; overtones f_3, f_5, f_7 = **573, 956, 1338 Hz**

(c) open: n = 20 000(0.90)/344 = 52.3 ⇒ **52nd harmonic**. Closed: n = 20 000(1.80)/344 = 104.7, but only odd harmonics exist ⇒ **103rd harmonic**.

Example 22.2.4

A pipe 90 cm long is open at both ends. How long must a pipe closed at one end be to have the same fundamental frequency?

Solution

$$\frac{v}{4l_c} = \frac{v}{2l_o} \Rightarrow l_c = \frac{l_o}{2} = \textbf{45 cm}$$

Example 22.2.5

A tube of air 1.5 m long is closed at one end. It sustains a standing wave at its third harmonic. What is the distance between a node and the adjacent antinode? (v = 331 m s^{-1})

Solution

Third harmonic: $l = 3\lambda/4 \Rightarrow \lambda = 4(1.5)/3 = 2.0$ m

node to adjacent antinode = $\lambda/4$ = **0.5 m**

Example 22.2.6

A pipe closed at one end produces a fundamental note of 300 Hz. The end correction is 8.00 mm and the speed of sound is 330 m s^{-1}. Calculate the length of the pipe.

Solution

$$f_1 = \frac{v}{4(l + c)} \Rightarrow l + c = \frac{330}{4(300)} = 0.275 \text{ m}$$

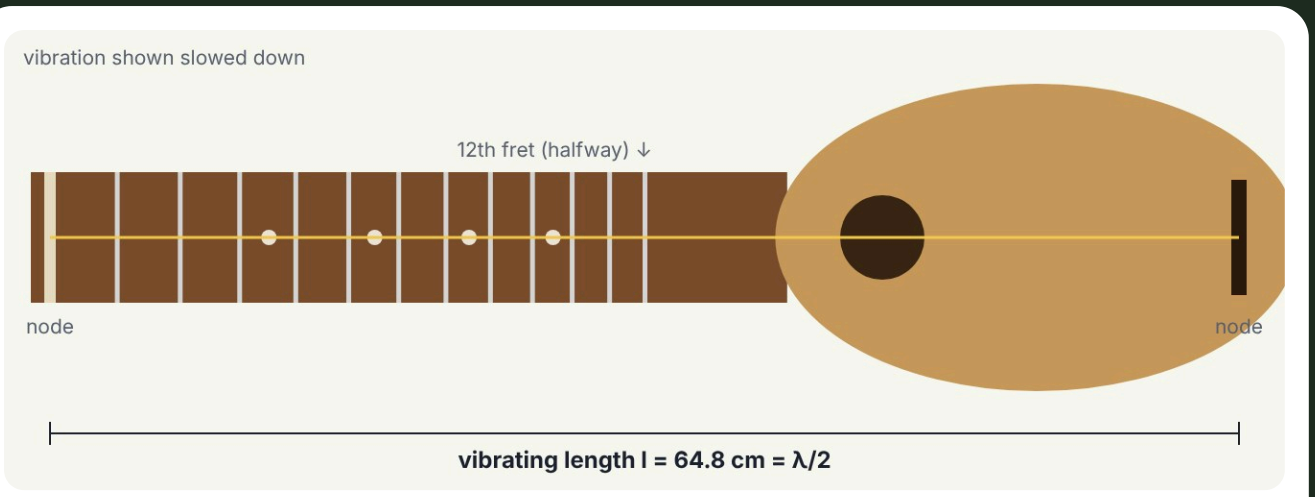
$$l = 0.275 - 0.008 = \mathbf{0.267\ m}$$

Real instruments: the guitar and the flute

- **Guitar**: each string is fixed at the bridge and at the nut, so it vibrates with nodes at both ends: $f_1 = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$. Thicker strings (larger μ) give lower notes; turning the tuning peg changes the tension T.
- Pressing a string against a **fret** shortens the vibrating length. The frets are placed so that each one raises the note by a semitone; the 12th fret is halfway along the string, so the note is one octave higher (double the frequency).
- **Flute**: a pipe open at both ends (the mouth hole and the far end), so there is an antinode at each end: $f_1 = \frac{v}{2l}$. Opening a finger hole makes the air column end there, so it is shorter and the note is higher.
- Blowing harder makes the flute sound its second harmonic (overblowing): the same fingering gives a note one octave higher.

Simulation Guitar and flute

Choose the guitar or the flute. On the guitar, pick a string, change the tension and press a fret. On the flute, choose a note and overblow. Watch the vibrating length and the stationary wave, and switch on the sound to hear the note.



frequency f	note	wavelength λ	wave speed on the string v = √(T/μ)
330 Hz	**E4**	**129.6 cm**	**427 m s^{-1}**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 22 to use the simulation.

Example 22.2.7

The thinnest string of a guitar has a vibrating length of 0.648 m and a mass per unit length of 4.0×10^{-4} kg m^{-1}. It is tuned to a fundamental frequency of 330 Hz. Find (a) the speed of the transverse wave on the string, (b) the tension in the string, (c) the frequency when the string is pressed at the 12th fret, halfway along.

Solution

(a) Fundamental: $l = \lambda/2 \Rightarrow \lambda = 1.296$ m; $v = f\lambda = (330)(1.296) =$ **428 m s^{-1}**

(b) $v = \sqrt{\dfrac{T}{\mu}} \Rightarrow T = v^2\mu = (428)^2(4.0 \times 10^{-4}) =$ **73 N**

(c) $f \propto \dfrac{1}{l}$: half the length gives **660 Hz**, one octave higher

Exercises: strings and pipes

1. The distance between the first and the fourth antinodes of a stationary wave on a string is 15 cm. What is the speed of the wave if it vibrates at 50 Hz?

   Answer

   $3 \times \lambda/2 = 15$ cm $\Rightarrow \lambda = 0.10$ m; $v = 5$ m s^{-1}

2. A string 2.0 m long is fixed at both ends. At 85.0 Hz a standing wave with five loops forms. Find (a) the wavelength, (b) the wave speed, (c) the fundamental frequency.

   Answer

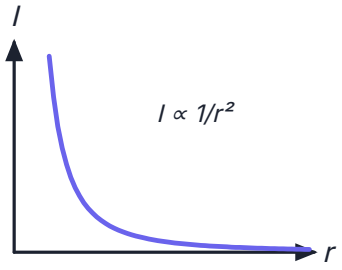
   (a) 0.8 m (b) 68 m s^{-1} (c) 17 Hz

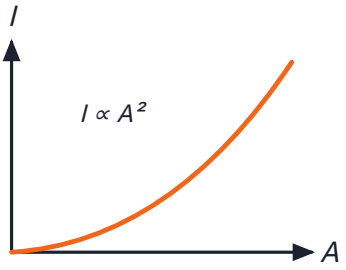
22.3 Intensity level of sound

Intensity of sound

is **the rate of energy transported per unit area perpendicular to the direction of propagation**: $I = \frac{E}{tA}$ $= \frac{P}{A}$, unit W m^{-2}.

For a point source radiating uniformly, $A = 4\pi r^2$, so $\frac{I_2}{I_1} = \frac{r_1^2}{r_2^2}$. The intensity (and so the loudness) depends on the amplitude, $I \propto A^2$, and on the distance, $I \propto 1/r^2$.





Example 22.3.1

A loudspeaker radiates sound uniformly in all directions. At 3 m the intensity is 0.4 W m^{-2}. Find the power of the loudspeaker and the intensity at 5 m.

Solution

$P = I(4\pi r^2) = 0.4 \times 4\pi(3)^2 =$ **45.2 W**

$I_2 = I_1(r_1/r_2)^2 = 0.4(3/5)^2 =$ **0.144 W m^{-2}**

Intensity level

- **Threshold of hearing**: the lowest intensity a normal ear can detect, $I_0 = 1.0 \times 10^{-12}$ W m^{-2}.
- **Threshold of pain**: about 1 W m^{-2}.
- Our sense of loudness follows the logarithm of the intensity, so a logarithmic scale is used.

Intensity level of sound, β

is **a logarithmic measure of the intensity of a sound compared with the threshold of hearing**.

$$\beta = 10 \log_{10} \frac{I}{I_0}$$

unit: decibel (dB). $I_0 = 1.0 \times 10^{-12}$ W m^{-2}. An increase of 10 dB means the intensity is 10 times greater; +3 dB means the intensity doubles.

Sound	Intensity level / dB
threshold of hearing	0
whisper	20
conversation, classroom	60
siren, rock concert	100
jet aeroplane, threshold of pain	120

Example 22.3.2

The intensity of a sound wave is 1 W m^{-2}. Calculate its intensity level.

Solution

$\beta = 10 \log_{10} (1/10^{-12}) = 10 \times 12 =$ **120 dB**

Example 22.3.3

The intensity level of one sound is 80 dB and of another is 20 dB. Find the ratio of their intensities.

Solution

$I_1/I_0 = 10^8$, $I_2/I_0 = 10^2 \Rightarrow I_1/I_2 =$ **10^6**

Shortcut: $\beta_1 - \beta_2 = 10 \log (I_1/I_2) \Rightarrow 60 = 10 \log (I_1/I_2)$

STPM 2022 · Section A

The sound intensity level of 88.0 dB is detected 30.0 cm from a point source. What is the energy received at that point in one second? [$I_0 = 1.00 \times 10^{-12}$ W m^{-2}]

A. 1.19×10^{-3} J

B. 7.14×10^{-4} J

C. 5.58×10^{-3} J

D. 1.78×10^{-3} J

Answer

Answer: B

$I = 10^{8.8} \times 10^{-12} = 6.31 \times 10^{-4}$ W m^{-2}; $P = I \times 4\pi r^2 = 6.31 \times 10^{-4} \times 4\pi(0.300)^2 = 7.14 \times 10^{-4}$ W, i.e. 7.14×10^{-4} J each second.

STPM 2021

A bomb blasts 12 m from a soldier. The area of his eardrum is 25 mm^2 and the sound energy it receives every second is 1.0×10^{-6} W. [$I_0 = 10^{-12}$ W m^{-2}] (a) Find the sound intensity level at the eardrum. [3] (b) Find the change in level if the soldier is 17 m from the blast. [3] (c) If the level increases by 3 dB, what happens to the intensity? [1]

Solution

(a) $I = P/A = 1.0 \times 10^{-6}/25 \times 10^{-6} = 0.040$ W m^{-2}; $\beta = 10 \log (0.040/10^{-12})$ = **106 dB**

(b) $I_2 = 0.040(12/17)^2 = 0.0199$ W m^{-2}; $\beta_2 = 103.0$ dB ⇒ a **decrease of 3.0 dB**

(c) $3 = 10 \log (I_2/I_1) \Rightarrow I_2/I_1 = 10^{0.3} \approx 2$: the intensity **doubles**.

22.4 Beats

Beats

are **the periodic variation in amplitude (loudness) at a point, caused by the superposition of two sound waves of slightly different frequencies**.

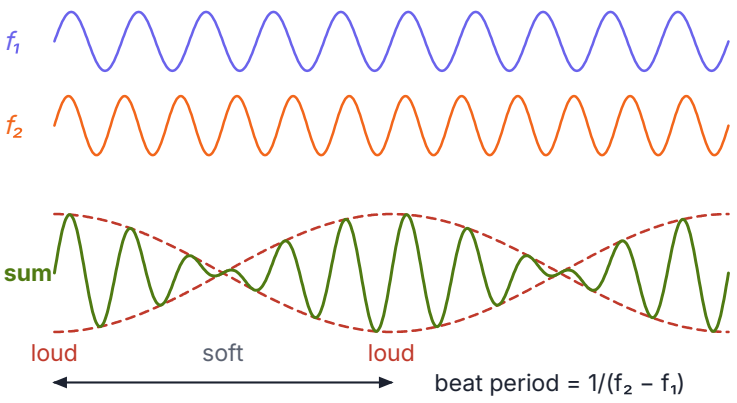


Figure 22.6: two waves of 50 Hz and 60 Hz and their sum. The loudness rises and falls 10 times per second.

- Where the two waves are in phase, they interfere constructively: the amplitude is maximum and a loud sound is heard.
- Where they are in antiphase, they interfere destructively: the amplitude is minimum and little or no sound is heard.
- The number of times per second the loudness rises and falls is the **beat frequency**.

Derivation of the beat frequency

$y_1 = A \sin 2\pi f_1 t, \quad y_2 = A \sin 2\pi f_2 t \quad (f_1 > f_2)$

$y = y_1 + y_2$, with $\sin P + \sin Q = 2 \cos \dfrac{P - Q}{2} \sin \dfrac{P + Q}{2}$

$y = [2A \cos 2\pi(\dfrac{f_1 - f_2}{2})t] \sin 2\pi(\dfrac{f_1 + f_2}{2})t$

The sound heard has frequency $(f_1 + f_2)/2$. Its amplitude $2A \cos 2\pi(\dfrac{f_1 - f_2}{2})t$ varies slowly.

The loudness is maximum whenever the cosine is +1 **or** −1, i.e. twice in each cycle of the cosine. So the beat frequency is twice $(f_1 - f_2)/2$.

$$f_{\text{beat}} = f_1 - f_2 = \frac{v}{\lambda_1} - \frac{v}{\lambda_2}$$

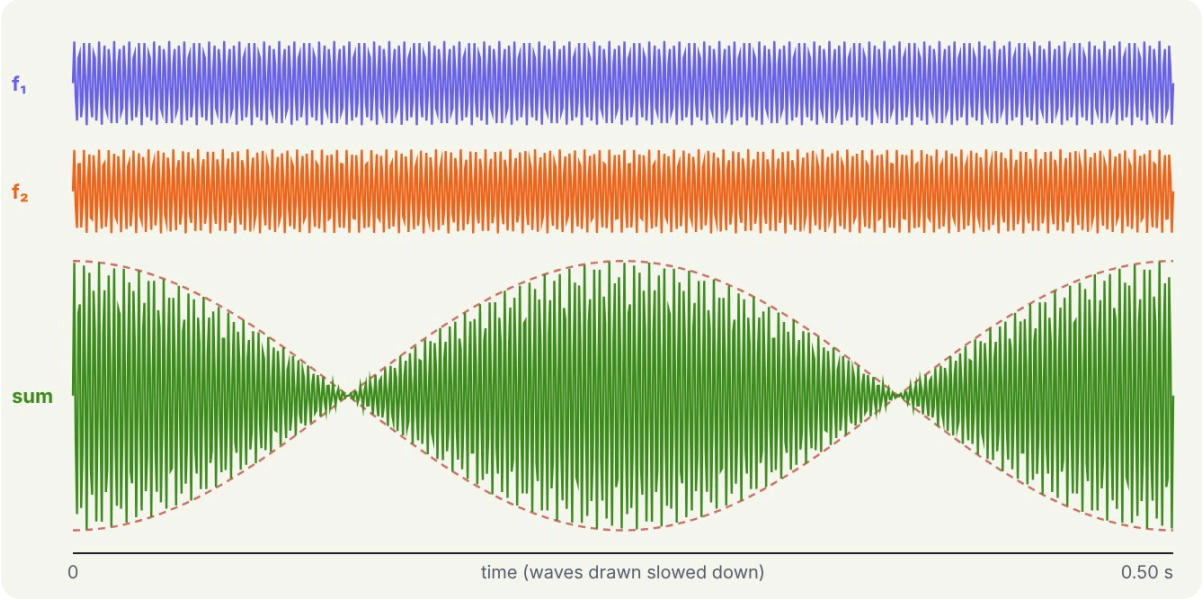
beat period $T_{\text{beat}} = 1/f_{\text{beat}}$. For 50 Hz and 60 Hz: 10 beats per second, $T_{\text{beat}} = 0.10$ s.

Teacher's note

Beats are used to tune instruments: the string is adjusted until the beats with a reference tuning fork slow down and disappear ($f_{\text{beat}} = 0$).

Simulation **Beats**

Set two frequencies that are close together. The green curve is their sum: its amplitude rises and falls at the beat frequency. Press Listen to hear the beats (use earphones or a speaker).



beat frequency $\lvert f_1 - f_2 \rvert$	beat period	frequency heard $(f_1 + f_2)/2$
4 Hz	**0.25 s**	**442.0 Hz**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 22 to use the simulation.

Example 22.4.1

Two sound waves of wavelengths 3.00 m and 2.50 m superpose. Both travel at 330 m s^{-1}. How many beats are produced each second?

Solution

$$f_{\text{beat}} = \frac{330}{2.50} - \frac{330}{3.00} = 132 - 110 = \mathbf{22\ Hz}, \text{ i.e. 22 beats per second}$$

STPM 2022 · Section A

A musician plucks the first two strings of a ukulele together and hears three beats per second. The first string is tuned to 350 Hz. The musician tightens the screw to stretch the second string, plucks the two strings again and hears two beats per second. What was the frequency of the second string before it was stretched?

A. 345 Hz

B. 347 Hz

C. 351 Hz

D. 353 Hz

Answer

Answer: B

At first $|350 - f_2| = 3$, so f_2 = 347 Hz or 353 Hz. Tightening increases the tension, so f_2 increases. The beat frequency falls from 3 to 2, so f_2 moved closer to 350 Hz: it was **347 Hz** (now 348 Hz).

22.5 Doppler effect

Doppler effect

is **the change in the observed frequency of a wave because of the relative motion between the source and the observer**.

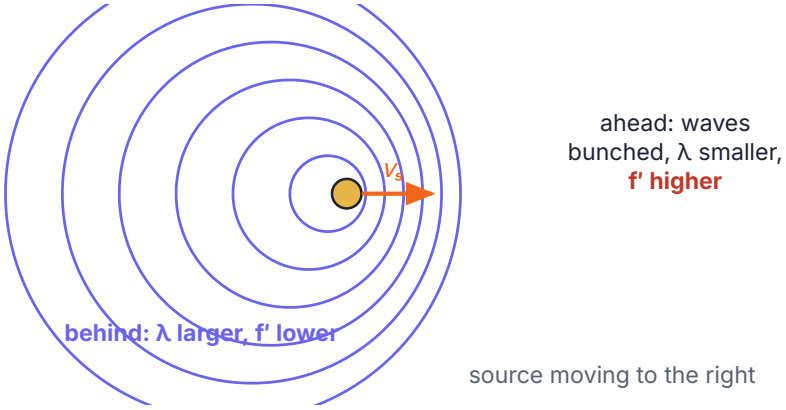


Figure 22.7: a moving source. The wavefronts are closer together in front of the source and further apart behind it.

- Source at rest: every observer hears the source frequency f_s.
- Source moving **towards** the observer: the wavelength reaching the observer is shorter, so the observed frequency is **higher** ($f' > f_s$).
- Source moving **away**: the wavelength is longer, so the observed frequency is **lower** ($f' < f_s$).

Derivation

Moving source (speed u_s, towards the observer): in one period $T = 1/f$, the source moves $u_s T$ while the wave moves vT. The wavelength ahead is $\lambda' = (v - u_s)/f$.

$$f' = v/\lambda' = \frac{v}{v - u_s} f \quad \text{(away: } v + u_s\text{)}$$

Moving observer (speed u_o, towards the source): the waves pass the observer at relative speed $v + u_o$; the wavelength is unchanged, $\lambda = v/f$.

$$f' = \frac{v + u_o}{\lambda} = \frac{v + u_o}{v} f \quad \text{(away: } v - u_o\text{)}$$

$$f' = \frac{v \pm u_o}{v \mp u_s} f$$

v = speed of sound; u_o, u_s = speeds of observer and source along the line joining them. Upper signs: moving **towards** each other (f' increases). Lower signs: moving **away** (f' decreases).

Teacher's note

Decide the sign from the physics, not from memory: "towards" must make f' bigger (bigger numerator or smaller denominator); "away" must make f' smaller.

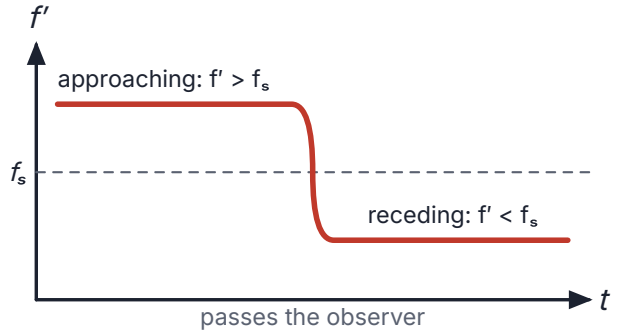
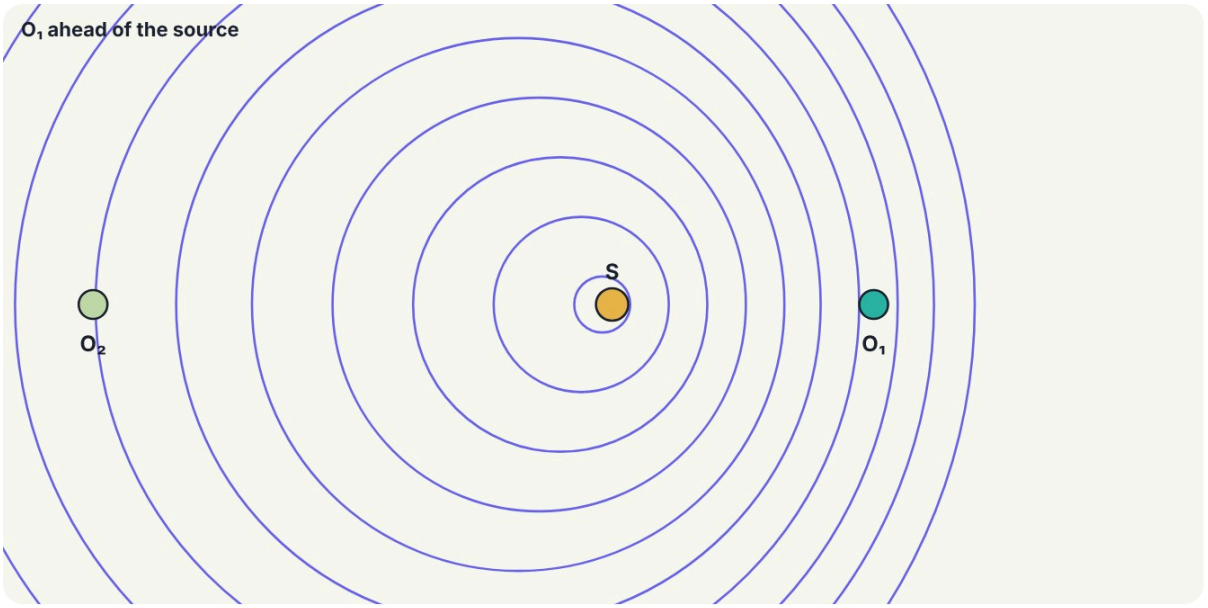


Figure 22.8: the frequency heard by a stationary observer as a source approaches, passes and moves away at constant speed.

- While the source approaches, the observed frequency is constant and higher than f_s; it drops to f_s as the source passes; it is then constant and lower while the source moves away.
- The **loudness** does change: as the source comes closer, r decreases and $I \propto 1/r^2$ increases. Do not confuse loudness (intensity) with pitch (frequency).

Simulation Doppler effect

Move the source, or the observer, and watch the wavefronts. Compare the frequency heard by the observer in front of the source with the one behind it.



source frequency f_s	O_1 hears	O_2 (behind, at rest) hears
400 Hz	**618 Hz**	**296 Hz**

formula

$$f' = f(v \pm u_o)/(v \mp u_s),\ v = 340\ \text{m/s}$$

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 22 to use the simulation.

Example 22.5.1

A factory whistle emits sound at 900 Hz. What frequency is heard by a person in a car travelling at 70 km h^{-1} (a) away from the whistle, (b) towards it? (v = 340 m s^{-1})

Solution

u_o = 70 km h^{-1} = 19.4 m s^{-1}

(a) $f' = \dfrac{340 - 19.4}{340}(900) =$ **849 Hz**

(b) $f' = \dfrac{340 + 19.4}{340}(900) =$ **951 Hz**

Example 22.5.2

An observer moves along the line joining two identical sources, each of frequency 500 Hz. The observer hears 4.0 beats per second. Find the speed of the observer. (v = 340 m s^{-1})

Solution

Moving towards one source: $f_1' = \dfrac{v + u_o}{v} f$ (higher). Moving away from the other: $f_2' = \dfrac{v - u_o}{v} f$ (lower).

$$f_{beat} = f_1' - f_2' = \frac{2fu_o}{v} \Rightarrow 4.0 = \frac{2(500)u_o}{340}$$

u_o = **1.36 m s^{-1}**

Example 22.5.3

Two identical sources S_1 and S_2, each of frequency 500 Hz, move towards a stationary observer at 20 m s^{-1} and 23 m s^{-1}. Find the beat frequency heard. (v = 340 m s^{-1})

Solution

$$f_1' = 500 \times \frac{340}{340 - 20} = 531.25 \text{ Hz}; \quad f_2' = 500 \times \frac{340}{340 - 23} = 536.28 \text{ Hz}$$

f_{beat} = 536.28 − 531.25 = **5.03 Hz**

STPM 2022

An ambulance moves at 42 m s^{-1} and sounds its siren at 960 Hz. A bus in front of the ambulance stops at the roadside. [v = 330 m s^{-1}] (i) Find the frequency of the siren heard by the passengers in the bus. [2] (ii) The sound is reflected by the bus. Find the frequency and the wavelength of the reflected sound heard by the ambulance driver. [2] (iii) What change in frequency do the passengers hear after the ambulance overtakes the bus? [1]

Solution

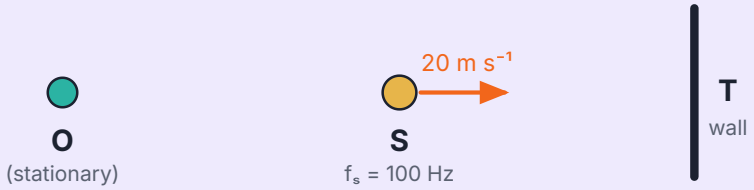
(i) $f' = 960 \times \dfrac{330}{330 - 42} =$ **1100 Hz**

(ii) The bus reflects 1100 Hz as a stationary source; the driver moves towards it: $f' = 1100 \times \dfrac{330 + 42}{330} =$ **1240 Hz**. The reflected wave in the air has $\lambda = v/f' = 330/1100 =$ **0.300 m** (a moving observer changes the frequency heard, not the wavelength in the air).

(iii) The frequency heard **decreases** (the siren now moves away).

Exercise: Doppler effect and beats

1. O is a stationary observer. Source S moves away from O towards a wall T at 20 m s^{-1}; its frequency is 100 Hz. (v = 340 m s^{-1}) Find (a) the wavelength in front of and behind the source (ignore the wall), (b) the frequency O hears directly from S, (c) the frequency O hears after reflection from T, (d) the beat frequency O hears.



Answer

(a) in front: (340 − 20)/100 = 3.2 m; behind: (340 + 20)/100 = 3.6 m (b) 100 × 340/360 = 94 Hz (c) 100 × 340/320 = 106 Hz (d) 106 − 94 = 12 Hz

Chapter 23 · Semester 3

GEOMETRICAL OPTICS

Notes by Cikgu Suhaili

33 worked examples 4 simulations Exercises with answers

Geometrical optics: introduction

Geometrical optics studies light on the assumption that it travels in straight lines called **rays**. A collection of rays is a **beam**. Two laws describe how light changes direction at a surface: the law of reflection and the law of refraction.

Real image	Virtual image
formed by light rays that actually pass through it	the rays only appear to come from it; no light passes through it
can be formed on a screen	cannot be formed on a screen
drawn with solid lines	drawn with dashed lines

> **Law of reflection**
>
> the angle of reflection equals the angle of incidence, and the incident ray, the reflected ray and the normal lie in the same plane. (Angles are measured from the normal, not from the surface.)

A **plane mirror** forms an image that is virtual, upright, the same size ($m = +1$), as far behind the mirror as the object is in front ($v = u$), and laterally inverted. This is why “AMBULANCE” is written back to front on the front of an ambulance: it reads correctly in a driver’s mirror.

23.1 Spherical mirrors

A spherical mirror is a reflecting surface that is part of a sphere. It is **concave** if the reflection is on the inner surface and **convex** if it is on the outer surface.



Figure 23.1: C = centre of curvature, P = pole (vertex), F = focal point, CP = principal axis, AB = aperture, f = focal length (FP), r = radius of curvature (CP).

Focal point of a concave mirror

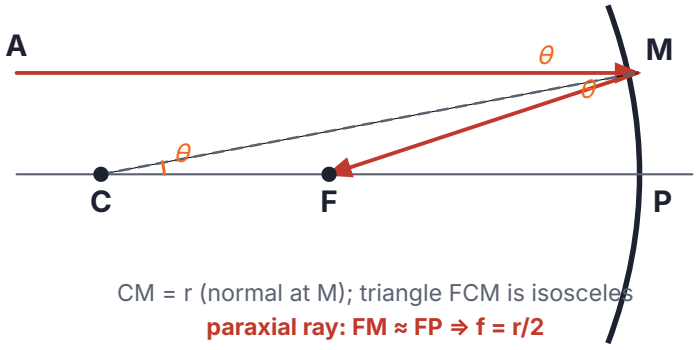
the point on the principal axis **through which rays parallel and close to the principal axis pass after reflection**.

Focal point of a convex mirror

the point on the principal axis **from which rays parallel and close to the principal axis appear to diverge after reflection**.

This is true only for **paraxial rays** (rays close to the principal axis). Wide rays are focused nearer the mirror (spherical aberration).

Relation between f and r



CM = r (normal at M); triangle FCM is isosceles

paraxial ray: FM ≈ FP ⇒ f = r/2

Figure 23.2: a paraxial ray AM, parallel to the axis, reflects through F.

CM is the normal at M, so the angle of incidence = angle of reflection = θ.

AM ∥ CP, so ∠MCF = θ (alternate angles). Triangle FCM has two equal angles, so it is isosceles: FC = FM.

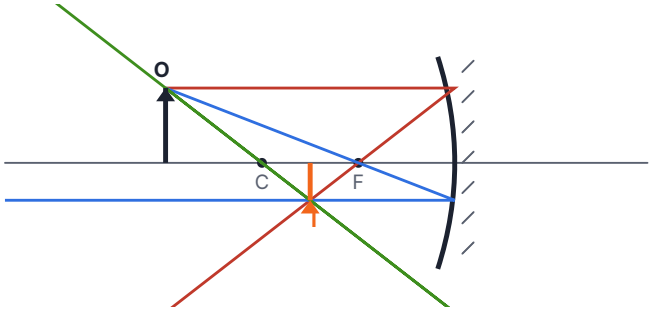
For a paraxial ray, M is very close to P, so FM ≈ FP. Then FC = FP, i.e. F is the midpoint of CP.

$$f = \frac{r}{2}$$

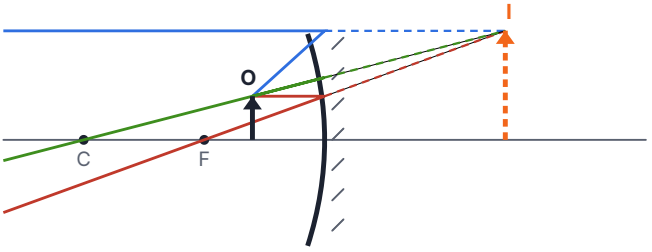
Ray diagrams

- **Ray 1**: a ray parallel to the principal axis is reflected through F (concave), or as if it came from F (convex).
- **Ray 2**: a ray through F (or heading towards F, convex) is reflected parallel to the axis.
- **Ray 3**: a ray through C (or heading towards C, convex) is reflected back along the same path.
- Any two rays from the top of the object locate the top of the image.

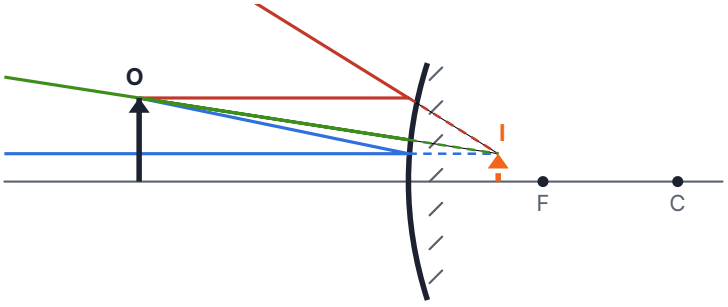
Object position (concave mirror)	Image position	Nature
at infinity	at F	real, inverted, diminished (a point)
beyond C	between C and F	real, inverted, diminished
at C	at C	real, inverted, same size
between C and F	beyond C	real, inverted, magnified
at F	at infinity	—
between F and P	behind the mirror	virtual, upright, magnified



Concave mirror, object beyond C.



Concave mirror, object between F and P: virtual, upright, magnified.



Convex mirror: the image is always virtual, upright and diminished, between P and F behind the mirror.

The mirror equation

Using similar triangles in the ray diagram (object distance OP = u, image distance IP = v, CP = r):

Ray to P: $\frac{h}{u} = \frac{h'}{v} \Rightarrow \frac{h'}{h} = \frac{v}{u}$ … (1)

Ray through C: $\frac{h}{u-r} = \frac{h'}{r-v} \Rightarrow \frac{h'}{h} = \frac{r-v}{u-r}$ … (2)

(1) = (2): $v(u-r) = u(r-v) \Rightarrow 2uv = r(u+v) \Rightarrow \frac{2}{r} = \frac{1}{u} + \frac{1}{v}$

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} = \frac{2}{r} \qquad m = \frac{h'}{h} = -\frac{v}{u}$$

m = linear magnification

Quantity	Positive (+)	Negative (−)
object distance u	real object	virtual object
image distance v	real image (in front of the mirror)	virtual image (behind the mirror)
focal length f, radius r	concave mirror	convex mirror
magnification m	upright image	inverted image

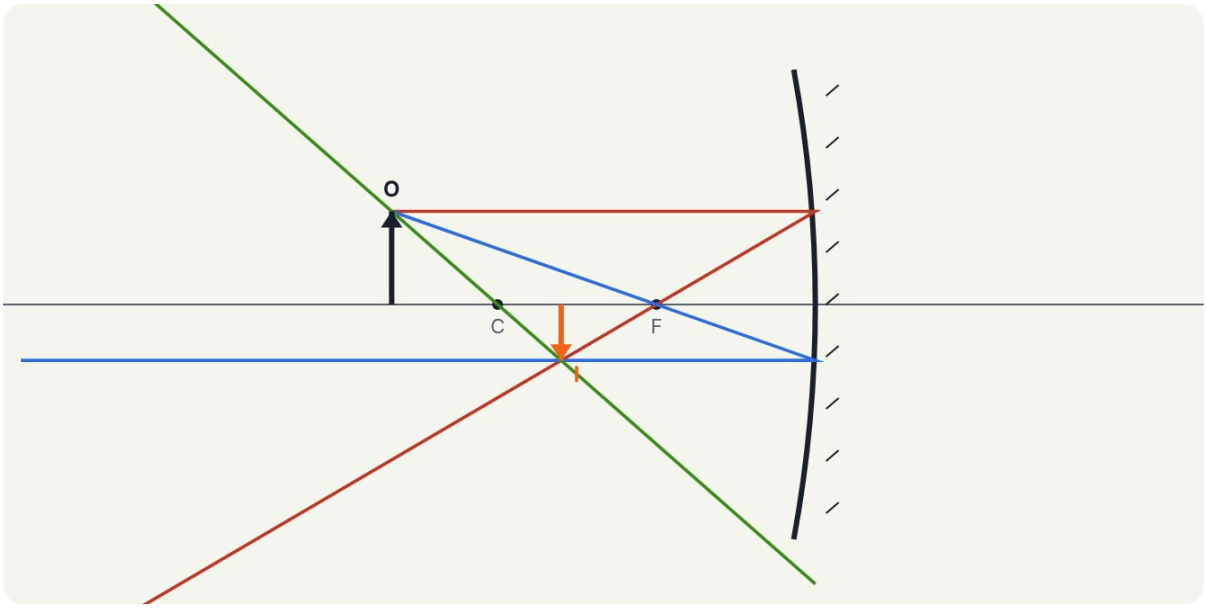
Teacher's note

Write the sign of f first (concave +, convex −), solve for v, then describe the image from the signs: $v > 0$ real, $v < 0$ virtual; $m > 0$ upright, $m < 0$ inverted; $|m| > 1$ magnified.

Simulation

Ray diagram for a spherical mirror

Choose a concave or convex mirror and drag the object along the axis. The three principal rays are drawn and the image appears where they meet. Watch the image change from real and inverted to virtual and upright as the object passes F.



f	image distance v	magnification m	image
15.0 cm	**24.0 cm**	**−0.60**	**real, inverted, diminished**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 23 to use the simulation.

Example 23.1.1

An object 6 cm high is 30 cm in front of a convex mirror of radius 40 cm. Find the position and height of its image.

Solution

Convex: $r = -40$ cm; $\frac{2}{r} = \frac{1}{u} + \frac{1}{v} \Rightarrow \frac{2}{-40} = \frac{1}{30} + \frac{1}{v} \Rightarrow v =$ **−12 cm** (virtual, 12 cm behind the mirror)

$m = -v/u = 12/30 = 0.40 \Rightarrow h' = 0.40 \times 6 =$ **2.4 cm** (upright)

Example 23.1.2

An object is 15 cm from (a) a concave mirror, (b) a convex mirror, each of radius of curvature 20 cm. Find the image position and magnification in each case.

Solution

(a) $f = +10$ cm: $\frac{1}{v} = \frac{1}{10} - \frac{1}{15} \Rightarrow v = \mathbf{+30\ cm}$ (real, 30 cm in front); $m = -30/15 = \mathbf{-2}$ (inverted, twice as high)

(b) $f = -10$ cm: $\frac{1}{v} = \frac{1}{-10} - \frac{1}{15} \Rightarrow v = \mathbf{-6.0\ cm}$ (virtual, 6.0 cm behind); $m = 6/15 = \mathbf{+0.40}$ (upright, two-fifths as high)

Example 23.1.3

What is the focal length of a convex mirror that forms an image one-sixth the size of an object 12 cm from the mirror?

Solution

Convex mirror: image virtual and upright, $m = +1/6 \Rightarrow v = -u/6 = -2$ cm

$\frac{1}{f} = \frac{1}{12} + \frac{1}{-2} \Rightarrow f = \mathbf{-2.4\ cm}$

Example 23.1.4

An object 20 cm from a concave mirror forms a real image three times its size. (a) Find the focal length. (b) Where must the object be placed to give a virtual image three times its size?

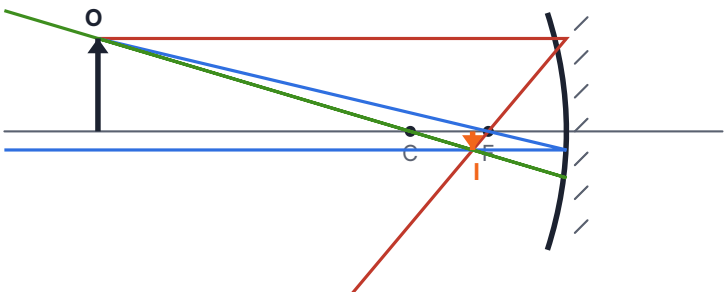
Solution

(a) real $\Rightarrow$ inverted, $m = -3 \Rightarrow v = 3u = 60$ cm; $\frac{1}{f} = \frac{1}{20} + \frac{1}{60} \Rightarrow f = \mathbf{+15\ cm}$

(b) virtual $\Rightarrow$ upright, $m = +3 \Rightarrow v = -3u$; $\frac{1}{u} - \frac{1}{3u} = \frac{1}{15} \Rightarrow u = \mathbf{10\ cm}$

Example 23.1.5

An object 2.0 cm high is 30 cm from a concave mirror of radius of curvature 10 cm. Find the position and the characteristics of the image.



Solution

$f = r/2 = +5$ cm; $\frac{1}{v} = \frac{1}{5} - \frac{1}{30} \Rightarrow v =$ **+6.0 cm**

$m = -6/30 = -0.20$, $h' = -0.40$ cm

The image is **real**, **inverted**, **diminished** (0.4 cm high), 6.0 cm in front of the mirror.

Exercises: spherical mirrors

1. A concave mirror has a focal length of 10 cm. Find the two positions where an object can be placed to give an image twice the height of the object.

   Answer

   15 cm (real, inverted image) and 5.0 cm (virtual, upright image)

2. A convex mirror of radius of curvature 40 cm forms an image half the height of the object. Find the object and image positions.

   Answer

   u = 20 cm; image 10 cm behind the mirror (v = −10 cm)

3. An object is 5.0 cm in front of a concave mirror of focal length 10.0 cm. Find the location of the image and its characteristics.

   Answer

   v = −10 cm (10 cm behind the mirror); m = +2: virtual, upright, twice the height

4. What kind of spherical mirror, and what radius, gives an upright image one-fifth as large as an object placed 15 cm in front of it?

   Answer

   m = +1/5 ⇒ v = −3.0 cm, f = −3.75 cm: a convex mirror of radius 7.5 cm (r = −7.5 cm)

5. A concave mirror forms an image, on a wall 3.0 m from the mirror, of a lamp filament 10 cm in front of the mirror. (a) What is the radius of curvature? (b) How high is the image if the filament is 5 mm high?

   Answer

   (a) f = 9.68 cm, r = 19.4 cm (b) m = −30, image 15 cm high (inverted)

6. What are the nature, size and location of the image of a 6 cm tall object 15 cm from a concave mirror of focal length 20 cm?

   Answer

   v = −60 cm (behind the mirror), m = +4: virtual, upright, 24 cm high

7. The magnification of a mirror is −0.333. Its image is formed on a card 540 mm from the mirror. Where is the object, and what is the focal length?

   Answer

   u = 1.62 m; f = +405 mm (concave)

23.2 Refraction at spherical surfaces

Refraction at a plane surface (revision)

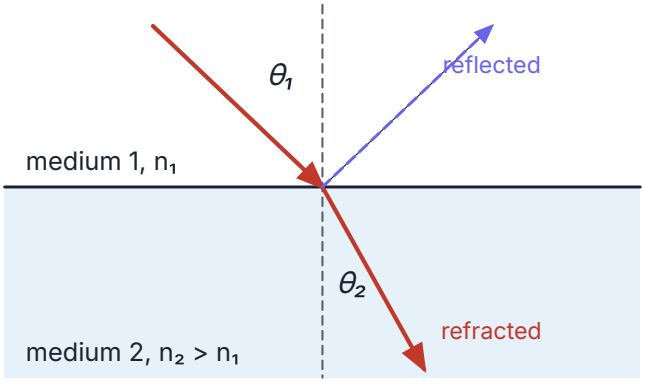


Figure 23.3: light going into a denser medium bends towards the normal.

- The incident ray, the refracted ray and the normal lie in the same plane.
- **Snell's law**: $\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}$ = constant.

- **Refractive index**: $n = \frac{c}{v}$ (speed in a vacuum / speed in the medium); $n \geq 1$, no unit.
- The frequency does not change across the boundary, but the speed and the wavelength do: $n_1\lambda_1 = n_2\lambda_2$, and $\lambda_n = \lambda_0/n$.

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Example 23.2.1

Light of wavelength 500 nm in air strikes a slab of transparent material at 40.0° to the normal. The refracted beam makes 26.0° with the normal. Find the refractive index and the wavelength in the material.

Solution

$$n_2 = \frac{(1)\sin 40.0°}{\sin 26.0°} = \mathbf{1.47}$$

$$\lambda_2 = \frac{n_1\lambda_1}{n_2} = \frac{500}{1.47} = \mathbf{340\ nm}$$

Example 23.2.2

Light travelling in air strikes crown glass ($n = 1.52$) at 30.0° to the normal. Find the angle of refraction.

Solution

$$\sin\theta_2 = \frac{\sin 30.0°}{1.52} = 0.329 \Rightarrow \theta_2 = \mathbf{19.2°}$$

Example 23.2.3

A layer of oil (n = 1.45) floats on water (n = 1.33). Light strikes the oil at 40.0° to the normal. Find the angle the ray makes with the normal in the water.

Solution

$n_{air} \sin 40.0° = n_{oil} \sin \theta_{oil} = n_{water} \sin \theta_{water}$ (parallel boundaries)

$\sin \theta_{water} = \sin 40.0°/1.33 \Rightarrow \theta_{water} =$ **28.9°** (the oil does not affect the final angle)

Refraction at a spherical surface

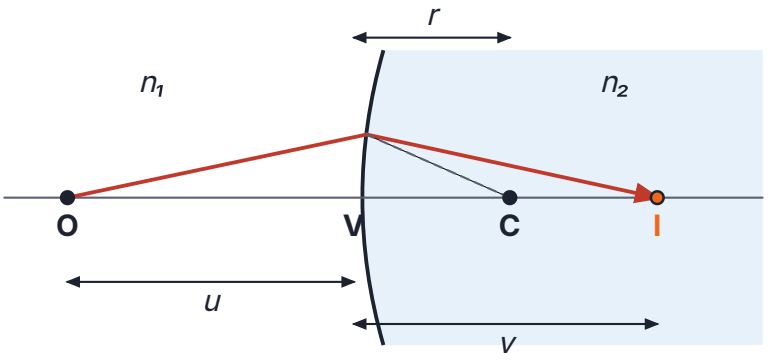


Figure 23.4: a point object O in medium n_1 forms an image I in medium n_2 by refraction at a spherical surface with centre C.

$$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r} \qquad m = \frac{h'}{h} = -\frac{n_1 v}{n_2 u}$$

n_1 = medium the light comes from; n_2 = medium the light goes into. (The derivation is not required.)

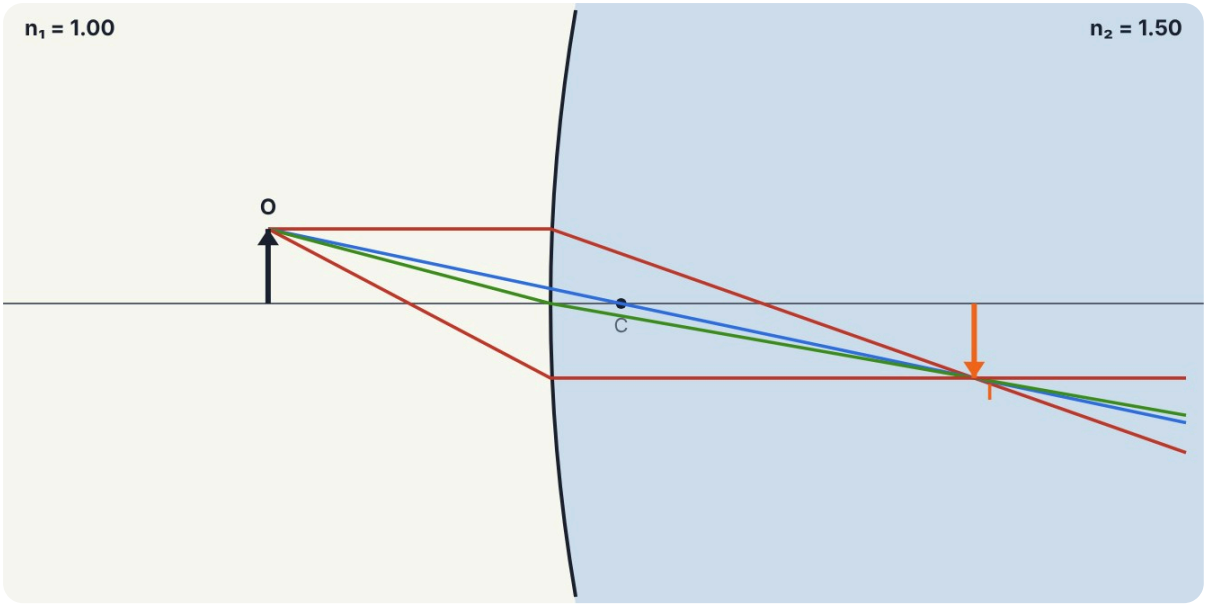
Quantity	Positive (+)	Negative (−)
u	real object (in front of the surface)	virtual object (behind the surface)
v	real image (behind the surface, in medium 2)	virtual image (in front of the surface, in medium 1)
r	centre of curvature behind the surface (surface convex to the incoming light)	centre of curvature in front of the surface (surface concave to the incoming light)
flat surface	$r \to \infty$, so the right-hand side is 0	

Some books use $|n_2 - n_1|/r$ with r positive when the centre of curvature is in the denser medium. Both forms are accepted in STPM. State the convention you use and never mix the two rules.

Simulation

Refraction at a spherical surface

Change the refractive indices, the radius of the surface and the object distance. The ray towards the centre of curvature crosses the surface without bending; the parallel ray is refracted. Try a fish in water seen from the air (flat surface).



$n_1/u + n_2/v = (n_2 - n_1)/r$	image distance v	magnification m	image
r = 10 cm	**60.0 cm**	**−1.00**	**real, inverted, same size**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 23 to use the simulation.

Example 23.2.4

A glass rod (n = 1.50) has one end ground to a hemispherical surface of radius 20 mm. A point object is on the axis, 80 mm to the left of the vertex, in air. Find (a) the image distance, (b) the magnification.

Solution

n_1 = 1, n_2 = 1.50, r = +20 mm (centre inside the glass), u = +80 mm

(a) $\frac{1}{80} + \frac{1.50}{v} = \frac{1.50 - 1}{20} \Rightarrow v =$ **+120 mm** (real, inside the rod)

(b) $m = -\frac{(1)(120)}{(1.50)(80)} =$ **−1.0**: same size, inverted

Example 23.2.5

The rod in Example 23.2.4 is placed in water (n = 1.33). Find the image distance and the magnification.

Solution

$\frac{1.33}{80} + \frac{1.50}{v} = \frac{1.50 - 1.33}{20} = 0.0085 \Rightarrow v =$ **−185 mm** (virtual, 185 mm in front of the vertex, in the water)

$m = -\frac{(1.33)(-185)}{(1.50)(80)} =$ **+2.05**: upright, magnified

Example 23.2.6

A coin is embedded in a spherical plastic paperweight of radius 3.0 cm (n = 1.50). The coin is 2.0 cm from the surface nearest the viewer. Find the position of its image.

Solution

Light goes from plastic to air: n_1 = 1.50, n_2 = 1.00, u = 2.0 cm. The centre of curvature is on the coin's side (in front of the surface), so r = −3.0 cm.

$\frac{1.50}{2.0} + \frac{1.00}{v} = \frac{1.00 - 1.50}{-3.0} = 0.1667 \Rightarrow v =$ **−1.7 cm**

Virtual image 1.7 cm inside the plastic: the coin looks nearer the surface than it is.

Example 23.2.7

A fish is 8.0 cm below the flat surface of water (n = 1.33). Find the apparent depth of the fish seen from directly above, and the magnification.

Solution

Flat surface: $r \to \infty$, so $\frac{1.33}{8.0} + \frac{1.00}{v} = 0 \Rightarrow v =$ **−6.0 cm** (virtual; apparent depth 6.0 cm)

$m = -\frac{(1.33)(-6.0)}{(1.00)(8.0)} =$ **+1.0**: upright, same size

In general, apparent depth = real depth/n.

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An object 5.0 mm high is 35.0 cm from the vertex of the spherical surface of a glass rod (n = 1.5). The image forms inside the rod, 50.0 cm from the vertex. (i) State the characteristics of the image. (ii) Find the radius of curvature. (iii) Where must the object be placed to give a virtual image 10.0 cm from the vertex?

Solution

(i) $m = -\frac{n_1 v}{n_2 u} = -\frac{50.0}{1.5 \times 35.0} = -0.952$: **real, inverted**, almost the same size ($h' \approx 4.8$ mm)

(ii) $\frac{1}{35.0} + \frac{1.5}{50.0} = \frac{0.5}{r} \Rightarrow 0.05857 = 0.5/r \Rightarrow r =$ **+8.54 cm**

(iii) v = −10.0 cm: $\frac{1}{u} - \frac{1.5}{10.0} = \frac{0.5}{8.54} \Rightarrow \frac{1}{u} = 0.2086 \Rightarrow u =$ **4.80 cm**

Exercise: refraction at spherical surfaces

1. A point object is 25.0 cm from the centre of a glass sphere of radius 5.0 cm (n = 1.50). Find the position of the image formed by (a) the first surface, (b) the second surface.

   Answer

   (a) u_1 = 20.0 cm, r = +5.0 cm: $\frac{1}{20.0} + \frac{1.50}{v_1} = \frac{0.50}{5.0} \Rightarrow v_1$ = +30.0 cm (20.0 cm beyond the sphere) (b) this is a virtual object for the back surface, u_2 = 10.0 − 30.0 = −20.0 cm, r = −5.0 cm: $\frac{1.50}{-20.0} + \frac{1}{v_2} = \frac{1 - 1.50}{-5.0} \Rightarrow v_2$ = +5.71 cm: a real image 5.71 cm from the back surface

23.3 Thin lenses

A lens is a transparent object with two refracting surfaces whose principal axes coincide. A **thin lens** is one whose thickness is small compared with its focal length.

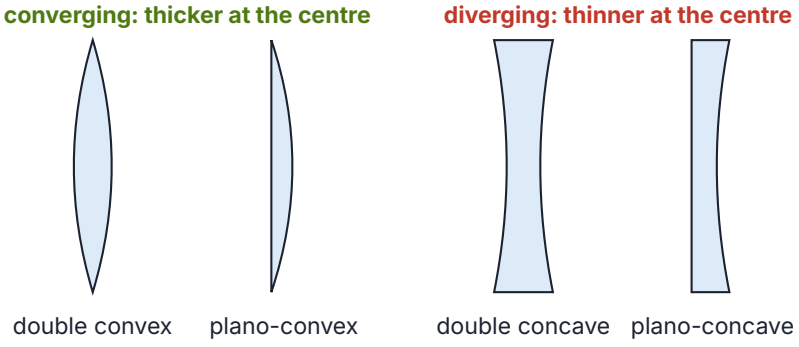


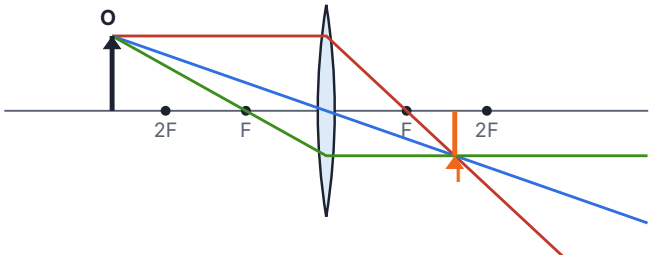
Figure 23.5: converging (convex) lenses make parallel rays converge; diverging (concave) lenses make them diverge.

- **Principal axis**: the line through the centre of the lens, perpendicular to both surfaces.
- **Focal point F**: the point where parallel paraxial rays meet after a converging lens, or from which they appear to diverge after a diverging lens. A lens has a focal point on each side.
- **Focal length f**: the distance between the focal point and the centre of the lens.

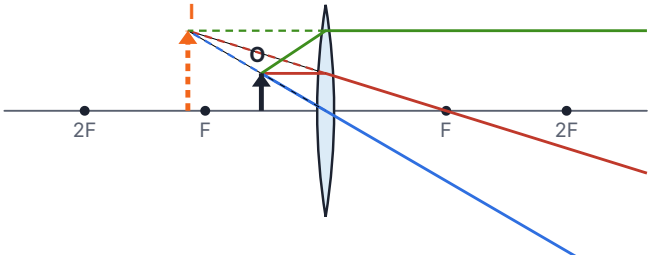
Ray diagrams for lenses

- **Ray 1**: a ray parallel to the axis passes through F on the other side (converging), or appears to come from F on the same side (diverging).
- **Ray 2**: a ray through F (or heading towards F on the far side, diverging) leaves parallel to the axis.
- **Ray 3**: a ray through the centre of the lens goes straight on.

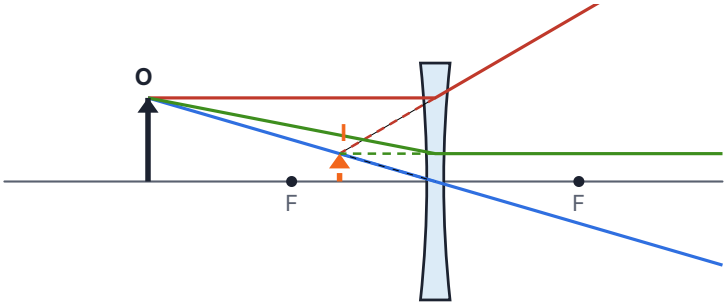
Object position (converging lens)	Image	Everyday use
beyond 2F	between F and 2F; real, inverted, diminished	camera, phone camera, the eye
at 2F	at 2F; real, inverted, same size	photocopier at 1 : 1
between F and 2F	beyond 2F; real, inverted, magnified	projector, enlarger
at F	at infinity	spotlight, searchlight
between F and the lens	on the same side; virtual, upright, magnified	magnifying glass



Converging lens, object beyond 2F.



Converging lens, object inside F: magnifying glass.



Diverging lens: the image is always virtual, upright and diminished, between the lens and F (door viewer, spectacles for short sight).

Thin lens formula and lensmaker's equation

Apply $\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$ to each surface of a thin lens of refractive index n_l in a medium of index n_m.

Surface 1 (medium → lens): $\frac{n_m}{u} + \frac{n_l}{v_1} = \frac{n_l - n_m}{r_1}$ … (1)

The image of surface 1 is the object for surface 2. For a thin lens it is a distance v_1 beyond surface 2, i.e. a virtual object: $u_2 = -v_1$.

Surface 2 (lens → medium): $\frac{n_l}{-v_1} + \frac{n_m}{v} = \frac{n_m - n_l}{r_2}$ … (2)

(1) + (2): $n_m(\frac{1}{u} + \frac{1}{v}) = (n_l - n_m)(\frac{1}{r_1} - \frac{1}{r_2})$

For an object at infinity, $v = f$. So the right-hand side divided by n_m is $1/f$, giving both results:

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \qquad m = -\frac{v}{u}$$

thin lens formula

$$\frac{1}{f} = (\frac{n_l}{n_m} - 1)(\frac{1}{r_1} - \frac{1}{r_2})$$

lensmaker's equation. In air, $n_m = 1$: $1/f = (n - 1)(1/r_1 - 1/r_2)$

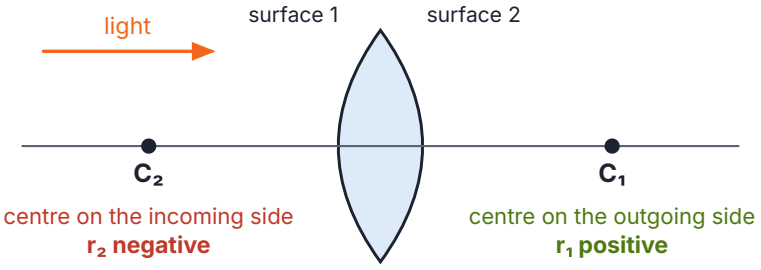


Figure 23.6: radius signs for a biconvex lens. r is positive when the centre of curvature is on the outgoing (back) side of that surface.

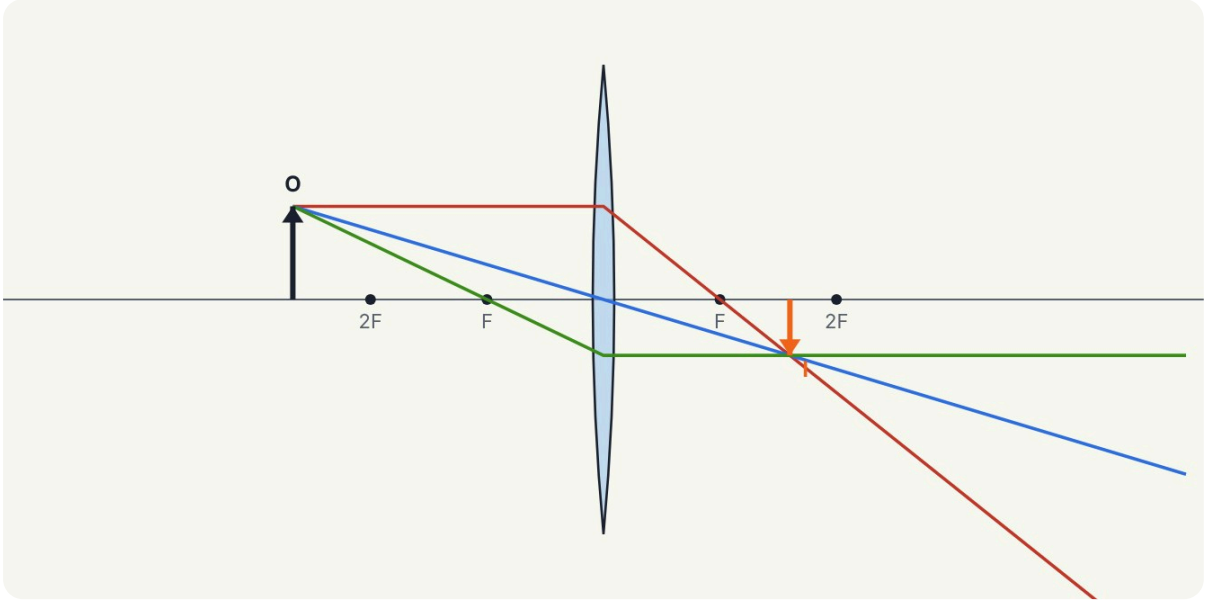
Quantity	Positive (+)	Negative (−)
f	converging lens	diverging lens
u	real object (in front of the lens)	virtual object (behind the lens)
v	real image (behind the lens)	virtual image (in front of the lens)
r	centre of curvature on the outgoing side	centre of curvature on the incoming side
m	upright image	inverted image

Biconvex: $r_1 = +$, $r_2 = -$. Biconcave: $r_1 = -$, $r_2 = +$. Flat face: $r = \infty$. Some books write $(1/r_1 + 1/r_2)$ with both radii as magnitudes for a biconvex lens; it is the same physics with a different convention. These notes use signed radii and the minus form throughout.

Simulation

Ray diagram for a thin lens

Choose a converging or diverging lens and move the object. The three principal rays find the image. Put the object at 2F, between F and 2F, and inside F, and compare with the table.



f	image distance v	magnification m	image
15.0 cm	**24.0 cm**	**−0.60**	**real, inverted, diminished**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 23 to use the simulation.

Example 23.3.1

An object is 10 cm from a converging lens of focal length 15 cm. Find the image position and its characteristics.

Solution

$\frac{1}{v} = \frac{1}{15} - \frac{1}{10} \Rightarrow v =$ **−30 cm** (virtual, on the same side as the object)

$m = -(-30)/10 =$ **+3.0**: upright, three times as large (a magnifying glass)

Example 23.3.2

Where must a small insect be placed if a diverging lens of focal length 25 cm is to form a virtual image 20 cm in front of the lens?

Solution

$f = -25$ cm, $v = -20$ cm: $\frac{1}{u} = \frac{1}{-25} - \frac{1}{-20} = \frac{1}{100} \Rightarrow u =$ **100 cm**

Example 23.3.3

A diverging meniscus lens has a focal length of −16 cm. The lens is held 10 cm from an object. Find the image position and the magnification.

Solution

$\frac{1}{v} = \frac{1}{-16} - \frac{1}{10} \Rightarrow v =$ **−6.15 cm**

$m = 6.15/10 =$ **+0.615** (virtual, upright, diminished)

Example 23.3.4

An object 450 mm from a converging lens forms a real image 900 mm from the lens. Find the focal length.

Solution

$\frac{1}{f} = \frac{1}{450} + \frac{1}{900} = \frac{3}{900} \Rightarrow f = 300$ mm = **30 cm**

Example 23.3.5

An object 6 cm high is 4 cm from a diverging meniscus lens of focal length −24 cm. Find the nature, size and location of the image.

Solution

$$\frac{1}{v} = \frac{1}{-24} - \frac{1}{4} \Rightarrow v = \mathbf{-3.43\ cm}$$

m = 3.43/4 = 0.857 ⇒ h' = 0.857 × 6 = **5.14 cm**; virtual and upright

Exercises: thin lens formula

1. An object 8 cm high is 30 cm from a converging lens of focal length 12 cm. Find the nature, size and location of the image.

   Answer

   v = +20 cm, m = −0.667: real, inverted, 5.33 cm high (h' = −5.33 cm)

2. A 1.70 m tall person stands 2.50 m in front of a camera whose converging lens has a focal length of 0.0500 m. Find the characteristics of the image on the sensor.

   Answer

   v = 0.0510 m, m = −0.0204: real, inverted, diminished, 0.0347 m high

3. An object is 18 cm in front of a diverging lens of focal length −12 cm. How far in front of the lens should the object be placed so that the size of its image is halved?

   Answer

   At 18 cm, m = 0.40; for m = 0.20: u = 48 cm

4. How far from a 50.0 mm focal length lens must an object be placed for a real image magnified 3.00×? And for a virtual image magnified 3.00×?

   Answer

   66.7 mm (real); 33.3 mm (virtual)

5. A lens focuses an object 22.5 cm away as an image 33.0 cm on the other side. What type of lens is it, what is its focal length, and is the image real or virtual?

   Answer

   converging, f = 13.4 cm, real

6. (a) A 2.70 cm high insect is 2.20 m from a lens of focal length 135 mm. Where is the image, how high is it, and what type is it? (b) Repeat for $f = -135$ mm.

Answer

(a) 144 mm behind the lens, real, inverted, 1.77 mm (b) 127 mm in front of the lens, virtual, upright, 1.56 mm

Using the lensmaker's equation

Example 23.3.6

A meniscus lens (n = 1.52) in air has radii r_1 = +22.4 cm and r_2 = +46.2 cm. Find its focal length and state the type of lens.

Solution

$\frac{1}{f} = (1.52 - 1)\left(\frac{1}{22.4} - \frac{1}{46.2}\right) = 0.52 \times 0.02300 \Rightarrow f =$ **+83.6 cm**

f is positive: a **converging** lens.

Example 23.3.7

Each face of a double convex lens has radius 20.0 cm; n = 1.50. Find the focal length (a) in air, (b) in carbon disulfide (n = 1.63).

Solution

r_1 = +20.0 cm, r_2 = −20.0 cm

(a) $\frac{1}{f} = (1.50 - 1)\left(\frac{1}{20.0} + \frac{1}{20.0}\right) \Rightarrow f =$ **+20.0 cm**

(b) $\frac{1}{f} = \left(\frac{1.50}{1.63} - 1\right)(0.100) = -0.00798 \Rightarrow f =$ **−125 cm**: in a denser liquid, the lens is **diverging**.

Example 23.3.8

A plano-convex lens of glass (n = 1.5) is to have a focal length of 40 cm. Find the radius of the curved surface.

Solution

Flat face first: $r_1 = \infty$. $\frac{1}{40} = (1.5 - 1)(0 - \frac{1}{r_2}) \Rightarrow r_2 = -20$ cm

radius of the curved surface = **20 cm**

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A thin biconvex lens (n = 1.5, both radii 10.0 cm) is immersed in a liquid of refractive index 1.65. Find the focal length and state how the lens behaves.

Solution

$\frac{1}{f} = (\frac{1.5}{1.65} - 1)(\frac{1}{10.0} + \frac{1}{10.0}) = (-0.0909)(0.200) \Rightarrow f =$ **−55 cm**

f is negative: in this liquid the lens **diverges** a parallel beam.

Power of a lens

Power of a lens

the reciprocal of its focal length in metres. Unit: dioptre (D) = m^{-1}. Converging lenses have positive power; diverging lenses have negative power.

$$P = \frac{1}{f}$$

Two thin lenses in contact

Lens 1: $\frac{1}{u} + \frac{1}{v_1} = \frac{1}{f_1}$. Its image is a virtual object for lens 2 (no gap): $u_2 = -v_1$.

Lens 2: $-\frac{1}{v_1} + \frac{1}{v} = \frac{1}{f_2}$. Adding: $\frac{1}{u} + \frac{1}{v} = \frac{1}{f_1} + \frac{1}{f_2}$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \qquad P = P_1 + P_2$$

Exercises: power and lenses in contact

1. A converging lens has a focal length of 12.5 cm. Calculate its power.

   Answer

   +8.0 D

2. A lens has a power of −4.0 D. State the type of lens and find its focal length.

   Answer

   diverging, f = −25 cm

3. A lens of power +6.0 D is in contact with a lens of power −2.0 D. Find the power and focal length of the combination.

   Answer

   +4.0 D, f = +25 cm

4. Two thin lenses of focal lengths +20.0 cm and +30.0 cm are in contact. Find the focal length and power of the combination.

   Answer

   f = +12.0 cm, P = +8.33 D

5. A double convex lens (n = 1.50) has r_1 = +15.0 cm and r_2 = −15.0 cm. Find its focal length and power in air.

   Answer

   f = +15.0 cm, P = +6.67 D

Combination of two lenses (separated)

Find the image I_1 formed by the first lens alone.

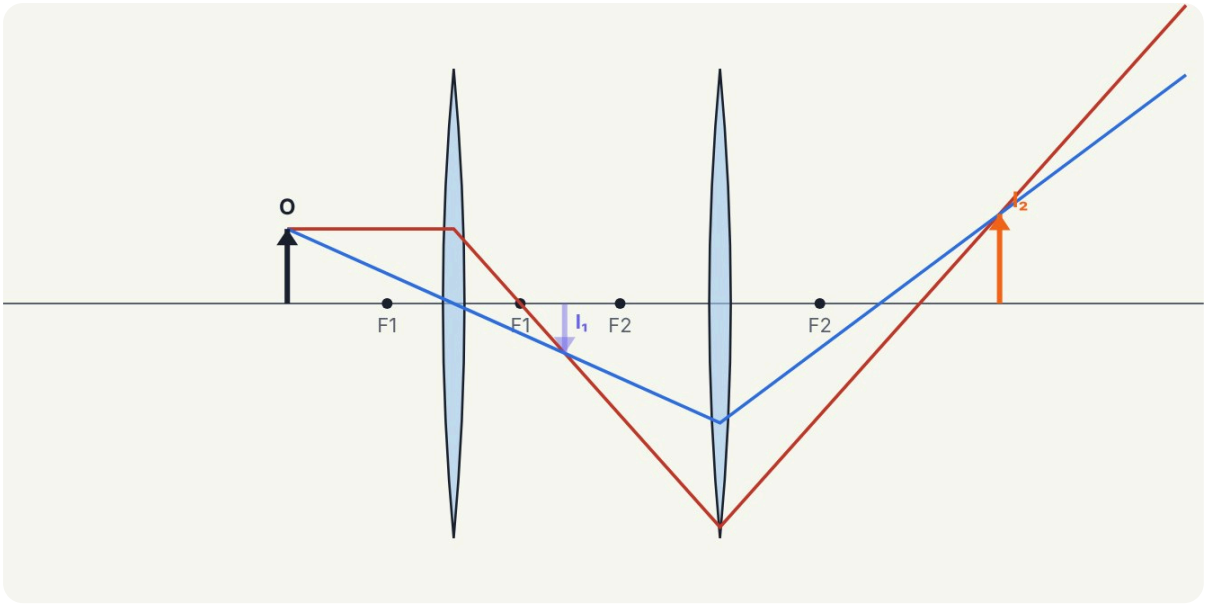
Treat I_1 as the object for the second lens: $u_2 = d - v_1$. If I_1 is in front of lens 2, u_2 is positive (real object). If I_1 falls beyond lens 2, u_2 is negative (virtual object).

Find the final image from lens 2. The overall magnification is $m = m_1 \times m_2$.

Simulation

Two-lens system

Set the two focal lengths, the separation and the object distance. The intermediate image I_1 (faint) becomes the object for the second lens. Make I_1 fall beyond lens 2 to see a virtual object.



lens 1: v_1	lens 2: $u_2 = d - v_1$	lens 2: v_2	$m = m_1 \times m_2$
16.7 cm	**23.3 cm**	**42.0 cm**	**1.20**

image
real, upright, magnified

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 23 to use the simulation.

Example 23.3.9

Two converging lenses, f_1 = 10.0 cm and f_2 = 20.0 cm, are 20.0 cm apart. An object is 30.0 cm in front of the first. Find the position and magnification of the final image.

Solution

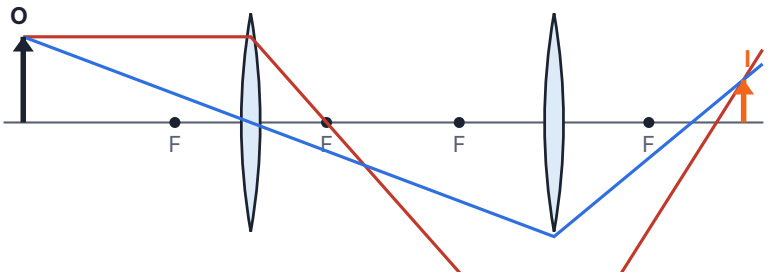
Lens 1: $\frac{1}{v_1} = \frac{1}{10.0} - \frac{1}{30.0} \Rightarrow v_1 = 15.0$ cm, $m_1 = -0.500$

Lens 2: $u_2 = 20.0 - 15.0 = 5.0$ cm; $\frac{1}{v_2} = \frac{1}{20.0} - \frac{1}{5.0} \Rightarrow v_2 =$ **−6.67 cm** (virtual, 6.67 cm in front of lens 2)

$m_2 = -(-6.67)/5.0 = +1.33$; $m = (-0.500)(+1.33) =$ **−0.667** (inverted)

Example 23.3.10

Two converging lenses, f_1 = 20.0 cm and f_2 = 25.0 cm, are 80.0 cm apart. An object is 60.0 cm in front of the first. Find the position and magnification of the final image.



Solution

Lens 1: $v_1 = 30.0$ cm, $m_1 = -0.500$

Lens 2: $u_2 = 80.0 - 30.0 = 50.0$ cm $\Rightarrow v_2 =$ **+50.0 cm** beyond lens 2 (real); $m_2 = -1.00$

$m = (-0.500)(-1.00) =$ **+0.500** (upright relative to the object)

Exercises: two lenses

1. A converging lens has focal length 0.080 m. An object is 0.040 m to its left. A second identical lens is 0.120 m to the right of the first. Where is the final image, relative to the second lens?

   Answer

   v_1 = −8.0 cm (virtual), u_2 = 20.0 cm ⇒ v_2 = +13.3 cm = 0.133 m to the right of lens 2; m = (+2.00)(−0.667) = −1.33

2. A coin is 20.0 cm to the left of a converging lens (f = 16.0 cm). A second identical lens is placed to the right so that the final image has the same size and orientation as the coin. Find the separation of the lenses.

   Answer

   v_1 = 80.0 cm, m_1 = −4.00; need m_2 = −0.250 ⇒ u_2 = 80.0 cm; separation = 160 cm

3. Two converging lenses 60 cm apart have focal lengths 10 cm and 15.0 cm. An object 6.0 cm high is 20 cm in front of the first. Find the location and size of the final image. Is it real or virtual?

   Answer

   v_1 = 20 cm, u_2 = 40 cm ⇒ v_2 = 24 cm beyond lens 2; m = (−1.00)(−0.600) = +0.600, h' = 3.6 cm; real

Combination of a lens and a spherical mirror

- Light passes through the lens, reflects from the mirror, then passes through the lens **again**. So there are three stages (lens → mirror → lens), and the three magnifications multiply.
- The image formed by the lens is the object for the mirror. If that image would fall behind the mirror, the mirror intercepts the rays before they meet: it is a **virtual object** and its object distance is negative.

Example 23.3.11

A converging lens (f = 10.0 cm) and a concave mirror (r = 30.0 cm) are 50.0 cm apart on a common axis. An object is 30.0 cm in front of the lens. Find the position and magnification of the final image.

Solution

Lens: $\frac{1}{v_1} = \frac{1}{10.0} - \frac{1}{30.0} \Rightarrow v_1 = 15.0$ cm, $m_1 = -0.500$

Mirror (f = +15.0 cm): $u_2 = 50.0 - 15.0 = 35.0$ cm $\Rightarrow v_2 = 26.25$ cm, $m_2 = -0.750$

Lens again: $u_3 = 50.0 - 26.25 = 23.75$ cm $\Rightarrow v_3$ = **17.3 cm** from the lens (on the object side, real); $m_3 = -0.727$

$m = (-0.500)(-0.750)(-0.727)$ = **−0.273** (inverted, diminished)

Exercises: lens and mirror

1. A converging lens (f = 15.0 cm) is 40.0 cm from a convex mirror of radius 20.0 cm on a common axis. An object is 30.0 cm in front of the lens. Find the position and magnification of the final image.

   Answer

   v_1 = 30.0 cm, m_1 = −1.00; mirror u_2 = 10.0 cm, f = −10.0 cm ⇒ v_2 = −5.0 cm, m_2 = +0.500; lens again u_3 = 45.0 cm ⇒ v_3 = 22.5 cm, m_3 = −0.500; final image 22.5 cm from the lens, m = +0.25 (upright)

2. A converging lens (f = 10.0 cm) and a concave mirror (f = 20.0 cm) are 80.0 cm apart. An object is 15.0 cm in front of the lens. Find the position and magnification of the final image.

   Answer

   v_1 = 30.0 cm (m_1 = −2.00); u_2 = 50.0 cm ⇒ v_2 = 33.3 cm (m_2 = −0.667); u_3 = 46.7 cm ⇒ v_3 = 12.7 cm (m_3 = −0.272); final image 12.7 cm from the lens, m = −0.36 (inverted)

STPM past-year questions

STPM 2013 · Paper 3 · Q7

A baby is in front of a concave mirror of radius of curvature R. An upright image twice the size of the baby is seen. What is the distance between the baby and the mirror?

A. $R/4$

B. $R/2$

C. $3R/4$

D. R

Answer

Answer: A

Upright ⇒ virtual, $m = +2 \Rightarrow v = -2u$. $\frac{1}{u} - \frac{1}{2u} = \frac{2}{R} \Rightarrow u = R/4$.

STPM 2013 · Paper 3 · Q8

A diverging lens and a converging lens, each of focal length 10.0 cm, are 10.0 cm apart, with the diverging lens in front. The arrangement is focused on a distant object. The image formed is

A. virtual, 5.0 cm behind the converging lens

B. virtual, 10.0 cm behind the diverging lens

C. real, 10.0 cm behind the converging lens

D. real, 20.0 cm behind the converging lens

Answer

Answer: D

Parallel rays give a virtual image at the focus of the diverging lens, 10.0 cm in front of it. So $u_2 = 20.0$ cm for the converging lens: $\frac{1}{20.0} + \frac{1}{v} = \frac{1}{10.0} \Rightarrow v = +20.0$ cm (real).

STPM 2014 · Paper 3 · Q7

A parallel beam is incident normally on a converging lens of focal length f_1 in contact with a diverging lens of focal length f_2. The focal length of the combination is f_3. If the image distances of the converging lens and of the diverging lens are v_1 and v_2, which pair (v_1, v_2) is correct?

A. (f_1, f_2)

B. (f_1, f_3)

C. (f_2, f_1)

D. (f_3, f_1)

Answer

Answer: B

The converging lens alone would focus the parallel beam at its own focus, so $v_1 = f_1$. Together the lenses act as one lens of focal length f_3, so the light converges at $v_2 = f_3$.

STPM 2015 · Paper 3 · Q7

An object is 5.0 cm in front of a concave mirror of radius of curvature 20.0 cm. Which statement about the image is true?

A. It is enlarged, upright and 10 cm behind the mirror.

B. It is enlarged, inverted and 10 cm in front of the mirror.

C. It is diminished, upright and 6.7 cm behind the mirror.

D. It is diminished, inverted and 6.7 cm in front of the mirror.

Answer

Answer: A

$f = +10$ cm: $\frac{1}{5.0} + \frac{1}{v} = \frac{1}{10} \Rightarrow v = -10$ cm; $m = +2$ (virtual, upright, enlarged).

STPM 2015 (repeat) · Paper 3 · Q7

A converging lens forms an inverted image twice the size of the object when the object is 30 cm from the lens. What is the new object distance if the image is also twice the size of the object but upright?

A. 5.0 cm

B. 10.0 cm

C. 15.0 cm

D. 20.0 cm

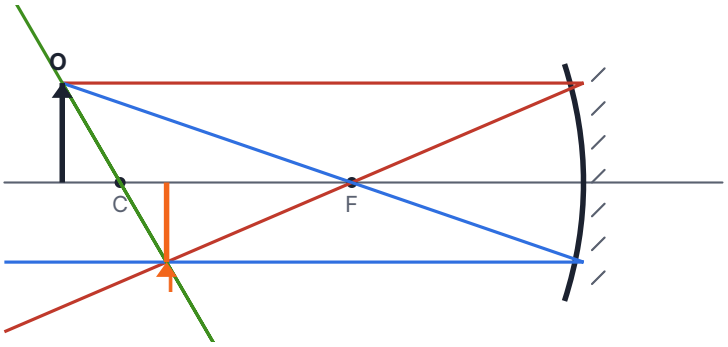
Answer

Answer: B

Inverted, $m = -2$ at $u = 30$ cm $\Rightarrow v = 60$ cm, $f = 20$ cm. Upright $\Rightarrow$ virtual, $v = -2u$: $\frac{1}{u} - \frac{1}{2u} = \frac{1}{20}$ $\Rightarrow u = 10.0$ cm.

STPM 2014 · Paper 3 · Q19(a)

A concave mirror has radius of curvature 40.0 cm. A nail 5.00 cm long stands upright on the principal axis, 45.0 cm in front of the mirror. (i) Calculate the image distance. (ii) Find the size of the image. (iii) State two characteristics of the image.



Solution

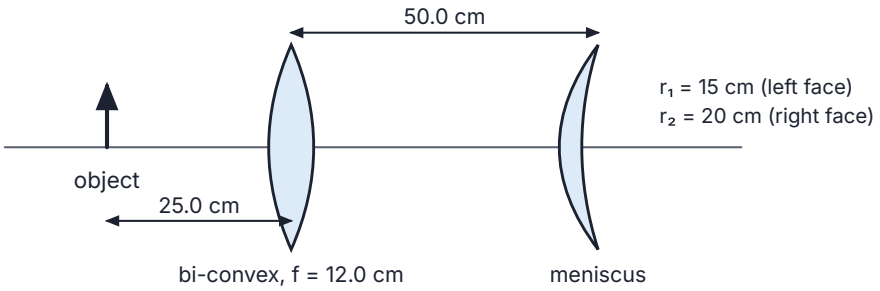
(i) $f = +20.0$ cm: $\frac{1}{v} = \frac{1}{20.0} - \frac{1}{45.0} \Rightarrow v =$ **+36.0 cm**

(ii) $m = -36.0/45.0 = -0.800 \Rightarrow$ image height = **4.00 cm**

(iii) real (v positive) and inverted (m negative); also diminished

STPM 2014 · Paper 3 · Q19(b)

An object is 25.0 cm in front of a biconvex lens of focal length 12.0 cm. A meniscus lens (radii 15.0 cm and 20.0 cm, n = 1.5) is 50.0 cm behind the biconvex lens. Determine (i) the focal length of the meniscus lens, (ii) the position of the final image, (iii) the magnification.



The arrangement in STPM 2014 Paper 3 Q19(b) (not to scale).

Solution

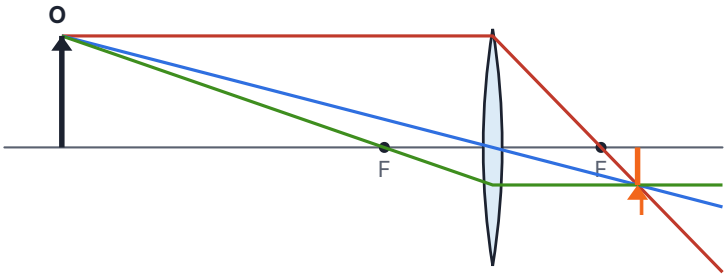
(i) From the figure, both faces of the meniscus bulge towards the incoming light, and the left face (radius 15.0 cm) is the more curved. Both centres of curvature are on the outgoing side, so r_1 = +15.0 cm, r_2 = +20.0 cm: $\frac{1}{f} = 0.5(\frac{1}{15.0} - \frac{1}{20.0}) \Rightarrow f =$ **+120 cm** (converging: thicker at the centre)

(ii) Biconvex lens: $\frac{1}{25.0} + \frac{1}{v_1} = \frac{1}{12.0} \Rightarrow v_1$ = +23.08 cm. Meniscus: u_2 = 50.0 − 23.08 = 26.92 cm $\Rightarrow \frac{1}{v_2} = \frac{1}{120} - \frac{1}{26.92} \Rightarrow v_2 =$ **−34.7 cm**: a virtual image 34.7 cm in front of the meniscus lens

(iii) $m = (-\frac{23.08}{25.0})(-\frac{-34.71}{26.92}) =$ **−1.19** (inverted, slightly enlarged)

STPM 2015 · Paper 3 · Q19

(a) An object is 30.0 cm in front of a converging lens of focal length 7.55 cm. Find the image position, the magnification and the characteristics of the image. (b) A 10.0 cm object is 15.0 cm in front of a convex mirror of radius 40.0 cm. Find the image position and height. (c) A spider 2.00 cm across hangs 200 cm from a wall. A lens forms an image of the spider on the wall half its size. What lens is needed, and where?



Solution

(a) $\frac{1}{v} = \frac{1}{7.55} - \frac{1}{30.0} \Rightarrow v =$ **+10.1 cm**; $m = -10.1/30.0 =$ **−0.336**: real, inverted, diminished

(b) $f = -20.0$ cm: $\frac{1}{v} = \frac{1}{-20.0} - \frac{1}{15.0} \Rightarrow v =$ **−8.57 cm** (behind the mirror); $h' = 10.0 \times 8.57/15.0$ = **5.71 cm**: virtual, upright, diminished

(c) The image is on a screen, so it is real: a **converging** lens. $|m| = v/u = ½ \Rightarrow u = 2v$, and $u + v =$ 200 cm $\Rightarrow v = 66.7$ cm from the wall, $u = 133$ cm; $\frac{1}{f} = \frac{1}{133} + \frac{1}{66.7} \Rightarrow f =$ **44.4 cm**

Chapter 24 · Semester 3

WAVE OPTICS

Notes by Cikgu Suhaili

21 worked examples 6 simulations Exercises with answers

24.1 Huygens's principle

Physical (wave) optics studies the wave nature of light: interference, diffraction and polarisation.

- A **wavefront** is a line or surface on which all points vibrate in the same phase. Wavefronts can be plane or spherical.
- A **ray** is a line perpendicular to the wavefronts; it shows the direction of travel of the wave.

> **Huygens's principle**
>
> states that **every point on a wavefront acts as a source of secondary wavelets that spread out in the forward direction at the same speed as the wave. The new wavefront is the surface tangent to all these wavelets**.

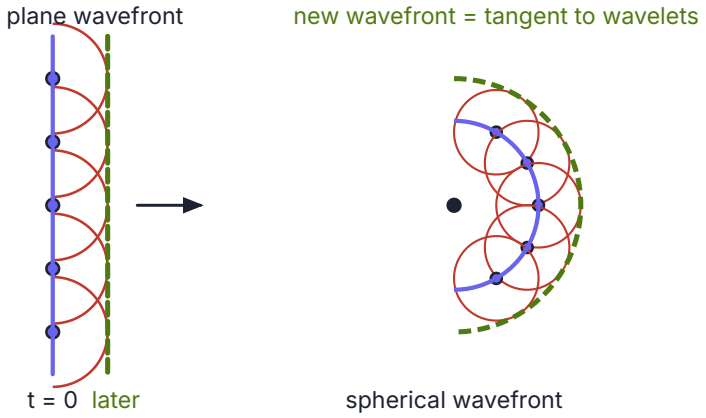


Figure 24.1: constructing the new wavefront from the secondary wavelets, for a plane and a spherical wavefront.

Using Huygens's principle

Phenomenon	Explanation with wavelets
Reflection	Points on the wavefront reach the mirror one after another. Their wavelets spread back from the mirror; the tangent to them is the reflected wavefront, at an equal angle ($i = r$).
Refraction	Wavelets in the second medium travel at a different speed, so in the same time they have a different radius. The tangent wavefront changes direction, giving Snell's law, $\sin\theta_1/\sin\theta_2 = v_1/v_2$.
Diffraction	Only the part of the wavefront in the gap produces wavelets. The wavelets at the edges spread into the shadow, so the new wavefront bends round the edges. The effect is greatest when the gap is about the size of λ.
Interference	Wavelets from two (or more) coherent sources overlap. Where crests meet crests the light is bright; where crests meet troughs it is dark.

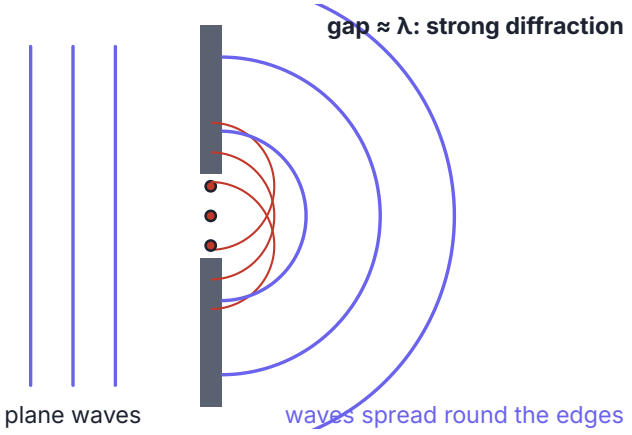
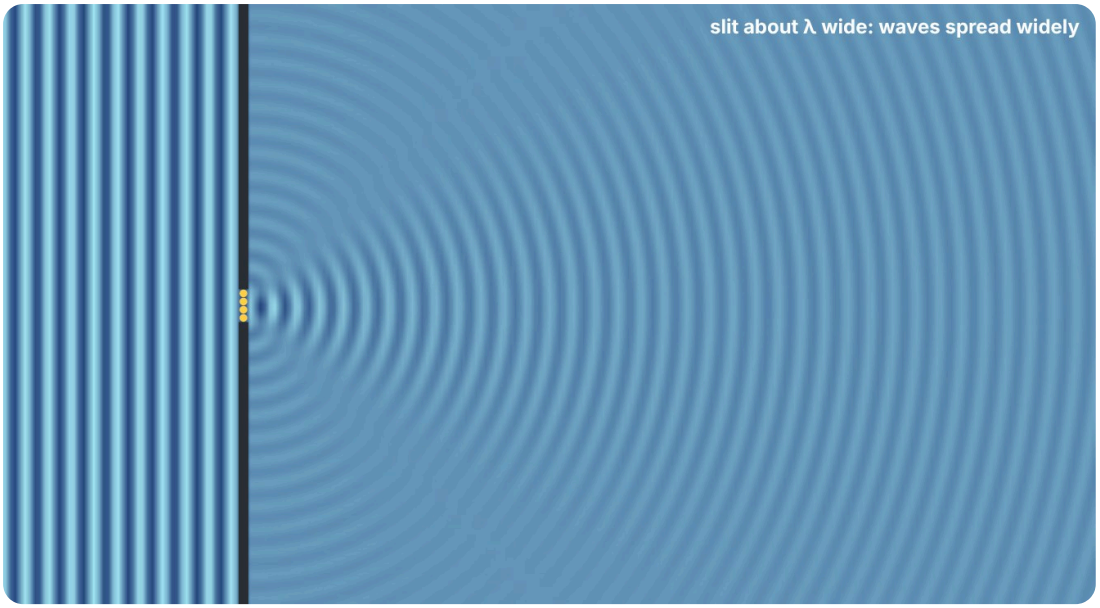


Figure 24.2: diffraction at a gap explained by Huygens's wavelets.

Simulation

Huygens wavelets: diffraction and interference

Each dot in the gap is a source of wavelets. With one slit you see diffraction: narrow the slit and the waves spread more. With two slits the wavelets from both gaps interfere and give bright and dark directions.



This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 24 to use the simulation.

24.2 Interference

Coherent sources

are sources that **emit waves of the same frequency (wavelength) with a constant phase difference**.

- Two separate lamps are not coherent: each atom emits at random, so the phase difference changes all the time. Coherent sources are made by splitting light from one source: by two slits (division of wavefront) or by partial reflection at a thin film (division of amplitude).
- A clear interference pattern also needs waves of roughly the same amplitude, so that the dark fringes are really dark.

Constructive interference	Destructive interference
the waves meet in phase (crest on crest)	the waves meet in antiphase (crest on trough)
resultant amplitude = sum of the amplitudes: bright	resultant amplitude = difference (zero if equal): dark

Path difference and optical path difference

- **Path difference**: the difference between the distances travelled by the two waves from their sources to the point.
- A path difference of λ is a phase difference of 2π; a path difference of λ/2 is a phase difference of π. In general, Δφ = (2π/λ) × path difference.
- **Optical path** = refractive index × geometric path, nL. In a medium of index n the wavelength is λ/n, so a length L holds as many waves as a length nL in a vacuum.
- **Optical path difference** (OPD): the difference between the optical paths of the two waves. Use the OPD whenever the light goes through different media.

constructive: OPD = $m\lambda$ destructive: OPD = $(m + \tfrac{1}{2})\lambda$

m = 0, 1, 2, … for two coherent sources in phase; λ = wavelength in a vacuum

Example 24.2.1

Light of wavelength 600 nm passes through a glass plate (n = 1.50) of thickness 2.0 μm, while a second beam travels the same distance in air. Find the optical path difference and the phase difference.

Solution

OPD = $(n - 1)t$ = $(0.50)(2.0 \times 10^{-6})$ = **1.0×10^{-6} m**

Δφ = 2π × OPD/λ = 2π × $(1.0 \times 10^{-6})/(600 \times 10^{-9})$ = **3.33π rad** (1.67 cycles)

24.3 Two-slit interference pattern (Young's experiment)

In 1801 Thomas Young showed interference of light with two slits. It proved that light is a wave, and it lets us measure its wavelength.

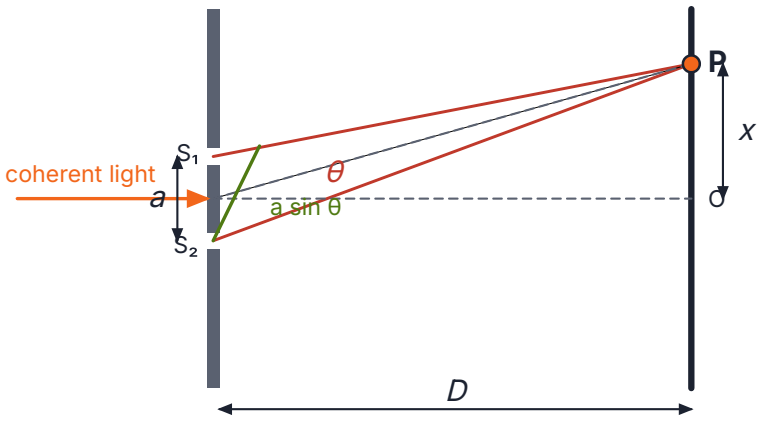


Figure 24.3: the two slits S_1 and S_2 act as coherent sources. The path difference to P is a sin θ.

- Monochromatic light from one source falls on two narrow slits S_1 and S_2 a distance a apart. By diffraction, the light spreads out from each slit, and the two slits act as **coherent sources** in phase.
- The waves overlap on a screen a distance D away. Where the path difference is a whole number of wavelengths they interfere constructively: a **bright fringe**. Where it is an odd number of half wavelengths: a **dark fringe**.
- The result is a series of equally spaced, equally bright fringes.

Derivation of the fringe separation

For a distant screen ($D \gg a$) the two rays are almost parallel. Path difference $S_2P - S_1P = a \sin\theta$.

Bright fringes: $a \sin\theta = m\lambda$; dark fringes: $a \sin\theta = (m + \tfrac{1}{2})\lambda$, $m = 0, 1, 2, \ldots$

θ is small, so $\sin\theta \approx \tan\theta = \dfrac{x}{D}$, where x is the distance of the fringe from the central maximum O.

Bright: $x_m = \dfrac{m\lambda D}{a}$; dark: $x_m = (m + \tfrac{1}{2})\dfrac{\lambda D}{a}$

Separation of adjacent bright (or dark) fringes: $\Delta x = x_{m+1} - x_m = \dfrac{\lambda D}{a}$

$$\Delta x = \frac{\lambda D}{a}$$

Δx = fringe separation; λ = wavelength; D = slit-to-screen distance; a = slit separation (centre to centre). The central bright fringe is $m = 0$; the first dark fringe on each side is $m = 0$ in the dark-fringe formula.

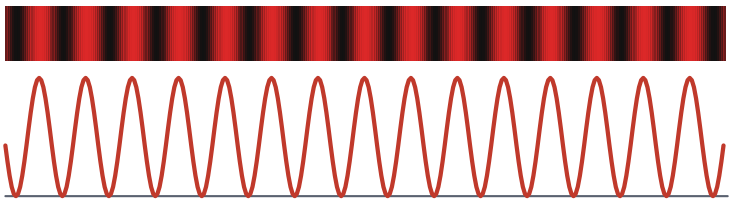


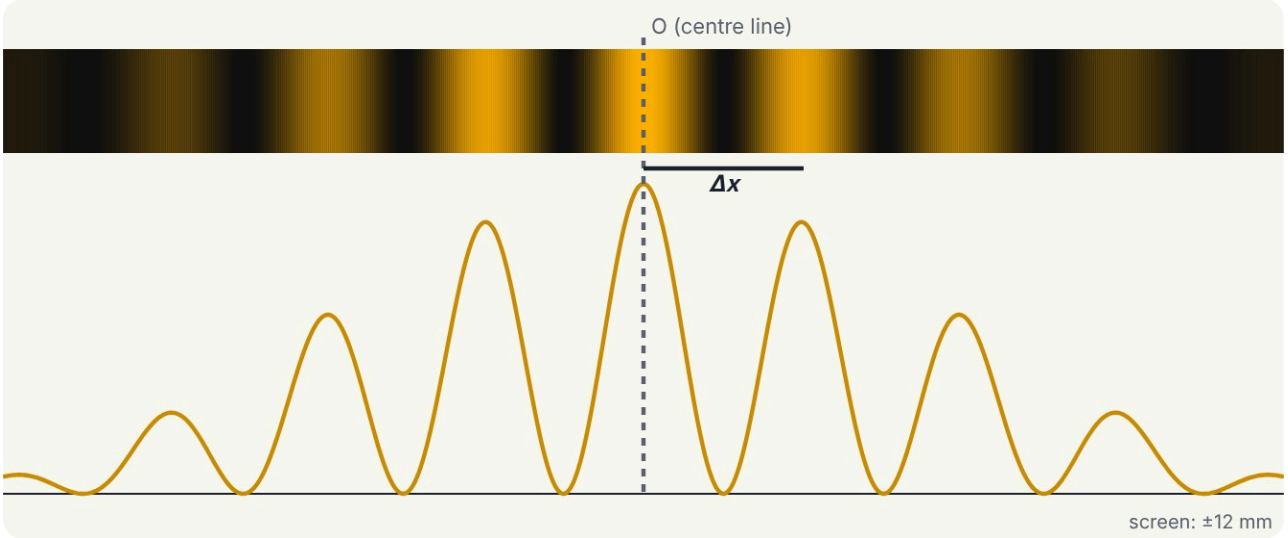
Figure 24.4: the two-slit pattern. All the bright fringes are about equally bright and equally spaced.

Change	Effect on $\Delta x = \lambda D/a$
longer wavelength (red instead of blue)	fringes further apart
screen further away (larger D)	fringes further apart
slits further apart (larger a)	fringes closer together
apparatus in water	λ becomes λ/n: fringes closer together, more fringes seen
white light	central fringe white; other fringes coloured (violet inside, red outside)
light from one slit delayed by $\lambda/2$	bright and dark fringes swap places; the separation does not change
thin transparent film over one slit	the whole pattern shifts towards that slit by $(n - 1)t/\lambda$ fringes

Simulation

Young's two-slit experiment

Change the wavelength, the slit separation and the screen distance, and measure the fringe separation. Then cover one slit, put the apparatus in water, or place a thin glass film over one slit and watch the pattern move.



λ in the medium	fringe separation Δx = λD/a	pattern shift by the film	colour
600 nm	**3.00 mm**	**none**	**orange**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 24 to use the simulation.

Example 24.3.1

Light of wavelength 550 nm falls on two slits 50 μm apart. The second-order bright fringe is 4.5 cm from the central maximum. How far is the screen from the slits?

Solution

$$x_2 = \frac{2\lambda D}{a} \Rightarrow 4.5 \times 10^{-2} = \frac{2(550 \times 10^{-9})D}{50 \times 10^{-6}} \Rightarrow D = \mathbf{2.0\ m}$$

Example 24.3.2

Light passes through two slits 0.050 mm apart. On a wall 1.0 m away, the second-order bright fringe is 2.4 cm from the central maximum. Find (a) the wavelength, (b) the distance between the second- and third-order bright fringes.

Solution

(a) $\lambda = \frac{x_2 a}{2D} = \frac{(2.4 \times 10^{-2})(0.050 \times 10^{-3})}{2(1.0)} = \mathbf{6.0 \times 10^{-7}\ m}$

(b) $\Delta x = \frac{\lambda D}{a} = \frac{(6.0 \times 10^{-7})(1.0)}{0.050 \times 10^{-3}} = \mathbf{1.2 \times 10^{-2}\ m}$

Example 24.3.3

Find the separation of two slits that gives the first dark fringe for 410 nm violet light at 45°.

Solution

First dark fringe: $a \sin\theta = \tfrac{1}{2}\lambda \Rightarrow a = \frac{0.5(410\ \text{nm})}{\sin 45°} = \mathbf{290\ nm}$

STPM past-year · Section C

Monochromatic light of frequency 6.00×10^{14} Hz is used in a Young's two-slit experiment. The fringe separation is 0.75 mm and the screen is 1.80 m from the slits. (i) Calculate the separation of the slits. [3] (ii) A thin film of glass (n = 1.50) is placed in front of one slit, and the original central bright fringe becomes the 10th bright fringe. Find the thickness of the film. [3] (iii) If one slit is covered, a single-slit pattern is seen with a central bright fringe 1.0 cm wide. Find the width of the slit. [3]

Solution

$\lambda = c/f = 3.00 \times 10^8 / 6.00 \times 10^{14} = 5.00 \times 10^{-7}$ m (do not put the frequency into the formula!)

(i) $a = \dfrac{\lambda D}{\Delta x} = \dfrac{(5.00 \times 10^{-7})(1.80)}{0.75 \times 10^{-3}} = \mathbf{1.20 \times 10^{-3}\ m}$

(ii) The film adds an optical path difference $(n - 1)t$. The central point is now the 10th bright fringe: $(n - 1)t = 10\lambda \Rightarrow t = \dfrac{10(5.00 \times 10^{-7})}{0.50} = \mathbf{1.00 \times 10^{-5}\ m}$

(iii) This is now single-slit diffraction: central width $= 2\lambda D/a \Rightarrow a = \dfrac{2(5.00 \times 10^{-7})(1.80)}{1.0 \times 10^{-2}} = \mathbf{1.80 \times 10^{-4}\ m}$

Teacher's note

Examiners' report: many candidates used the thin-film formula in (ii) instead of the optical path difference $(n - 1)t$, and used the two-slit formula in (iii). Read which arrangement the question describes.

24.4 Interference in a thin film

The colours of soap bubbles and oil films come from interference between light reflected from the **top** and the **bottom** surfaces of a thin transparent film (division of amplitude).

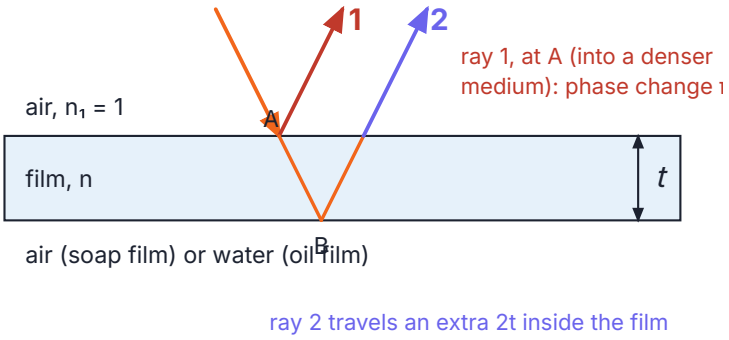


Figure 24.5: light incident almost normally on a film of thickness t and refractive index n.

Two things to consider

- **The extra path**: ray 2 travels an extra distance $2t$ inside the film, an optical path difference of $\mathbf{2}nt$.
- **Phase change on reflection**: light reflected at a boundary with a medium of **higher** refractive index undergoes a phase change of π (equivalent to an extra λ/2). Reflection at a boundary with a **lower** refractive index gives no phase change.

Situation	Phase changes	Bright (constructive)	Dark (destructive)
soap film in air; oil film (n = 1.45) on water (n = 1.33)	one (at the top surface only)	$2nt = (m + ½)\lambda$	$2nt = m\lambda$
coating (n = 1.38) on glass (n = 1.50)	two (at both surfaces), so they cancel	$2nt = m\lambda$	$2nt = (m + ½)\lambda$

λ is the wavelength in air (vacuum). Writing $2nt$ is the same as comparing $2t$ with $\lambda_n = \lambda/n$, the wavelength inside the film.

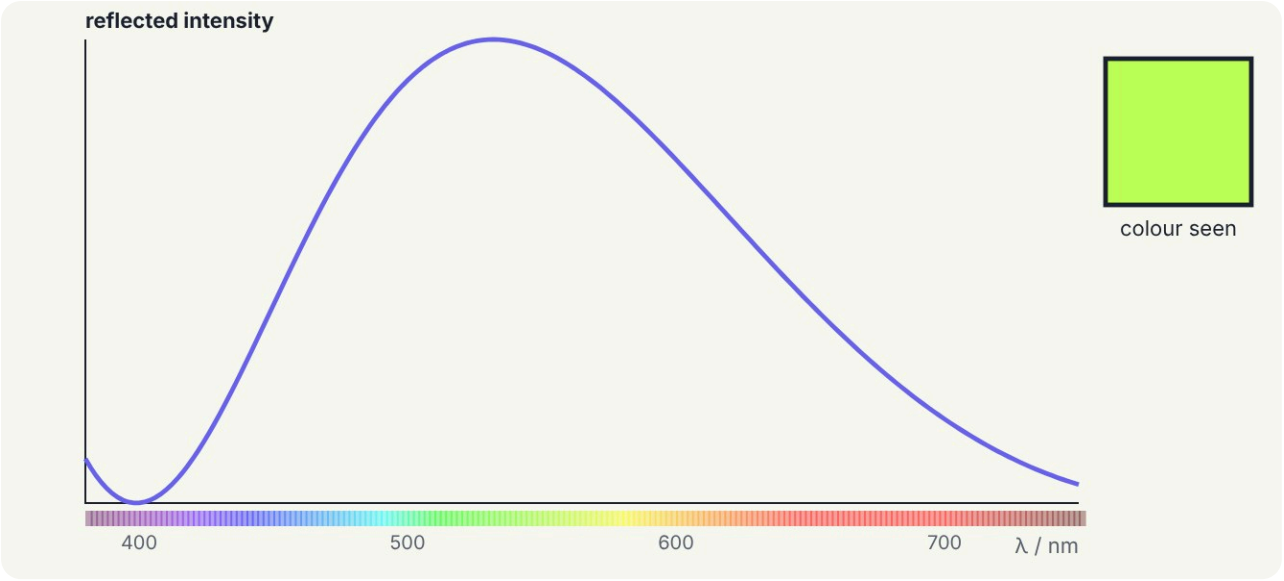
Teacher's note

Count the phase changes first, then choose the formula. With one phase change, $2nt = m\lambda$ is **dark**, the opposite of the two-slit rule. A very thin film ($t \to 0$) looks dark in reflected light: this is why a soap film turns black just before it bursts.

Simulation

Colours of a thin film

Change the thickness and refractive index of the film, and choose a soap film in air or an anti-reflection coating on glass. The graph shows which wavelengths are reflected strongly; the colour swatch is what you see.



optical path difference 2nt	bright when	strongest visible reflection	note
798 nm	**2nt = (m + ½)λ**	**532 nm**	

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 24 to use the simulation.

Example 24.4.1

A soap film (n = 1.33) in air is lit normally with white light. What is the minimum thickness that strongly reflects green light of wavelength 532 nm?

Solution

One phase change (at the top surface). Bright: $2nt = (m + ½)\lambda$

Minimum thickness, $m = 0$: $t = \frac{\lambda}{4n} = \frac{532}{4(1.33)} =$ **100 nm**

Example 24.4.2

Glass lenses (n = 1.50) are coated with a thin layer of magnesium fluoride (n = 1.38) to reduce reflection of light of wavelength 550 nm. Find the minimum thickness of the coating.

Solution

Both reflections are from a denser medium (air → MgF_2, MgF_2 → glass): two phase changes, which cancel.

For no reflection, destructive: $2nt = (m + ½)\lambda$. Minimum, $m = 0$: $t = \frac{\lambda}{4n} = \frac{550}{4(1.38)} =$ **99.6 nm**

Example 24.4.3

A thin oil film (n = 1.45) floats on water (n = 1.33). Viewed from above at normal incidence, it strongly reflects light of 600 nm and no other visible wavelength. What is the minimum thickness of the film?

Solution

Top: air → oil (phase change). Bottom: oil → water, a less dense medium (no phase change). One phase change.

Bright: $2nt = (m + ½)\lambda \Rightarrow t_{min} = \frac{\lambda}{4n} = \frac{600}{4(1.45)} =$ **103 nm**

24.5 Diffraction by a single slit

Diffraction

is **the spreading of waves when they pass through an aperture or round the edge of an obstacle**. It is the result of interference between wavelets from different points on the same wavefront.

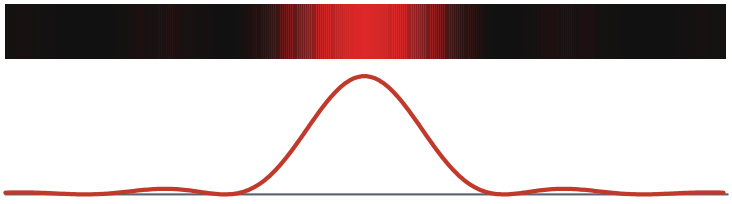


Figure 24.6: the single-slit pattern. The central maximum is twice as wide as the others and much brighter.

Explaining the pattern

- **Central maximum**: straight ahead (θ = 0), all the wavelets from the slit travel the same distance and arrive in phase, so they interfere constructively.
- **First minimum**: divide the slit into two halves. If the bottom edge is λ further from the screen than the top edge, each wavelet in the top half is λ/2 ahead of the matching wavelet in the bottom half. They cancel in pairs, so no light arrives.
- Second minimum: divide the slit into four parts; neighbouring parts cancel. In general minima occur at $a \sin\theta = m\lambda$, $m = \pm 1, \pm 2, \ldots$
- Between the minima there are weaker secondary maxima, at about $a \sin\theta \approx (m + \tfrac{1}{2})\lambda$.

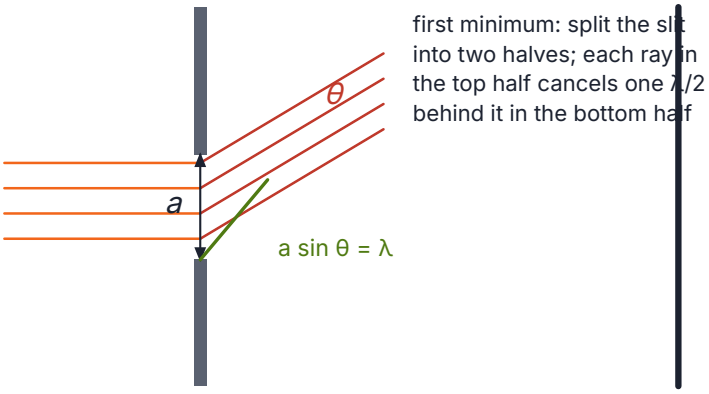


Figure 24.7: the first minimum of a single slit of width a.

$$a \sin\theta = \lambda \quad \text{(first minimum)} \qquad \sin\theta = \frac{\lambda}{a}$$

a = slit width. Position on a screen: $x = m\lambda D/a$ for the mth minimum; width of the central maximum = $2\lambda D/a$.

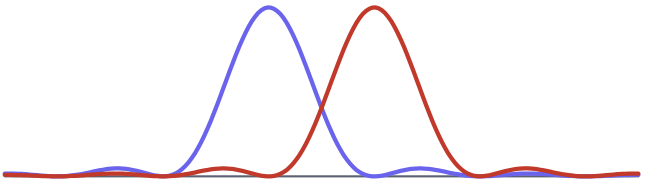
	Two-slit interference	Single-slit diffraction
formula	$a \sin\theta = m\lambda$ gives a **bright** fringe (a = slit separation)	$a \sin\theta = m\lambda$ gives a **dark** fringe (a = slit width)
brightness	the bright fringes are almost equally bright	the central maximum is much brighter than the others
width	all fringes have equal width	the central maximum is twice as wide as the others
cause	two coherent sources	wavelets from one slit cancel each other in pairs

Teacher's note

The same-looking equation $a \sin\theta = m\lambda$ means bright for two slits and a grating, but dark for a single slit. Always ask how many slits are interfering.

Resolving power

- Because of diffraction, the image of a point source formed through an aperture is not a point but a small spread-out pattern. Two close sources may overlap so much that they cannot be seen as separate.
- **Rayleigh's criterion**: two sources are just resolved when the central maximum of one diffraction pattern falls on the first minimum of the other.
- The smaller the angle θ between two sources that can still be resolved, the greater the resolving power.



st resolved: the central maximum of one falls on the first minimum of the

Figure 24.8: two sources just resolved.

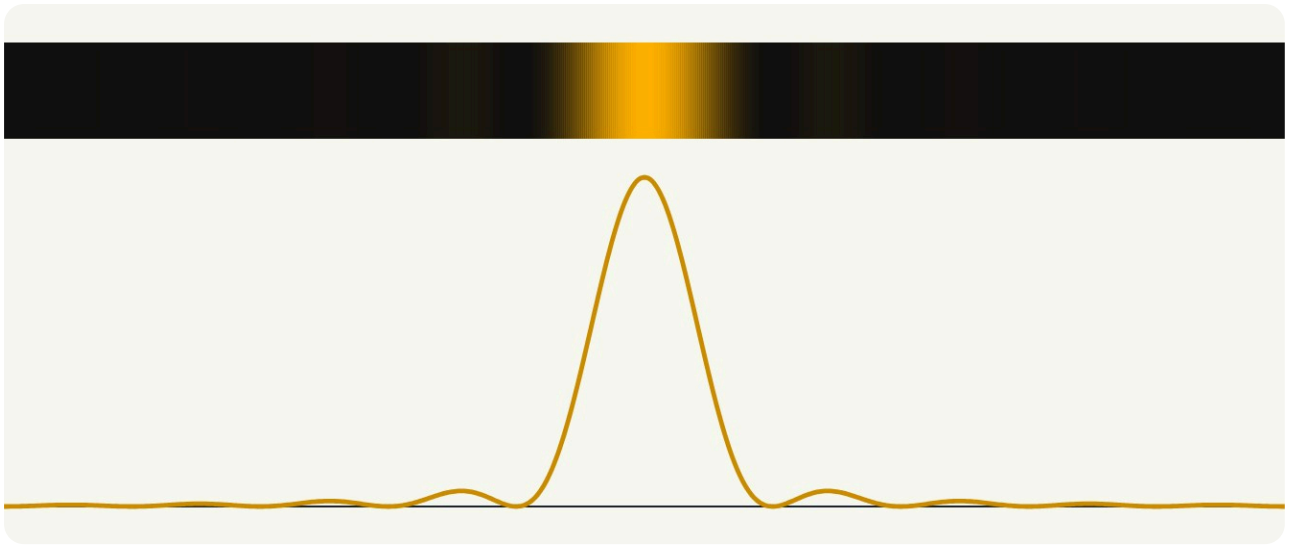
$$\theta = \frac{\lambda}{a} \text{ (slit)} \qquad \theta = \frac{1.22\lambda}{a} \text{ (circular aperture of diameter } a\text{)}$$

θ in radians = minimum angle of resolution. A larger aperture or a shorter wavelength gives a smaller θ, i.e. better resolution. Blue light resolves finer detail than red.

Simulation

Single-slit diffraction and resolution

Change the slit width and the wavelength and watch the central maximum. Then switch on a second source and move the two sources closer until they are only just resolved.



first minimum: sin θ = λ/a	central maximum width (D = 1 m)	minimum resolvable angle θ = λ/a
6.00 mrad	**12.0 mm**	**6.00 mrad**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 24 to use the simulation.

Example 24.5.1

Light of wavelength 550 nm falls on a single slit and produces its second minimum at 45.0°. Find (a) the slit width, (b) the angle of the first minimum.

Solution

(a) $a \sin\theta = 2\lambda \Rightarrow a = \dfrac{2(550 \times 10^{-9})}{\sin 45.0°} = \mathbf{1.56 \times 10^{-6}\ m}$

(b) $\sin\theta_1 = \lambda/a = 0.354 \Rightarrow \theta_1 = \mathbf{20.7°}$

Example 24.5.2

A single slit of width 0.140 mm is lit with monochromatic light. On a screen 2.00 m away, the second dark band is 16.0 mm from the centre of the central bright band. Find the wavelength.

Solution

$$x_2 = \frac{2\lambda D}{a} \Rightarrow \lambda = \frac{(16.0 \times 10^{-3})(0.140 \times 10^{-3})}{2(2.00)} = \mathbf{560\ nm}$$

Example 24.5.3

Light of wavelength 450 nm falls on a slit. On a screen 2.0 m away the central maximum is 6.0 cm wide. Find (a) the slit width, (b) the approximate distance from the centre to the third bright fringe.

Solution

(a) half-width x_1 = 3.0 cm = $\lambda D/a \Rightarrow a = \frac{(450 \times 10^{-9})(2.0)}{3.0 \times 10^{-2}} = \mathbf{3.0 \times 10^{-5}\ m}$

(b) $x \approx (3 + ½)\frac{\lambda D}{a}$ = 3.5 × 3.0 cm = **0.105 m**

Example 24.5.4

The pupil of the eye has a diameter of 3.0 mm. Find the minimum angle of resolution for light of 550 nm, and the smallest separation of two points 25 m away that can be seen as separate.

Solution

$$\theta = \frac{1.22\lambda}{a} = \frac{1.22(550 \times 10^{-9})}{3.0 \times 10^{-3}} = \mathbf{2.2 \times 10^{-4}\ rad}$$

separation = $r\theta$ = 25 × 2.2 × 10^{-4} = **5.6 mm**

24.6 Diffraction gratings

A **diffraction grating** has a very large number of equally spaced, parallel, narrow slits. A transmission grating can be made by ruling many fine lines on glass; a CD acts as a reflection grating.

- **Grating spacing** d = distance between the centres of adjacent slits = $1/N$, where N = number of lines per unit length. e.g. 5000 lines per cm: d = 1/5000 cm = 2.0×10^{-6} m.

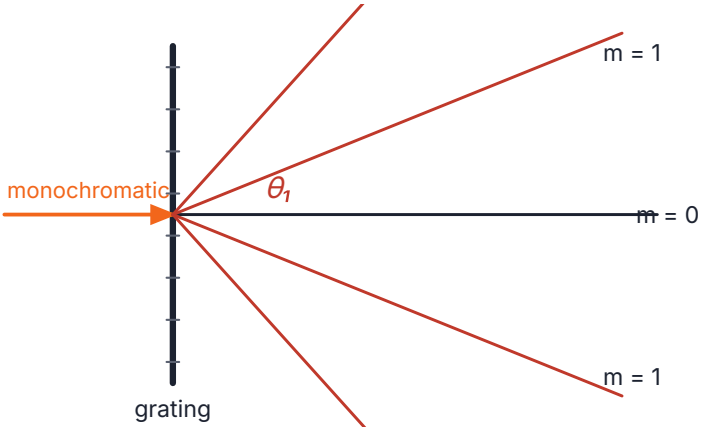


Figure 24.9: monochromatic light gives sharp bright lines (orders) at definite angles.

Formation of the pattern and the grating equation

- Monochromatic light falls normally on the grating. By Huygens's principle each slit acts as a source of secondary wavelets.
- In most directions the wavelets from the many slits cancel. Only in directions where the path difference between wavelets from **adjacent** slits is a whole number of wavelengths do they all interfere constructively.
- This gives very sharp, bright lines (principal maxima) with almost complete darkness between them, much sharper than with two slits.

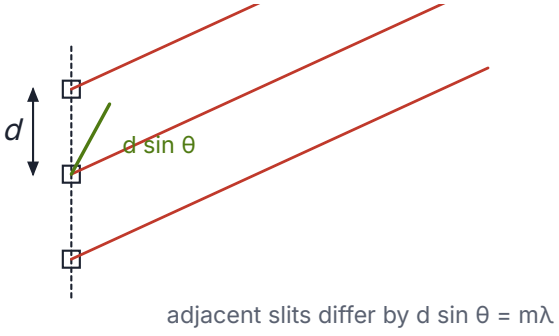


Figure 24.10: the path difference between adjacent slits.

$$d \sin\theta = m\lambda$$

m = 0, 1, 2, … (order); since sin θ ≤ 1, the highest order is $m_{max} \leq d/\lambda$ (take the whole number below)

Spectrum of white light

- The zero order (θ = 0) is white, because every wavelength has θ = 0 there.
- In every other order, each wavelength is diffracted at a different angle (sin θ = $m\lambda/d$), so the colours separate into a spectrum: violet nearest the centre, red furthest out.
- Higher orders are more spread out and may overlap (e.g. the red of the 2nd order with the violet of the 3rd).

Measuring wavelength with a grating

Set up a spectrometer so that a parallel beam of the monochromatic light falls normally on the grating.

Turn the telescope to find the first-order line on each side, and read the angles. θ = half the angle between them (this removes a small error in setting the grating).

Calculate λ = d sin θ. Repeat for the second order and take an average.

Simulation

Diffraction grating

Change the number of lines per mm and the wavelength, or choose white light. See the angles of each order and the highest order possible.



d = 1/N	highest order m ≤ d/λ	number of lines seen	order 1	order 2	order 3
3.333 µm	**6 (d/λ = 6.06)**	**13**	**9.5°**	**19.3°**	**29.7°**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 24 to use the simulation.

Example 24.6.1

A grating with 300 lines per mm is illuminated normally with monochromatic light. The second-order maximum is at 18.9°. Find the wavelength.

Solution

d = 1/300 mm = 3.33×10^{-6} m

$$\lambda = \frac{d \sin\theta}{m} = \frac{(3.33 \times 10^{-6}) \sin 18.9°}{2} = \mathbf{5.40 \times 10^{-7}\ m}$$

Example 24.6.2

Find the first- and second-order angles for light of 400 nm and 700 nm with a grating of 10 000 lines per cm.

Solution

$d = 1.0 \times 10^{-6}$ m

400 nm: $\theta_1 = \sin^{-1} 0.400 =$ **23.6°**; $\theta_2 = \sin^{-1} 0.800 =$ **53.1°**

700 nm: $\theta_1 = \sin^{-1} 0.700 =$ **44.4°**; second order: $\sin\theta = 1.4 > 1$, so there is **no** second order.

Example 24.6.3

The first-order line of 550 nm light is at 12° from a grating. How far apart are the slits?

Solution

$$d = \frac{\lambda}{\sin 12°} = \frac{550 \times 10^{-9}}{0.208} = \mathbf{2.64 \times 10^{-6}\ m}$$

Example 24.6.4

How many bright lines are produced when a grating of spacing 2.00×10^{-6} m is illuminated normally with light of 6.44×10^{-7} m?

Solution

$$m \le \frac{d}{\lambda} = 3.11 \Rightarrow m_{max} = 3$$

Lines: orders 0, ±1, ±2, ±3 = **7** bright lines

STPM past-year · Section C

Explain, with the aid of a diagram, the production of bright fringes in a diffraction grating by using Huygens's principle. [4]

Solution

- Every point on the wavefront at each slit acts as a source of secondary wavelets.
- The new wavefront is the line touching the wavelets; it bends round the edges of each slit (diffraction).
- The diffracted waves from all the slits overlap and interfere.
- Where the path difference between adjacent slits is $m\lambda$, the waves are in phase and give a bright fringe. (Draw at least three slits with circular wavelets, and new wavefronts in two different directions.)

24.7 Polarisation

- Light is a transverse electromagnetic wave: its electric field E vibrates perpendicular to the direction of travel.
- In **unpolarised** light (sunlight, a lamp) the electric field vibrates in all directions perpendicular to the direction of travel.
- In **plane-polarised** light the electric field vibrates in one plane only.
- Only transverse waves can be polarised. Polarisation is evidence that light is a transverse wave; sound (longitudinal) cannot be polarised.

Polarisation by a polariser

- A polariser (e.g. Polaroid sheet) has long-chain molecules aligned in one direction, giving it a **transmission axis**. Only the component of E parallel to the axis passes; the perpendicular component is absorbed.
- Unpolarised light through one polariser becomes plane-polarised, and its intensity is halved: $I = I_0/2$.

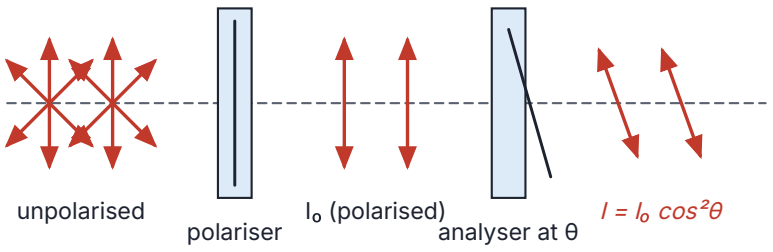


Figure 24.11: a polariser and an analyser.

Malus's law

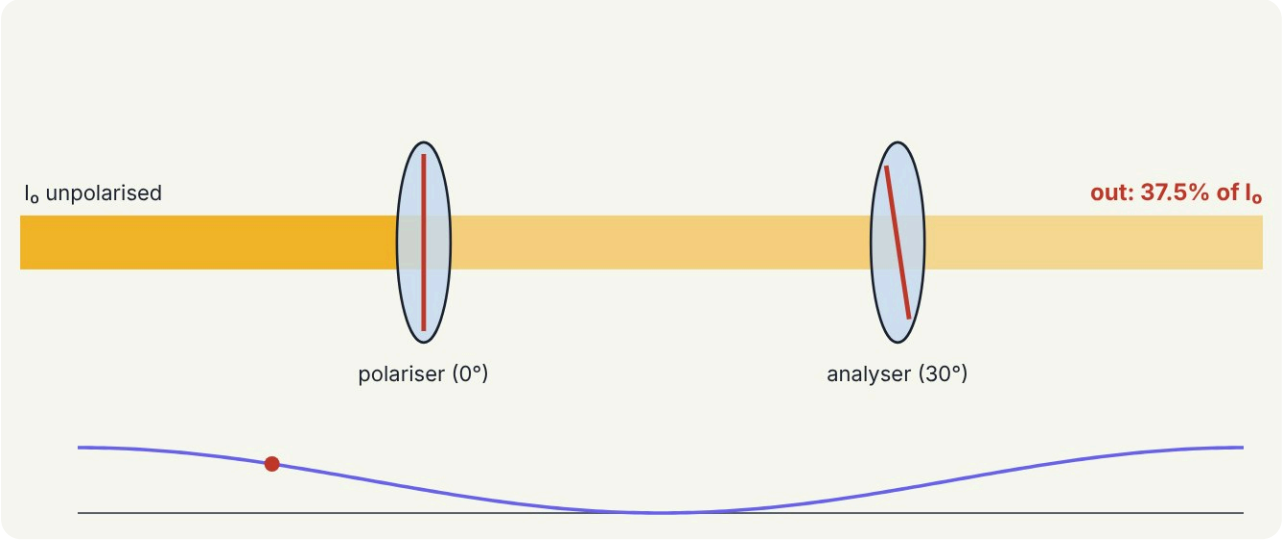
states that **when plane-polarised light of intensity I_0 passes through an analyser whose transmission axis is at angle θ to the plane of polarisation, the transmitted intensity is $I = I_0 \cos^2 \theta$.**

$$I = I_0 \cos^2 \theta$$

θ = 0°: all the light passes; θ = 90° (crossed): no light passes. The amplitude transmitted is $E_0 \cos \theta$, and $I \propto E^2$.

Simulation Polariser and analyser

Turn the analyser and watch the transmitted intensity follow $I = I_o \cos^2\theta$. Add a third polariser between two crossed ones: light gets through again.



after the first polariser	$I = I_o \cos^2\theta$	transmitted
½ I_o	**0.750 × (½ I_o)**	**0.375 I_o**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 24 to use the simulation.

Example 24.7.1

Unpolarised light of intensity I_0 passes through two polarisers whose axes are at 30° to each other. Find the transmitted intensity.

Solution

After the first polariser: $I_1 = I_0/2$

After the second: $I = I_1 \cos^2 30° = (I_0/2)(0.75) = \mathbf{0.375 I_0}$

Example 24.7.2

Two polarisers are crossed (90°). A third polariser is placed between them with its axis at 45° to the first. Unpolarised light of intensity I_0 enters. Find the intensity that emerges.

Solution

$I_1 = I_0/2; \quad I_2 = I_1 \cos^2 45° = I_0/4; \quad I_3 = I_2 \cos^2 45° = \mathbf{I_0/8}$

Without the middle polariser no light would pass.

Polarisation by reflection: Brewster's law

- Unpolarised light reflected from glass or water is partially polarised.
- At one angle of incidence, the **Brewster angle** θ_B, the reflected light is completely plane-polarised (vibrating parallel to the surface). At this angle the reflected and refracted rays are perpendicular.

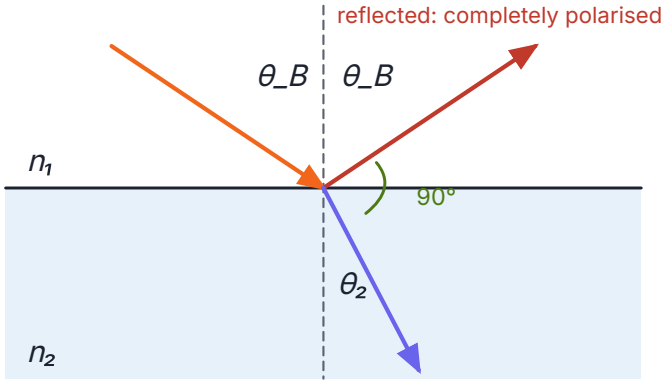


Figure 24.12: at the Brewster angle the reflected and refracted rays are at 90°.

Reflected ⊥ refracted: $\theta_B + 90° + \theta_2 = 180° \Rightarrow \theta_2 = 90° - \theta_B$

Snell's law: $n_1 \sin\theta_B = n_2 \sin(90° - \theta_B) = n_2 \cos\theta_B$

$$\tan\theta_B = \frac{n_2}{n_1}$$

Brewster's law

Example 24.7.3

Unpolarised light in air strikes glass (n = 1.50). Find the Brewster angle and the angle of refraction.

Solution

$\tan\theta_B = 1.50 \Rightarrow \theta_B =$ **56.3°**

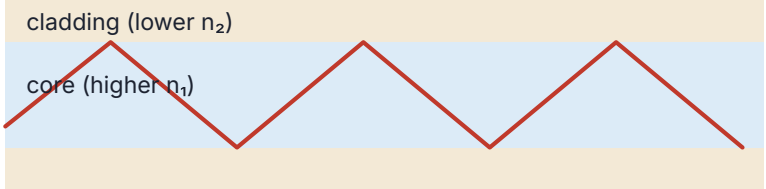
$\theta_2 = 90° - 56.3° =$ **33.7°** (check: sin 56.3° = 1.50 sin 33.7° ✓)

Uses of polarisation

- Polaroid sunglasses cut glare: light reflected from water and roads is mostly horizontally polarised, and the lenses block it.
- LCD screens use two polarisers with liquid crystal between them to control each pixel.
- Photography filters reduce reflections and darken blue sky; stress analysis shows strain patterns in transparent models; 3-D cinema glasses send a different image to each eye.

24.8 Optical waveguides

- **Total internal reflection**: when light in a denser medium hits the boundary with a less dense medium at an angle greater than the critical angle c ($\sin c = n_2/n_1$), all the light is reflected.
- An optical fibre has a glass **core** of higher refractive index n_1 surrounded by a **cladding** of lower index n_2.
- Light entering within the acceptance angle strikes the core–cladding boundary at more than the critical angle and is totally internally reflected again and again. It is guided along the core with very little loss, even round gentle bends.
- The cladding protects the core surface and stops light leaking into a neighbouring fibre.



t hits the core–cladding boundary above the critical angle: total internal reflec

Figure 24.13: light guided along an optical fibre by repeated total internal reflection.

Application	How fibres are used
Telecommunications	telephone, television and internet signals travel as light pulses: high bandwidth, low loss, no electrical interference
Medicine	endoscopes let doctors see inside the body; fibres also carry laser light for surgery
Sensing	changes in the light show changes in temperature, pressure or strain
Others	decorative lighting, inspection cameras in hard-to-reach places

Example 24.8.1

An optical fibre has a core of refractive index 1.50 and cladding of refractive index 1.45. Find the critical angle at the core–cladding boundary.

Solution

$$\sin c = \frac{1.45}{1.50} = 0.967 \Rightarrow c = \mathbf{75.2°}$$

Light must hit the boundary at more than 75.2° to the normal to be guided.

Summary

Situation	Formula	$m\lambda$ means	$(m + ½)\lambda$ means
Two slits (Young)	$a \sin\theta = m\lambda$, $\Delta x = \lambda D/a$	bright	dark
Diffraction grating	$d \sin\theta = m\lambda$	bright line	—
Single slit	$a \sin\theta = m\lambda$	**dark**	bright (approx.)
Thin film, one phase change	$2nt$	**dark**	bright
Thin film, two phase changes	$2nt$	bright	dark
Polarisation	$I = I_0 \cos^2\theta$; $\tan\theta_B = n_2/n_1$	—	—

Exercises

1. State two differences between a two-slit interference pattern and a single-slit diffraction pattern.

   Answer

   Two-slit: bright fringes of almost equal intensity and equal width. Single slit: the central maximum is much brighter than the others and twice as wide.

2. In a Young's experiment with light of 589 nm, the slits are 0.20 mm apart and the screen is 1.5 m away. Find the fringe separation. What happens to it if the apparatus is put in water (n = 1.33)?

   Answer

   Δx = 4.4 mm; in water $\lambda \to \lambda/1.33$, so Δx = 3.3 mm

3. A grating has 600 lines per mm. Find the highest order for light of 500 nm.

   Answer

   d/λ = 3.33 ⇒ third order

4. Unpolarised light passes through two polarisers. The transmitted intensity is 1/8 of the incident intensity. Find the angle between the axes.

   Answer

   $\cos^2\theta = 1/4 \Rightarrow \theta = 60°$

5. Find the Brewster angle for light in air reflected from water (n = 1.33).

   Answer

   53.1°

Chapter 25 · Semester 3

QUANTUM PHYSICS

Notes by Cikgu Suhaili

20 worked examples 6 simulations Exercises with answers

25.1 Photons

Planck's quantum theory

- A **black body** is an ideal object that absorbs all the electromagnetic radiation falling on it, and is also the best possible emitter.
- The spectrum of a hot body shows that (a) the total energy emitted (area under the curve) increases with temperature, and (b) the peak wavelength becomes shorter as the temperature increases.
- Classical physics assumed energy is continuous and predicted that the intensity rises without limit at short wavelengths. This disagreed with experiment.

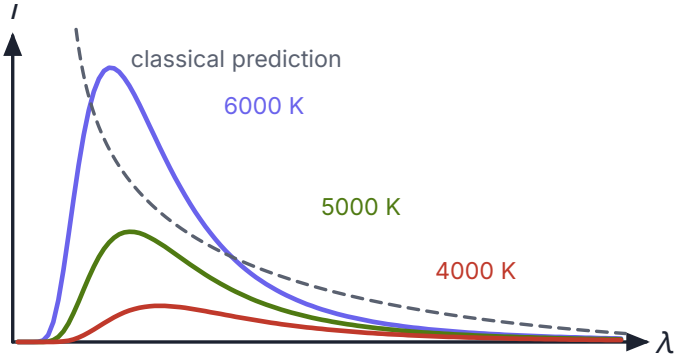


Figure 25.1: black-body spectra and the failed classical prediction.

Planck's quantum theory

electromagnetic radiation is emitted or absorbed in discrete packets of energy called quanta. The energy of one quantum is proportional to the frequency of the radiation: $E = hf$.

- An oscillator can only have energies 0, hf, $2hf$, $3hf$, ...: the energy is **quantised**.
- Einstein (1905) proposed that light itself travels as quanta called **photons**. A photon has no mass, travels at c in a vacuum and carries energy hf.
- **Electronvolt**: the kinetic energy gained by an electron accelerated through a potential difference of 1 V. 1 eV = 1.60×10^{-19} J.

$$E = hf = \frac{hc}{\lambda}$$

$h = 6.63 \times 10^{-34}$ J s (Planck constant)

Classical electromagnetic wave	Photon (quantum theory)
energy depends on the intensity ($\propto$ amplitude^2)	energy of one photon depends on frequency: $E = hf$
energy is continuous and spread through the wave	energy comes in discrete packets

Example 25.1.1

Find the energy of a photon of blue light of wavelength 450 nm, in joules and electronvolts.

Solution

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{450 \times 10^{-9}} = \mathbf{4.42 \times 10^{-19}\ J} = 4.42 \times 10^{-19}/1.60 \times 10^{-19} = \mathbf{2.76\ eV}$$

The photoelectric effect

Photoelectric effect

is **the emission of electrons from a metal surface when electromagnetic radiation of high enough frequency falls on it**. The emitted electrons are called **photoelectrons**.

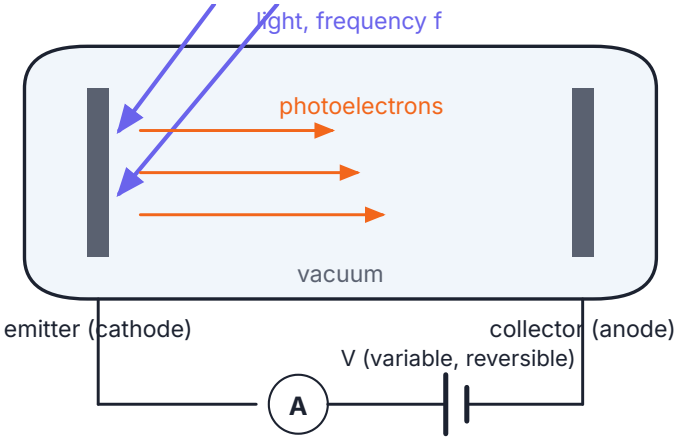


Figure 25.2: apparatus for the photoelectric effect.

- Monochromatic light falls on the emitter (cathode) in an evacuated tube. Photoelectrons travel to the collector (anode) and a photocurrent I flows.
- As the positive voltage increases, more photoelectrons reach the collector, until all of them do: the current reaches a maximum, the **saturation current**.

- Even at $V = 0$ a small current flows, because some photoelectrons have enough kinetic energy to reach the collector.
- If the supply is reversed, the collector repels the electrons. The current falls to zero at the **stopping potential** V_s.

> **Stopping potential, V_s**
>
> is **the minimum reverse potential difference needed to stop the most energetic photoelectrons from reaching the collector**.

$$eV_s = \frac{1}{2}mv_{max}^2$$

the work done against the field stops the fastest electrons

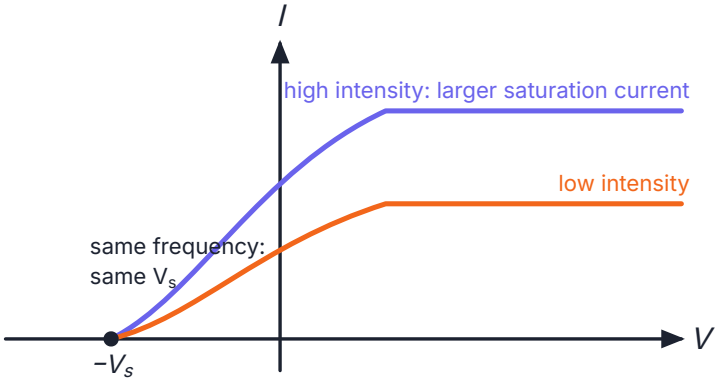


Figure 25.3: current against voltage for two intensities of the same frequency.

Important observations

- No electrons are emitted if the frequency is below a **threshold frequency** f_0, however intense the light.
- Above f_0, emission is **instantaneous**, even for very weak light.
- The maximum kinetic energy of the photoelectrons (and so V_s) increases with frequency, but does **not** depend on intensity.
- The photocurrent (number of electrons per second) is proportional to the intensity.

Wave theory predicts	Experiment (explained by photons)
any frequency can eject electrons if the light is intense enough	no emission below the threshold frequency, however intense the light
the maximum kinetic energy increases with intensity	the maximum kinetic energy depends only on frequency
with weak light there should be a delay while energy builds up	emission is instantaneous

Einstein's photoelectric equation

- An electron absorbs the energy of **one** photon, all at once.
- Some energy, at least the **work function** W, is needed to free the electron from the metal. The rest becomes kinetic energy.
- **Work function** W: the minimum energy needed to remove an electron from the surface of the metal. **Threshold frequency** f_0: the minimum frequency that can eject an electron, $W = hf_0$. Threshold wavelength $\lambda_0 = c/f_0$ is the maximum wavelength.
- More intense light means more photons per second, so more electrons per second (more current), but each photon still has the same energy.

$$hf = W + \frac{1}{2}mv_{max}^2 = W + eV_s$$

$$W = hf_0$$

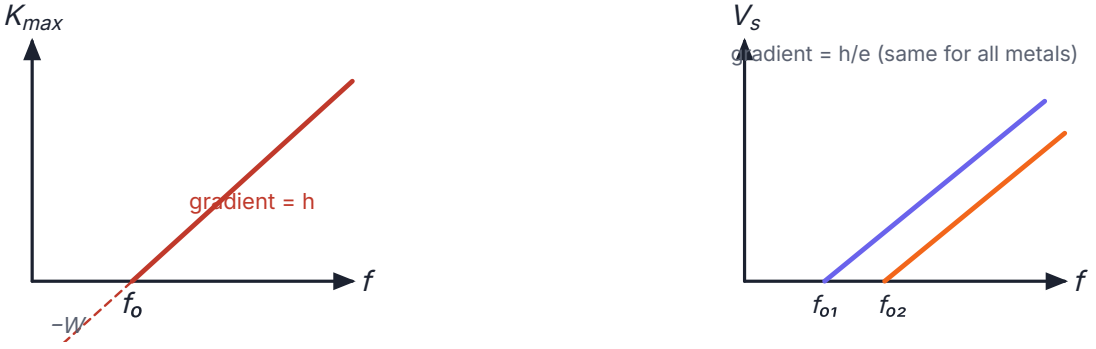
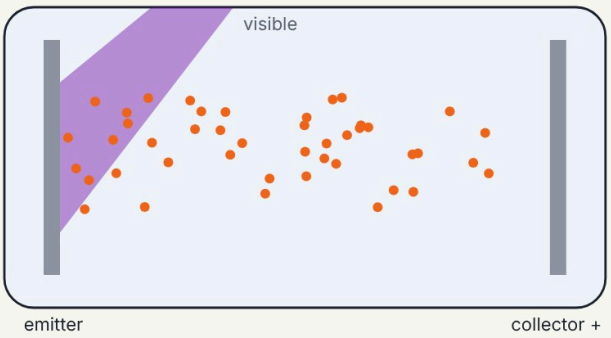


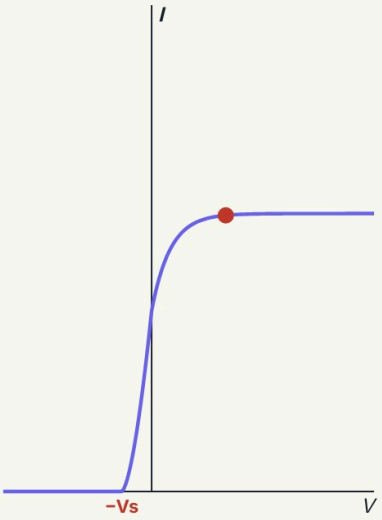
Figure 25.4: K_max against f is a straight line of gradient h; V_s against f has gradient h/e for every metal, with different threshold frequencies.

Simulation

The photoelectric effect

Choose a metal and change the wavelength, intensity and voltage. Below the threshold frequency no electrons come out, however bright the light. Find the stopping potential by making the voltage negative until the current is zero.





photon energy hf	work function W	threshold wavelength λ_o	K_max = hf − W
3.11 eV	**2.30 eV**	**540 nm**	**0.81 eV**

stopping potential V_s	photocurrent
0.81 V	**60 (relative)**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 25 to use the simulation.

Example 25.1.2

The work function of silver is 4.74 eV. Find the minimum frequency of light that can eject electrons from silver.

Solution

$$f_0 = \frac{W}{h} = \frac{4.74 \times 1.60 \times 10^{-19}}{6.63 \times 10^{-34}} = \mathbf{1.14 \times 10^{15}\ Hz}$$

Example 25.1.3

Find the maximum kinetic energy of electrons ejected from calcium (W = 2.71 eV) by violet light of 420 nm.

Solution

$E = hc/\lambda = 4.74 \times 10^{-19}$ J = 2.96 eV

$K_{max} = 2.96 - 2.71 =$ **0.25 eV**

Example 25.1.4

Sodium has a work function of 2.30 eV. Find (a) its threshold frequency, (b) the maximum speed of photoelectrons for light of 500 nm, (c) the stopping potential.

Solution

(a) $f_0 = \dfrac{2.30 \times 1.60 \times 10^{-19}}{6.63 \times 10^{-34}} =$ **5.55×10^{14} Hz**

(b) $hc/\lambda = 3.98 \times 10^{-19}$ J; $K_{max} = 3.98 \times 10^{-19} - 3.68 \times 10^{-19} = 2.98 \times 10^{-20}$ J; $v = \sqrt{2K/m} =$ **2.56×10^5 m s^{-1}**

(c) $V_s = K_{max}/e =$ **0.186 V**

Example 25.1.5

A graph of K_{max} against frequency for a metal cuts the frequency axis at 4.8×10^{14} Hz. Find (a) the work function, (b) the threshold wavelength.

Solution

(a) $W = hf_0 = (6.63 \times 10^{-34})(4.8 \times 10^{14}) =$ **3.18×10^{-19} J**

(b) $\lambda_0 = c/f_0 =$ **6.25×10^{-7} m**

Photovoltaic effect in a solar cell

- The **photovoltaic effect** is the production of an e.m.f. across a p–n junction of a semiconductor when it absorbs light.
- A photon with energy at least equal to the band gap is absorbed near the junction and creates an electron–hole pair.
- The electric field at the junction separates the electrons (to the n side) and holes (to the p side) before they recombine, so an e.m.f. appears across the cell and drives a current in an external circuit.
- Unlike the photoelectric effect, no electron leaves the material: the charge carriers are collected inside the cell. Solar cells convert light energy directly into electrical energy with no moving parts; many cells form a solar panel.

25.2 Wave–particle duality

Light behaves as a wave in interference and diffraction, and as a particle (photons) in the photoelectric effect. De Broglie suggested that particles might also behave as waves.

de Broglie's hypothesis

a moving particle has a wave associated with it, whose wavelength is $\lambda = h/p$, where $p = mv$ is the momentum of the particle.

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

for a photon, $p = E/c = h/\lambda$ as well

An electron accelerated from rest through V: $eV = \tfrac{1}{2}mv^2 = \dfrac{p^2}{2m} \Rightarrow p = \sqrt{2meV}$

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{h}{\sqrt{2mK}}$$

K = kinetic energy

Electron diffraction

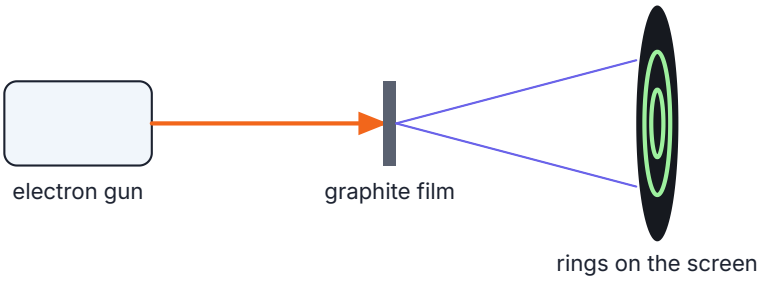


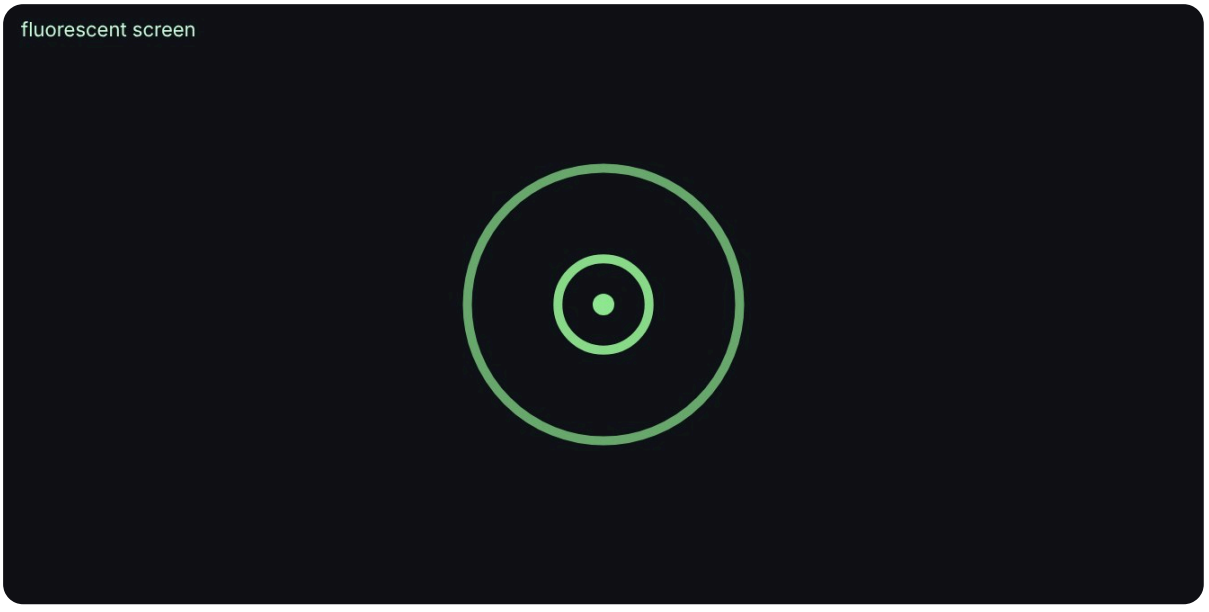
Figure 25.5: electrons passing through a thin graphite film form diffraction rings.

- Electrons are accelerated through a few kilovolts and pass through a thin graphite film.
- The rows of carbon atoms act like a diffraction grating, because their spacing is similar to the electron wavelength.
- Bright rings appear on the fluorescent screen: a **diffraction pattern**. Diffraction is a wave property, so this is evidence that electrons behave as waves.
- When the accelerating voltage increases, the rings become smaller: the electrons are faster, so $\lambda = h/mv$ is shorter, as de Broglie predicted. (Davisson and Germer first showed electron diffraction in 1927.)

Simulation

Electron diffraction

Change the accelerating voltage. A higher voltage gives faster electrons, a shorter de Broglie wavelength and smaller diffraction rings.



electron speed v = √(2eV/m)
3.75 × 10⁷ m/s

de Broglie λ = h/√(2meV)
19.4 pm

ring size
larger V → shorter λ → smaller rings

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 25 to use the simulation.

The electron microscope

- The resolving power of a microscope is limited by diffraction: the minimum resolvable detail is about the wavelength used.
- Electrons accelerated through tens of kilovolts have wavelengths of about 10^{-11} m, thousands of times shorter than visible light (about 5×10^{-7} m).
- So an electron microscope has a much **higher resolving power** than an optical microscope and can show much finer detail.

Example 25.2.1

Find the momentum and energy of a photon of wavelength 5.0×10^{-7} m.

Solution

$p = h/\lambda = 6.63 \times 10^{-34}/5.0 \times 10^{-7} =$ **1.33×10^{-27} kg m s^{-1}**

$E = hc/\lambda =$ **3.98×10^{-19} J**

Example 25.2.2

An electron and a photon both have a wavelength of 0.25 nm. Find the momentum and energy (in eV) of each.

Solution

Both: $p = h/\lambda = 6.63 \times 10^{-34}/0.25 \times 10^{-9} =$ **2.65×10^{-24} kg m s^{-1}**

Electron: $K = \dfrac{p^2}{2m} = \dfrac{(2.65 \times 10^{-24})^2}{2(9.11 \times 10^{-31})} = 3.86 \times 10^{-18}$ J = **24.1 eV**

Photon: $E = pc = hc/\lambda = 7.96 \times 10^{-16}$ J = **4.97×10^{3} eV**

The same wavelength needs far less energy for an electron than for a photon.

Example 25.2.3

An electron is accelerated from rest through 1200 V. Find its de Broglie wavelength.

Solution

$p = \sqrt{2meV} = \sqrt{2(9.11 \times 10^{-31})(1.60 \times 10^{-19})(1200)} = 1.87 \times 10^{-23}$ kg m s^{-1}

$\lambda = h/p =$ **3.55×10^{-11} m**

Example 25.2.4

An electron and a proton have the same kinetic energy. Find the ratio of the de Broglie wavelength of the electron to that of the proton. ($m_p = 1.67 \times 10^{-27}$ kg)

Solution

$\lambda = h/\sqrt{2mK} \Rightarrow \frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}} = \sqrt{\frac{1.67 \times 10^{-27}}{9.11 \times 10^{-31}}} =$ **42.8**

25.3 Atomic structure

Bohr's postulates for the hydrogen atom

The electron moves in certain circular orbits round the nucleus **without radiating energy** (stationary states).

In these orbits the angular momentum of the electron is a whole number times $h/2\pi$: $mvr = \frac{nh}{2\pi}$, $n = 1, 2, 3, \ldots$ (principal quantum number).

Radiation is emitted or absorbed only when the electron moves from one orbit to another. The photon energy is the difference in energy: $hf = E_i - E_f$.

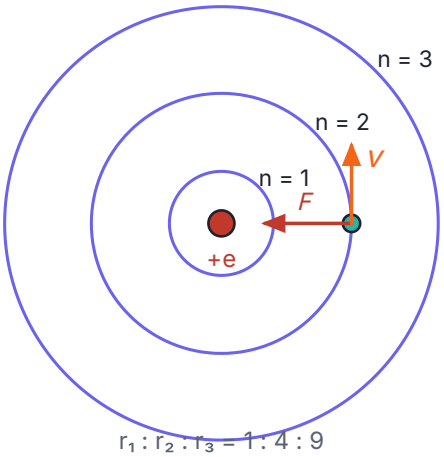


Figure 25.6: the Coulomb force provides the centripetal force on the orbiting electron.

Radius of the orbits

Coulomb force = centripetal force: $\frac{e^2}{4\pi\varepsilon_0 r^2} = \frac{mv^2}{r}$ ... (1)

From postulate 2: $v = \frac{nh}{2\pi mr}$ ... (2)

Substitute (2) in (1): $\frac{e^2}{4\pi\varepsilon_0 r} = m\frac{n^2h^2}{4\pi^2m^2r^2} \Rightarrow r = \frac{n^2h^2\varepsilon_0}{\pi me^2}$

$$r_n = \frac{n^2h^2\varepsilon_0}{\pi me^2} = n^2a_0$$

$a_0 = r_1 = 5.29 \times 10^{-11}$ m (the Bohr radius). The radii are in the ratio 1 : 4 : 9 : ...

Energy levels

From (1): kinetic energy $K = \frac{1}{2}mv^2 = \frac{e^2}{8\pi\varepsilon_0 r}$

Electric potential energy $U = -\frac{e^2}{4\pi\varepsilon_0 r}$

Total energy $E = K + U = -\frac{e^2}{8\pi\varepsilon_0 r}$; substitute r_n:

$$E_n = -\frac{e^4 m}{8\varepsilon_0^2 h^2 n^2} = -\frac{13.6}{n^2}\ \text{eV}$$

the negative sign means the electron is bound: energy must be supplied to remove it to infinity, where $E = 0$. $E_1 = -13.6$ eV $= -2.18 \times 10^{-18}$ J.

- **Ground state** ($n = 1$): the lowest energy level. **Excited states**: $n = 2, 3, 4, \ldots$
- **Excitation energy**: the energy needed to move an electron from the ground state to a higher level, e.g. $E_2 - E_1 = 10.2$ eV.
- **Ionisation energy**: the minimum energy needed to remove an electron from the ground state completely (to $n = \infty$): 13.6 eV for hydrogen.

Emission and absorption line spectra

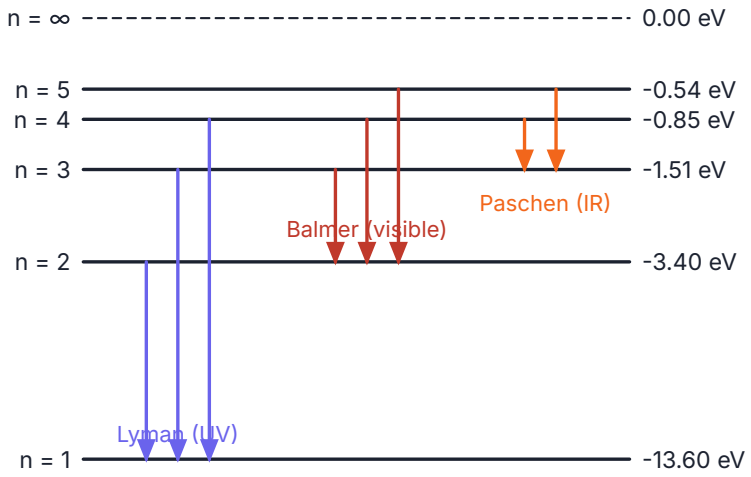


Figure 25.7: energy levels of hydrogen and three series of emission lines.

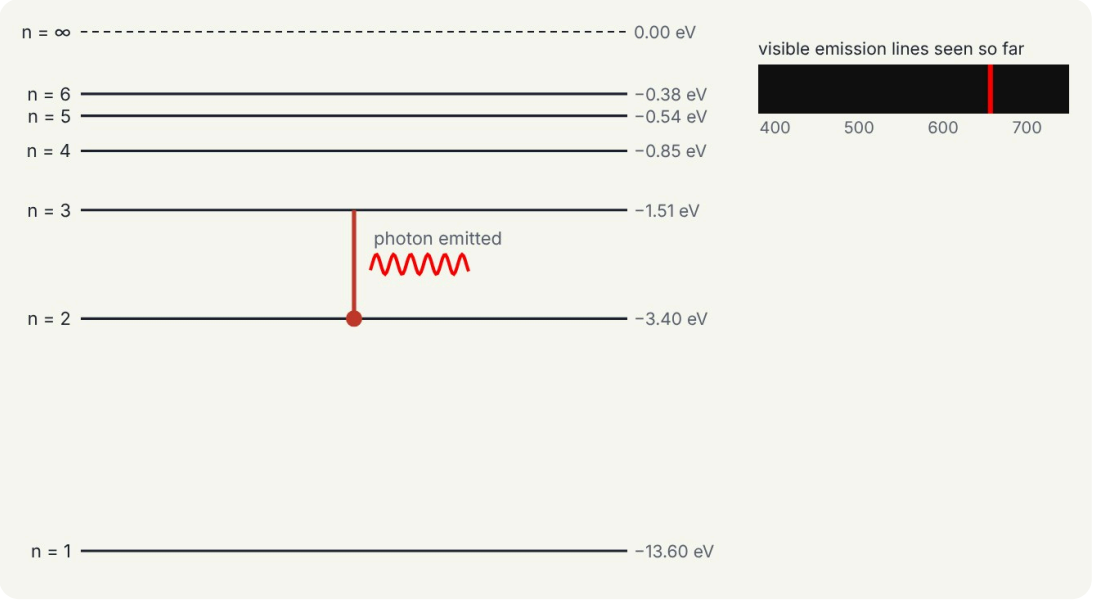
- **Emission**: atoms of a low-pressure gas are excited (by heating or by an electric discharge). When an electron falls from a higher level to a lower level, it emits a photon of energy $hf = E_i - E_f$. Only certain energy differences exist, so only certain wavelengths are emitted: **bright lines on a dark background**.
- **Absorption**: white light passes through a cool gas. Photons whose energy exactly equals an energy difference are absorbed, raising electrons to higher levels: **dark lines on a continuous spectrum**, at the same wavelengths as the emission lines.
- Lyman series: transitions ending at $n = 1$ (ultraviolet). Balmer series: ending at $n = 2$ (visible). Paschen: ending at $n = 3$ (infrared). Brackett ($n = 4$) and Pfund ($n = 5$): infrared.

$$hf = \frac{hc}{\lambda} = E_i - E_f$$

Simulation

Energy levels and line spectra of hydrogen

Pick the starting and final levels. A fall emits a photon; a jump up absorbs one. The wavelength and series are shown, and visible lines appear in the spectrum.



ΔE = E_i − E_f	photon wavelength	process	series
1.89 eV	**656 nm**	**emission**	**Balmer (visible/near UV)**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 25 to use the simulation.

Example 25.3.1

(a) Find the radius of the n = 4 orbit of hydrogen. (b) An excited hydrogen atom has a radius of 0.847 nm. Which state is this?

Solution

(a) $r_4 = 4^2(5.29 \times 10^{-11}) = \mathbf{8.46 \times 10^{-10}\ m}$

(b) $n^2 = 0.847 \times 10^{-9}/5.29 \times 10^{-11} = 16 \Rightarrow \mathbf{n = 4}$

Example 25.3.2

An electron in hydrogen falls from $E_2 = -3.40$ eV to the ground state, $E_1 = -13.6$ eV. Find the energy and wavelength of the emitted photon.

Solution

$\Delta E = -3.40 - (-13.6) = 10.2$ eV = $\mathbf{1.63 \times 10^{-18}\ J}$

$\lambda = hc/\Delta E = \mathbf{1.22 \times 10^{-7}\ m}$ (ultraviolet, Lyman series)

Example 25.3.3

Find the frequency of the photon needed to ionise a hydrogen atom in the ground state.

Solution

$E = 0 - (-13.6\text{ eV}) = 2.18 \times 10^{-18}$ J; $f = E/h = \mathbf{3.29 \times 10^{15}\ Hz}$

Example 25.3.4

An electron in a hydrogen atom moves from the ground state to the 4th excited state. Find the energy absorbed.

Solution

4th excited state: $n = 5$

$\Delta E = -\frac{13.6}{25} - (-13.6) = \mathbf{13.1\ eV} = 2.09 \times 10^{-18}$ J

Example 25.3.5

Find the shortest and longest wavelengths in the Balmer series.

Solution

Shortest: $\infty \to 2$, $\Delta E = 3.40$ eV $\Rightarrow \lambda = hc/\Delta E =$ **366 nm**

Longest: $3 \to 2$, $\Delta E = 3.40 - 1.51 = 1.89$ eV $\Rightarrow \lambda =$ **658 nm** (red)

Example 25.3.6

Hydrogen emits 121.5 nm and 102.4 nm when electrons fall from the 1st and 2nd excited states to the ground state. Find (a) the photon energies, (b) the wavelength emitted when an electron falls from the 2nd excited state to the 1st excited state.

Solution

(a) $E = hc/\lambda$: **1.64×10^{-18} J** and **1.94×10^{-18} J**

(b) $\Delta E = 1.942 \times 10^{-18} - 1.637 \times 10^{-18} = 3.05 \times 10^{-19}$ J $\Rightarrow \lambda =$ **652 nm** (or directly $1/\lambda = 1/102.4 - 1/121.5$)

Example 25.3.7

A hydrogen atom in the $n = 3$ state falls to the ground state and emits one photon. Find (a) its wavelength, (b) the type of radiation, (c) its momentum.

Solution

(a) $\Delta E = 13.6(1 - 1/9) = 12.1$ eV $= 1.94 \times 10^{-18}$ J $\Rightarrow \lambda =$ **1.03×10^{-7} m**

(b) **ultraviolet**

(c) $p = h/\lambda =$ **6.44×10^{-27} kg m s^{-1}**

25.4 X-rays

X-rays are electromagnetic waves of very short wavelength (about 10^{-12} m to 10^{-8} m), discovered by Röntgen in 1895. They travel in straight lines, penetrate matter, are not deflected by electric or magnetic fields, ionise gases, affect photographic film, cause fluorescence and can be diffracted.

Production of X-rays in an X-ray tube

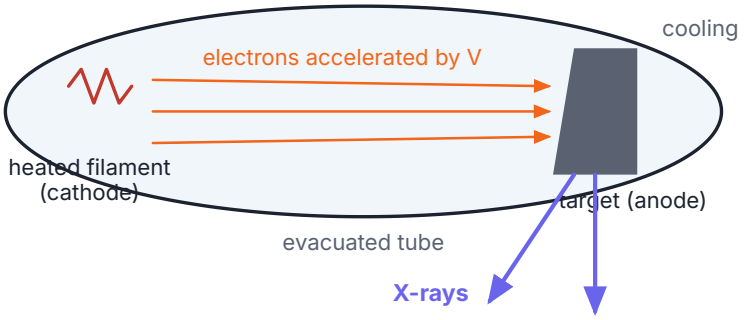


Figure 25.8: an X-ray tube.

- A heated filament (cathode) emits electrons by **thermionic emission**.
- A high voltage (tens of kV) accelerates them across an **evacuated** tube, so they do not collide with gas molecules.
- They strike a **target** of a heavy metal with a high melting point (tungsten or molybdenum). Less than 1% of their kinetic energy becomes X-rays; the rest becomes heat, so the target is cooled.
- The tube current (filament current) controls the **number** of electrons, so the **intensity** of the X-rays. The accelerating voltage controls the electron energy, so the **penetrating power** (hardness).

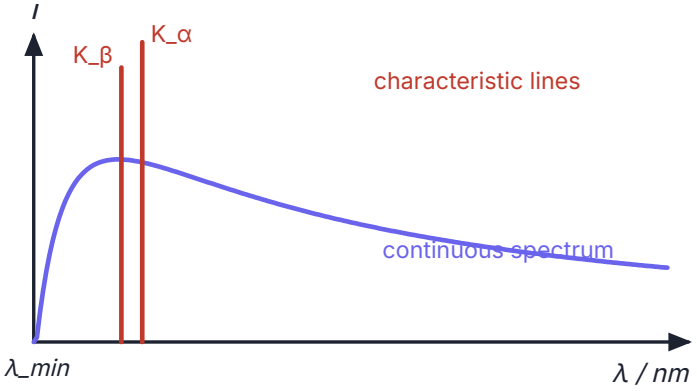


Figure 25.9: an X-ray spectrum: a continuous spectrum with a sharp cut-off at λ_min, and characteristic lines.

Continuous spectrum (braking radiation, bremsstrahlung)

- Electrons are decelerated by the nuclei of the target atoms; the kinetic energy lost is emitted as an X-ray photon.
- An electron can lose any amount of energy in a collision, so all wavelengths above a minimum are produced.
- The shortest wavelength comes from an electron that loses **all** its kinetic energy in one collision.

Characteristic line spectrum

- A fast electron knocks out an electron from an inner shell (e.g. K shell) of a target atom, leaving a vacancy.
- An electron from a higher shell falls into the vacancy and emits an X-ray photon of energy equal to the difference of the two energy levels.
- The energies are fixed for each element, so the lines are characteristic of the target metal. Kα: L → K; Kβ: M → K; Lα: M → L, ...

Minimum wavelength

Kinetic energy of the electron = eV. If all of it becomes one photon: $eV = hf_{max} = \frac{hc}{\lambda_{min}}$

$$\lambda_{min} = \frac{hc}{eV}$$

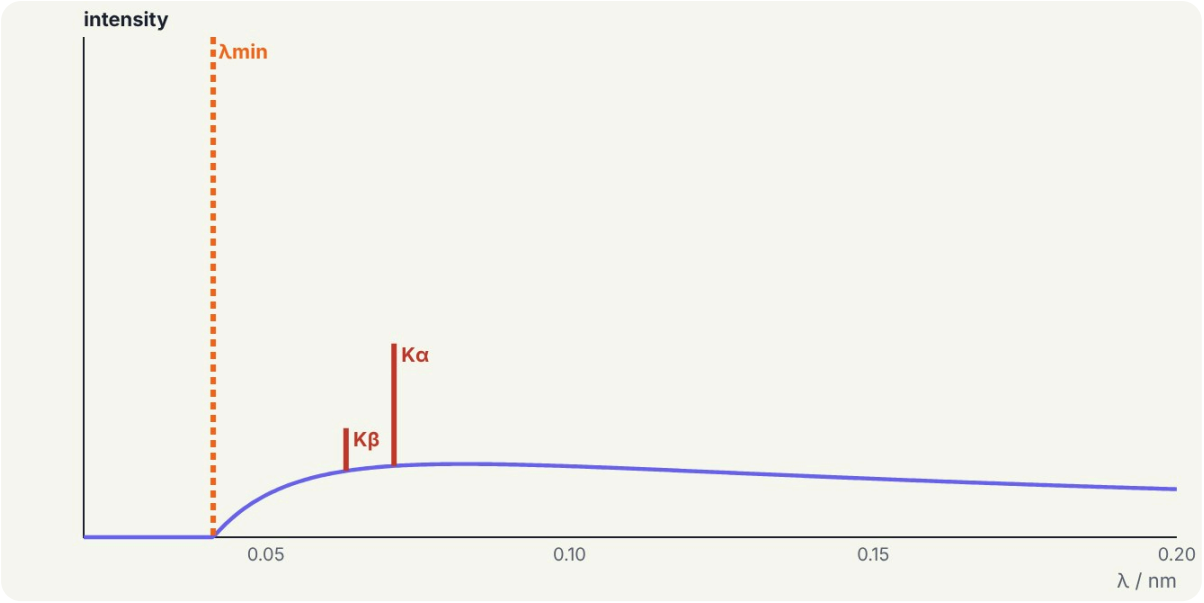
depends only on the accelerating voltage, not on the target metal

Change	Continuous spectrum	Characteristic lines
tube current increased	intensity increases; λ_{min} unchanged	more intense; same wavelengths
accelerating voltage increased	intensity increases; λ_{min} decreases	same wavelengths (may appear once eV is large enough)
target of higher atomic number	λ_{min} unchanged (more intense)	lines move to shorter wavelengths

Simulation

X-ray spectrum

Change the accelerating voltage, the tube current and the target. Watch the cut-off wavelength λmin = hc/eV move, and see the characteristic lines appear once the electrons have enough energy to knock out K-shell electrons.



λmin = hc/eV	max photon energy	characteristic K lines
41.3 pm	**30.0 keV**	**present (eV > K-shell binding energy 20 keV)**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 25 to use the simulation.

Example 25.4.1

Find the shortest wavelength of X-rays from a tube operated at 50 kV.

Solution

$$\lambda_{min} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(1.60 \times 10^{-19})(50 \times 10^{3})} = \mathbf{2.49 \times 10^{-11}\ m}$$

Example 25.4.2

The shortest wavelength of X-rays from a tube is 0.019 nm. What is the voltage across the tube?

Solution

$$V = \frac{hc}{e\lambda_{min}} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(1.60 \times 10^{-19})(0.019 \times 10^{-9})} = \mathbf{65.4\ kV}$$

X-ray diffraction and Bragg's law

- The regularly spaced atoms of a crystal act like a diffraction grating for X-rays, because the spacing of the atomic planes (about 10^{-10} m) is similar to the X-ray wavelength.
- X-rays are partly reflected by each plane of atoms. Waves reflected from adjacent planes interfere constructively when their path difference is a whole number of wavelengths.

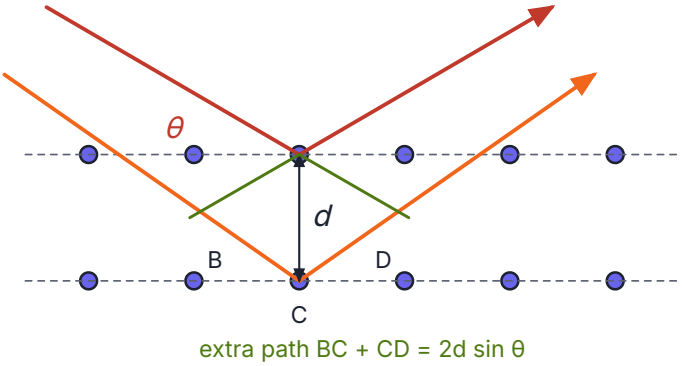


Figure 25.10: reflection from two adjacent atomic planes a distance d apart. θ is the glancing angle, measured from the plane.

The ray reflected from the lower plane travels an extra distance BC + CD = d sin θ + d sin θ = 2d sin θ.

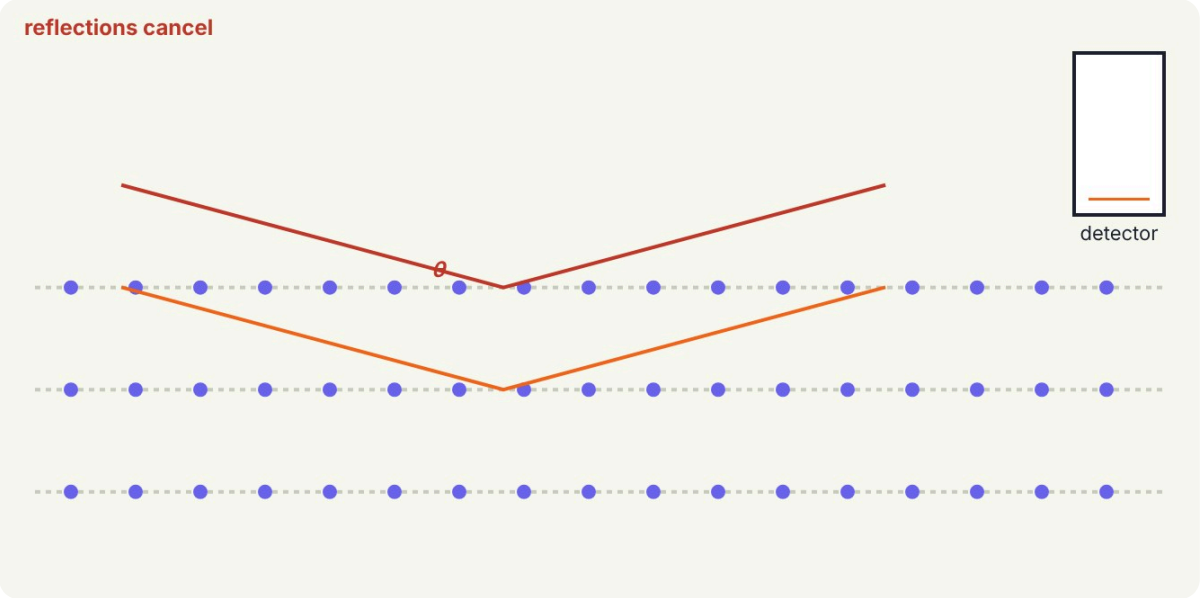
Constructive interference: path difference = $m\lambda$.

$$2d \sin\theta = m\lambda$$

Bragg's law; d = spacing of atomic planes; θ = glancing angle (between the ray and the plane, not the normal); m = 1, 2, 3, ...

Simulation Bragg's law

Change the glancing angle, the plane spacing and the wavelength. The detector reading is high only when 2d sin θ is a whole number of wavelengths.



path difference 2d sin θ	2d sin θ / λ	Bragg angles (m = 1, 2)
0.145 nm	**0.94**	**16.0°, 33.4°**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 25 to use the simulation.

Example 25.4.3

X-rays of wavelength 0.36×10^{-10} m give a first-order reflection from a crystal at a glancing angle of 4.8°. Find the spacing of the atomic planes.

Solution

$$d = \frac{m\lambda}{2 \sin \theta} = \frac{0.36 \times 10^{-10}}{2 \sin 4.8°} = \mathbf{2.15 \times 10^{-10}\ m}$$

Exercise: Bragg's law

1. X-rays are diffracted by a crystal with plane spacing 2.65×10^{-10} m. The first-order reflection is at 8.0°. Find the wavelength.

Answer

7.38×10^{-11} m

Applications of X-rays

Field	Uses
Medicine	radiography of bones and organs; CT scans; radiotherapy to destroy cancer cells
Industry	non-destructive testing of welds, castings and pipes; X-ray crystallography of materials
Security	baggage and cargo scanning at airports
Research	crystal structures of DNA and proteins; X-ray astronomy

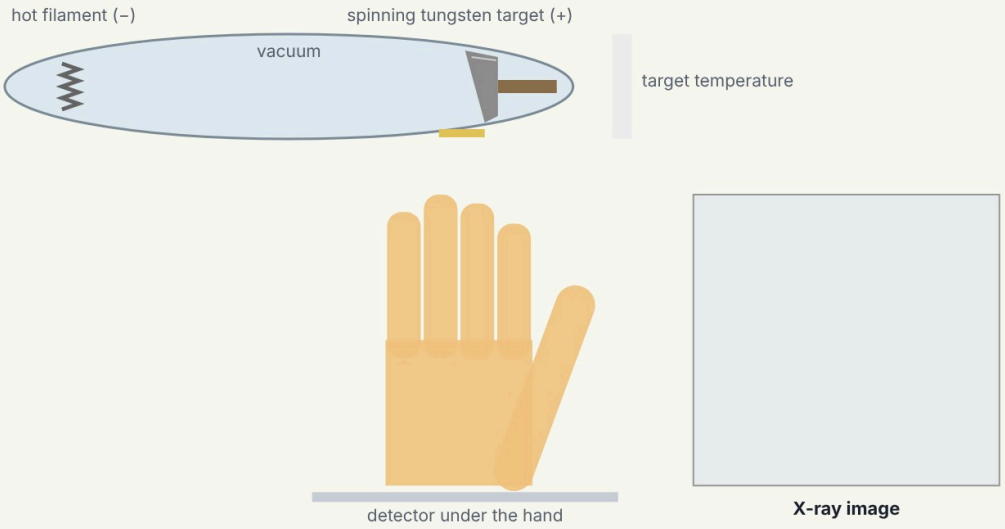
Inside a hospital X-ray machine

- The tube current (a few mA) is set by heating the filament: a hotter filament emits more electrons per second, so more X-ray photons are produced and the picture is brighter. It does not change λ_{min}.
- The tube voltage (about 50–120 kV) sets the energy of each electron, eV, and so the most energetic photons. Higher voltage gives more penetrating ("harder") X-rays.
- Bone contains calcium, which has a higher atomic number than soft tissue, so it absorbs X-rays much more strongly. Where X-rays reach the detector it turns dark; behind bone it stays light. That is why bones look white on an X-ray image.
- About 99% of the electrons' energy becomes heat in the target, so the anode is made of tungsten and is often a spinning disc, so the heat is spread around its rim.

Simulation

Taking an X-ray picture of a hand

Turn on the tube and watch an X-ray image build up. The filament current controls how many electrons (and photons) there are each second; the voltage controls the energy of each one. At low voltage even the soft tissue blocks most X-rays; at high voltage the bones become less clear.



electrons per second n = I/e	energy of each electron eV	shortest wavelength λmin = hc/eV
3.13 × 10¹⁶	**70 keV**	**17.8 pm**

power supplied IV	heat in the target (≈ 99%)
350 W	**347 W**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 25 to use the simulation.

Example 25.4.3

A hospital X-ray tube operates at 80 kV with a tube current of 5.0 mA. Find (a) the number of electrons striking the target each second, (b) the power supplied to the tube, (c) the rate of heat production in the target if 1% of the energy becomes X-rays, (d) the minimum wavelength of the X-rays.

Solution

(a) $n = \dfrac{I}{e} = \dfrac{5.0 \times 10^{-3}}{1.60 \times 10^{-19}} = \mathbf{3.1 \times 10^{16}\ s^{-1}}$

(b) $P = IV = (5.0 \times 10^{-3})(80 \times 10^{3}) = \mathbf{400\ W}$

(c) heat = 0.99 × 400 = **396 W** (only 4 W becomes X-rays)

(d) $\lambda_{min} = \dfrac{hc}{eV} = \dfrac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(1.60 \times 10^{-19})(80 \times 10^{3})} = \mathbf{1.55 \times 10^{-11}\ m}$

25.5 Nanoscience

Nanoscience

is **the study of structures and materials on the scale of about 1 nm to 100 nm** (1 nm = 10^{-9} m, a few atoms across). Nanotechnology is the design and making of devices at this scale.

- At the nanoscale, materials can behave very differently from the bulk: a large fraction of the atoms are at the surface, and quantum effects become important. Colour, strength, conductivity and chemical reactivity can all change with size.
- Examples: carbon nanotubes and graphene (very strong, excellent conductors), quantum dots (nanocrystals whose colour depends on their size), nanoparticles of titanium dioxide and silver.

Applications in electronic devices

Device	How nanoscience helps
Microprocessors	transistors only a few nanometres wide: billions on one chip, faster and using less power
Memory and data storage	nanoscale memory cells and magnetic layers store far more data in a smaller space (flash memory, hard disks)
Displays	quantum-dot (QLED) screens give bright, pure colours; organic and flexible displays
Sensors	nanoscale sensors detect tiny amounts of gases, chemicals or biological molecules
Solar cells and batteries	nanostructured materials absorb more light and store more charge
Future electronics	graphene and carbon-nanotube transistors; single-electron devices

- Other applications: stain-resistant fabrics, scratch-resistant paint, self-cleaning glass, drug delivery and early detection of disease.
- Concerns: nanoparticles may be breathed in or absorbed through the skin, and their effects on health and the environment are not yet fully understood.

Summary and exercises

Subtopic	Key results
25.1	$E = hf$; $hf = W + \frac{1}{2}mv_{max}^2$; $eV_s = \frac{1}{2}mv_{max}^2$; $W = hf_0$
25.2	$\lambda = h/p = h/\sqrt{(2meV)}$
25.3	$r_n = n^2h^2\varepsilon_0/\pi me^2$; $E_n = -13.6/n^2$ eV; $hf = E_i - E_f$
25.4	$\lambda_{min} = hc/eV$; $2d \sin\theta = m\lambda$

Exercises

1. Ultraviolet light of wavelength 250 nm falls on a metal of work function 3.0 eV. Find the stopping potential.

   Answer

   $hc/\lambda = 4.97$ eV $\Rightarrow V_s = 1.97$ V

2. Explain why increasing the intensity of light of frequency below the threshold frequency does not cause photoelectric emission.

   Answer

   Each electron absorbs one photon. Below f_0 each photon has energy $hf < W$, so no single photon can free an electron. More intensity only means more photons, each still with too little energy.

3. Find the de Broglie wavelength of a 0.15 kg ball moving at 40 m s^{-1}, and explain why its wave nature is not observed.

   Answer

   $\lambda = 1.1 \times 10^{-34}$ m, far smaller than any gap or obstacle, so no diffraction can be seen.

4. Find the ionisation energy of a hydrogen atom in the $n = 2$ state.

   Answer

   3.40 eV

5. An X-ray tube operates at 30 kV. Find the maximum frequency of the X-rays.

   Answer

   $f_{max} = eV/h = 7.24 \times 10^{18}$ Hz

Chapter 26 · Semester 3

NUCLEAR PHYSICS

Notes by Cikgu Suhaili

20 worked examples 4 simulations Exercises with answers

26.1 Nucleus

The nucleus (about 10^{-15} m across) contains protons and neutrons, together called **nucleons**.

Discovery of protons and neutrons

- **Proton**: Rutherford (1919) bombarded nitrogen with alpha particles and found that hydrogen nuclei were knocked out. He identified this positively charged particle, present in every nucleus, as the proton.
- **Neutron**: Chadwick (1932) showed that the radiation produced when beryllium is bombarded with alpha particles consists of uncharged particles with a mass close to that of the proton: neutrons.
- (Experimental details are not required.)

Particle	Mass / kg	Charge / C
proton	1.672×10^{-27}	$+1.60 \times 10^{-19}$
neutron	1.675×10^{-27}	0

- **Proton number** (atomic number) Z: the number of protons in the nucleus.
- **Nucleon number** (mass number) A: the total number of protons and neutrons. Number of neutrons $N = A - Z$.
- A nuclide is written ${}^{A}_{Z}\mathrm{X}$, e.g. ${}^{56}_{26}\mathrm{Fe}$ has 26 protons and 30 neutrons.
- **Isotopes**: atoms of the same element with the same number of protons but different numbers of neutrons, e.g. hydrogen ${}^{1}_{1}\mathrm{H}$, deuterium ${}^{2}_{1}\mathrm{H}$ and tritium ${}^{3}_{1}\mathrm{H}$.

Example 26.1.1

Find the number of protons, neutrons and electrons in a neutral atom of each nuclide: $^{9}_{4}\text{Be}$, $^{23}_{11}\text{Na}$, $^{59}_{27}\text{Co}$, $^{133}_{55}\text{Cs}$, $^{238}_{92}\text{U}$.

Solution

Nuclide	Protons	Neutrons	Electrons
$^{9}_{4}\text{Be}$	4	5	4
$^{23}_{11}\text{Na}$	11	12	11
$^{59}_{27}\text{Co}$	27	32	27
$^{133}_{55}\text{Cs}$	55	78	55
$^{238}_{92}\text{U}$	92	146	92

The mass spectrometer

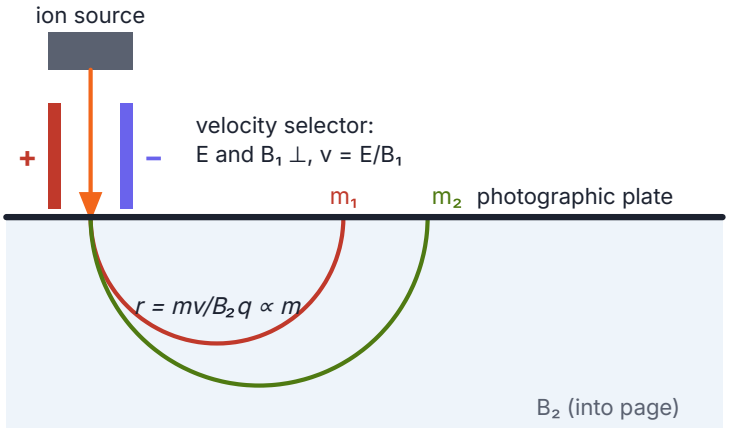


Figure 26.1: a Bainbridge mass spectrometer.

- Positive ions pass through a **velocity selector** with perpendicular fields E and B_1. Only ions with $qE = qvB_1$, i.e. $v = E/B_1$, go straight through.
- In the uniform field B_2 the ions move in semicircles: $qvB_2 = mv^2/r$, so $r = \dfrac{mE}{qB_1B_2} \propto m$.
- Isotopes of different mass land at different places on the plate: $m_1/m_2 = r_1/r_2$. It is used to measure atomic masses and the abundance of isotopes.

Mass defect and binding energy

Mass defect, Δm

the difference between the total mass of the separate nucleons and the mass of the nucleus.

$$\Delta m = Zm_p + (A - Z)m_n - M$$

Binding energy, E_B

the energy needed to separate a nucleus completely into its individual protons and neutrons. It is also the energy released when the nucleus is formed from its nucleons.

$$E = mc^2 \quad \Rightarrow \quad E_B = \Delta m\, c^2$$

Einstein's mass–energy equivalence: mass is a form of energy

Units: u and eV

- **Unified atomic mass unit**: 1 u = 1/12 of the mass of a carbon-12 atom = 1.66×10^{-27} kg.
- Its energy equivalent: $E = (1.66 \times 10^{-27})(3.00 \times 10^8)^2 = 1.49 \times 10^{-10}$ J = 931.5 MeV.

$$1\ \text{u} = 1.66 \times 10^{-27}\ \text{kg} \triangleq 931.5\ \text{MeV} \quad 1\ \text{eV} = 1.60 \times 10^{-19}\ \text{J}$$

so E_B (MeV) = Δm (u) × 931.5

Binding energy per nucleon

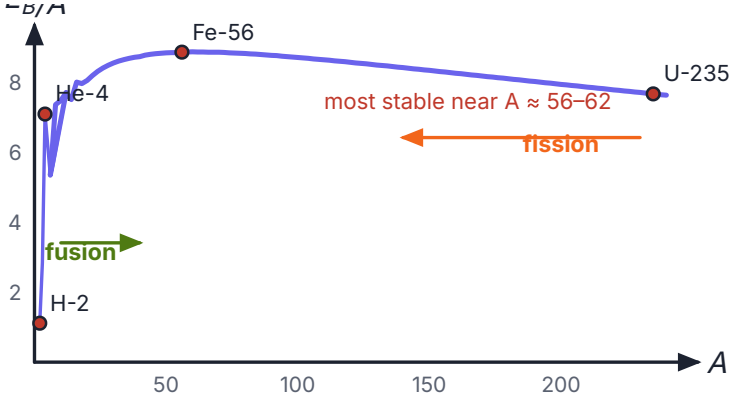


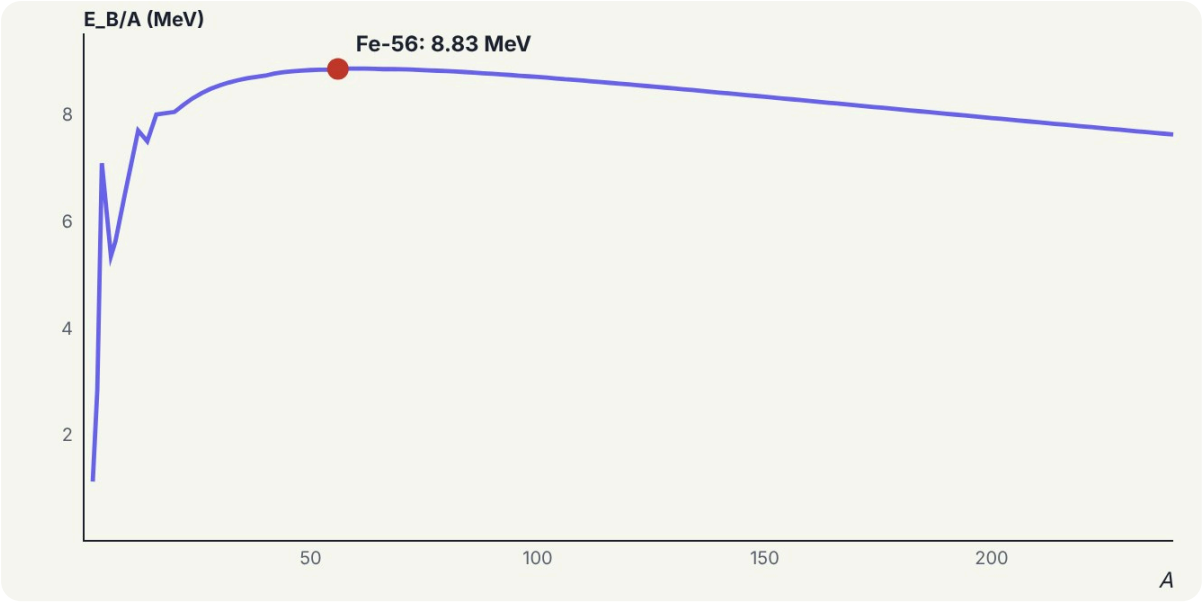
Figure 26.2: binding energy per nucleon against nucleon number.

- Binding energy per nucleon = E_B/A. It measures the **stability** of a nucleus: the larger it is, the more tightly bound and stable the nucleus.
- For light nuclei it rises rapidly with A (helium-4 is unusually stable, a peak at 7.1 MeV).
- It is largest, about 8.8 MeV, for $A \approx 56$–62 (iron and nickel): the most stable nuclei.
- For heavier nuclei it decreases slowly, to about 7.6 MeV for uranium.
- Energy is released when nuclei move towards the peak: light nuclei **fuse**, heavy nuclei **split** (fission).

Simulation

Binding energy per nucleon

Move along the curve to see how tightly bound each nucleus is. Then try a fission of uranium and a fusion of hydrogen: both move the nucleons up the curve and release energy.



binding energy per nucleon
8.83 MeV

total binding energy
495 MeV

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 26 to use the simulation.

Example 26.1.2

Find the binding energy of deuterium ${}^{2}_{1}\mathrm{H}$. (nucleus 2.013553 u, proton 1.007276 u, neutron 1.008665 u)

Solution

$\Delta m = 1.007276 + 1.008665 - 2.013553 = 0.002388$ u

$E_B = 0.002388 \times 931.5 =$ **2.22 MeV** $(= 3.57 \times 10^{-13}$ J)

Example 26.1.3

The binding energy of $^{20}_{10}\text{Ne}$ is 160.647 MeV. Find its atomic mass. (^{1}H atom 1.007825 u, neutron 1.008665 u)

Solution

Mass of 10 hydrogen atoms + 10 neutrons = 10(1.007825) + 10(1.008665) = 20.16490 u

Δm = 160.647/931.5 = 0.17246 u ⇒ M = 20.16490 − 0.17246 = **19.992 u**

Example 26.1.4

Find the binding energy per nucleon of $^{56}_{26}\text{Fe}$ (atomic mass 55.93494 u; ^{1}H 1.00782 u; neutron 1.00867 u).

Solution

Δm = 26(1.00782) + 30(1.00867) − 55.93494 = 0.52848 u

E_B = 0.52848 × 931.5 = 492.3 MeV ⇒ E_B/A = 492.3/56 = **8.79 MeV per nucleon**

Using atomic masses (with ^{1}H atoms instead of protons) is correct: the electron masses cancel.

Exercise: binding energy

1. Find the total binding energy and the binding energy per nucleon of $^{14}_{7}\text{N}$ (14.003074 u; ^{1}H 1.007825 u; neutron 1.008665 u).

   Answer

   Δm = 0.11236 u; 104.6 MeV; 7.47 MeV per nucleon

26.2 Radioactivity

Radioactivity

is **the spontaneous and random emission of radiation (α, β or γ) from unstable nuclei**, forming a new nucleus.

- **Spontaneous**: the decay happens on its own and is not affected by physical conditions (temperature, pressure) or chemical changes.
- **Random**: it is impossible to predict when a particular nucleus will decay; every nucleus has the same probability of decaying per unit time. Only the behaviour of a large number of nuclei can be predicted.

Types of radiation

	Alpha (α)	Beta (β^- / β^+)	Gamma (γ)
nature	helium nucleus ^4_2He	fast electron / positron	high-energy electromagnetic photon
charge	$+2e$	$-e$ / $+e$	0
deflected by E and B fields	yes	yes (opposite way)	no
ionising power	strong	moderate	weak
penetration	stopped by paper	stopped by a few mm of aluminium	reduced by thick lead or concrete

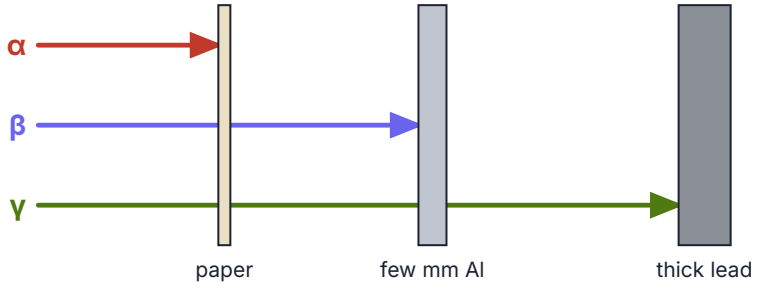


Figure 26.3: penetrating power of α, β and γ.

Decay	Equation	Change
α	$^A_Z\text{X} \rightarrow {}^{A-4}_{Z-2}\text{Y} + {}^4_2\text{He}$	loses 2 protons and 2 neutrons (heavy nuclei, $Z > 82$)
β^-	$^A_Z\text{X} \rightarrow {}^A_{Z+1}\text{Y} + {}^0_{-1}\text{e}$ + antineutrino	a neutron becomes a proton
β^+	$^A_Z\text{X} \rightarrow {}^A_{Z-1}\text{Y} + {}^0_{+1}\text{e}$ + neutrino	a proton becomes a neutron
γ	$^A_Z\text{X}^* \rightarrow {}^A_Z\text{X} + \gamma$	an excited nucleus drops to a lower state; A and Z unchanged

Example 26.2.1

Find the energy released in the alpha decay $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$. (238.050786 u, 234.043583 u, 4.002603 u)

Solution

Δm = 238.050786 − 234.043583 − 4.002603 = 0.004600 u

Q = 0.004600 × 931.5 = **4.28 MeV** (shared as kinetic energy of the α particle and the daughter nucleus)

Example 26.2.2

Find the energy released in the β^- decay of $^{213}_{83}Bi$. (nuclear masses: Bi 212.9944 u, Po 212.9928 u; electron 5.486×10^{-4} u)

Solution

$^{213}_{83}Bi \rightarrow {}^{213}_{84}Po + {}^{0}_{-1}e$

Q = (212.9944 − 212.9928 − 0.0005486)(931.5) = **0.979 MeV**

Example 26.2.3

Find the wavelength of a 0.186 MeV γ-ray photon emitted by radium-226.

Solution

$$\lambda = \frac{hc}{E} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(0.186 \times 10^{6})(1.60 \times 10^{-19})} = \mathbf{6.68 \times 10^{-12}\ m}$$

The radioactive decay law

Radioactive decay law

the rate of decay of a radioactive nuclide is directly proportional to the number of undecayed nuclei present.

$$A = -\frac{dN}{dt} = \lambda N$$

activity A = number of decays per second; unit becquerel (1 Bq = 1 decay s^{-1}); 1 curie = 3.7×10^{10} Bq. The minus sign shows that N decreases.

Decay constant, λ

the probability that a nucleus decays per unit time (unit s^{-1}). It is a constant for each nuclide; a larger λ means faster decay.

$\frac{dN}{N} = -\lambda\, dt$. Integrate from N_0 at $t = 0$ to N at time t:

$$\ln N - \ln N_0 = -\lambda t \Rightarrow \ln \frac{N}{N_0} = -\lambda t$$

$$N = N_0 e^{-\lambda t} \quad A = A_0 e^{-\lambda t}$$

also the mass of the radioactive sample: $m = m_0 e^{-\lambda t}$

Half-life, $T_{½}$

the time taken for the number of radioactive nuclei (or the activity) to fall to half of its original value.

At $t = T_{½}$, $N = N_0/2$: $\frac{N_0}{2} = N_0 e^{-\lambda T_{½}} \Rightarrow e^{\lambda T_{½}} = 2 \Rightarrow \lambda T_{½} = \ln 2$

$$\lambda = \frac{\ln 2}{T_{½}} = \frac{0.693}{T_{½}}$$

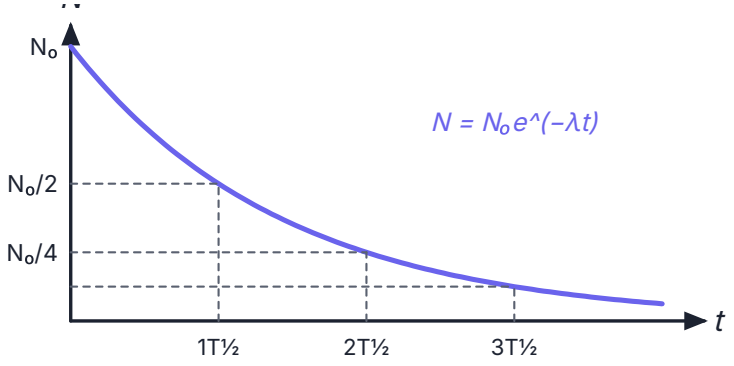


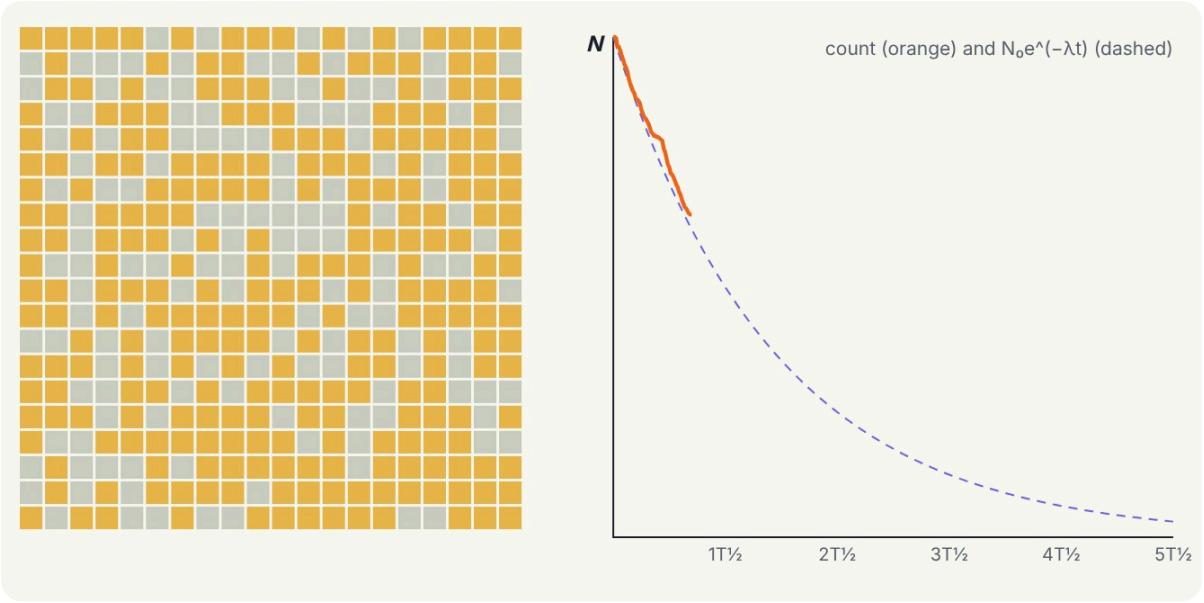
Figure 26.4: exponential decay. After each half-life the number left halves: N_0, $N_0/2$, $N_0/4$, ...

Teacher's note

After n half-lives, $N = N_0(½)^n$. To find the number of nuclei in a mass m: N = (m/molar mass) × N_A. Keep λ and t in matching units.

Simulation Random radioactive decay

Each square is a nucleus. Every tick each one has the same small chance of decaying, so you cannot tell which will go next, yet the total follows N = N₀e^(−λt). Change the half-life and the number of nuclei, and compare the count with the curve.



time	nuclei left (count)	N₀e^(−λt) predicts	λ = ln 2 / T½
2.8 s = 0.70 half-lives	**258**	**246**	**0.173 s⁻¹**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 26 to use the simulation.

Example 26.2.4

Krypton-87 has a half-life of 78 minutes. Find the activity of 10 μg of krypton-87 (in Bq and Ci).

Solution

$$N = \frac{10 \times 10^{-6}}{87} \times 6.02 \times 10^{23} = 6.92 \times 10^{16}$$

$$\lambda = \frac{0.693}{78 \times 60} = 1.48 \times 10^{-4}\ \text{s}^{-1}$$

$$A = \lambda N = \mathbf{1.02 \times 10^{13}\ Bq} = 1.02 \times 10^{13}/3.7 \times 10^{10} = \mathbf{277\ Ci}$$

Example 26.2.5

A sample has an activity of 9.00×10^{12} Bq and a half-life of 80.0 s. How long will it take for the activity to fall to 2.00×10^{12} Bq?

Solution

$$A = A_0 e^{-\lambda t} \Rightarrow t = \frac{\ln (A_0/A)}{\lambda} = \frac{\ln 4.5}{0.693/80.0} = \mathbf{174\ s}$$

Example 26.2.6

What mass of radium-227 has an activity of 1.0×10^6 Bq? (half-life 41 minutes)

Solution

$$N = \frac{A}{\lambda} = \frac{1.0 \times 10^6}{0.693/(41 \times 60)} = 3.55 \times 10^9$$

$$m = \frac{3.55 \times 10^9}{6.02 \times 10^{23}} \times 227\ \text{g} = \mathbf{1.34 \times 10^{-12}\ g}$$

Example 26.2.7

The half-life of potassium-45 is 17.3 minutes. How long does it take for 75% of the nuclei to decay?

Solution

25% remain: $N = 0.25N_0$ (two half-lives)

$$t = \frac{\ln 4}{\lambda} = 2T_{½} = \mathbf{34.6\ min}$$

Example 26.2.8

After 4 hours, 80% of the nuclei of a radioactive isotope have decayed. Find the half-life.

Solution

$$N = 0.20N_0 \Rightarrow \lambda = \frac{\ln 5}{4} = 0.402\ \text{h}^{-1}$$

$$T_{½} = \frac{0.693}{0.402} = \mathbf{1.72\ h}$$

Applications of radioisotopes as tracers

A **radioisotope** behaves chemically like the stable isotope, so it can be followed by detecting its radiation. Short half-lives are chosen so that the activity soon dies away.

Field	Use
Medical: blood volume	a small known activity is injected; after it mixes, the activity per unit volume of a blood sample gives the total volume
Medical: imaging	technetium-99m (γ emitter, half-life 6 h) collects in tumours, which then show up on a gamma camera; iodine-131 is used for the thyroid
Agriculture	phosphorus-32 in fertiliser shows how fast and where the plant takes it up; tracers also study pesticides and are used in the sterile insect technique
Industry	a short-lived tracer in an underground pipe: a Geiger counter above the ground detects higher activity where the pipe leaks; also thickness gauges and flow measurement

Example 26.2.9

A solution of sodium-24 (half-life 15 h) of activity 1.5×10^4 Bq is injected into a patient's blood. After 30 h, 1.0 cm^3 of blood has an activity of 0.50 Bq. Estimate the volume of blood.

Solution

After 30 h (2 half-lives) the total activity = $1.5 \times 10^4 \times ¼ = 3.75 \times 10^3$ Bq

The activity is spread evenly: $\frac{3.75 \times 10^3}{V} = \frac{0.50}{1.0\ cm^3} \Rightarrow V = 7.5 \times 10^3\ cm^3 = \mathbf{7.5 \times 10^{-3}\ m^3}$ (7.5 L)

26.3 Nuclear reactions

A nuclear reaction changes the nucleus: radioactive decay, bombardment by a particle, fission or fusion. For example, Rutherford's first artificial reaction: ${}^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}H$.

Conservation laws

- **Conservation of nucleon number**: the total nucleon number A before the reaction equals the total after.
- **Conservation of charge**: the total proton number (charge) Z before equals the total after.
- **Conservation of mass–energy**: the total mass-energy (rest mass energy + kinetic energy) before equals the total after. Any loss of mass appears as energy.

$$Q = \Delta m\, c^2, \quad \Delta m = \text{(total mass before)} - \text{(total mass after)}$$

$Q > 0$: energy released (exothermic). $Q < 0$: energy must be supplied, e.g. as kinetic energy of the bombarding particle (endothermic).

Example 26.3.1

Complete the reactions: (a) ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + ?$ (b) ${}^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + ?$ (c) ${}^{7}_{3}Li + {}^{1}_{1}H \rightarrow 2\ ?$ (d) ${}^{10}_{5}B + {}^{1}_{0}n \rightarrow {}^{7}_{3}Li + ?$ (e) ${}^{226}_{88}Ra \rightarrow ? + {}^{4}_{2}He$

Solution

(a) ${}^{4}_{2}He$ (α decay) (b) ${}^{1}_{1}H$ (a proton) (c) ${}^{4}_{2}He$ (d) ${}^{4}_{2}He$ (e) ${}^{222}_{86}Rn$

Check each with A and Z: e.g. (b) 14 + 4 = 17 + 1 and 7 + 2 = 8 + 1.

Example 26.3.2

When lithium-7 is bombarded with a proton, two alpha particles are produced. Find the energy released. (^{1}H 1.007825 u, ^{7}Li 7.016003 u, ^{4}He 4.002603 u)

Solution

$^{7}_{3}\text{Li} + {}^{1}_{1}\text{H} \rightarrow 2\,{}^{4}_{2}\text{He}$

$\Delta m = 7.016003 + 1.007825 - 2(4.002603) = 0.018622$ u

$Q = 0.018622 \times 931.5 =$ **17.3 MeV** (2.78×10^{-12} J), released

Example 26.3.3

A deuteron bombards carbon-13 and produces nitrogen-14. (a) Write the equation. (b) Find the energy released. (^{13}C 13.00335 u, ^{2}H 2.01410 u, ^{14}N 14.00307 u, n 1.00867 u)

Solution

(a) $^{13}_{6}\text{C} + {}^{2}_{1}\text{H} \rightarrow {}^{14}_{7}\text{N} + {}^{1}_{0}\text{n}$

(b) $\Delta m = 13.00335 + 2.01410 - 14.00307 - 1.00867 = 0.00571$ u $\Rightarrow Q =$ **5.32 MeV**

Nuclear fission

Nuclear fission

is **the splitting of a heavy nucleus into two lighter nuclei of roughly equal mass**, with the release of neutrons and energy.

$$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{236}_{92}\text{U}^{*} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3\,{}^{1}_{0}\text{n} + \text{energy}$$

- Fission of uranium-235 is usually **induced** by a slow (thermal) neutron. The excited U-236 nucleus splits within about 10^{-12} s.
- The fragments vary, mostly with A about 90–100 and 135–145.
- **Why energy is released**: the binding energy per nucleon of uranium is about 7.6 MeV, but that of the fragments is about 8.5 MeV. The products are more tightly bound, so they have less mass. Energy

released ≈ (8.5 − 7.6) × 236 ≈ **200 MeV** per fission, mostly as kinetic energy of the fragments and neutrons.

Example 26.3.4

Find the energy released when 10 kg of uranium-235 undergoes fission: $^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3{}^{1}_{0}\mathrm{n}$. (U 235.04 u, Ba 140.91 u, Kr 91.91 u, n 1.01 u)

Solution

Δm = (235.04 + 1.01) − (140.91 + 91.91 + 3 × 1.01) = 0.20 u ⇒ Q = **186 MeV** per fission

Number of nuclei = $\dfrac{10}{235 \times 10^{-3}} \times 6.02 \times 10^{23} = 2.56 \times 10^{25}$

Total = 186.3 MeV × 2.56×10^{25} = **4.77×10^{33} eV** = 7.6×10^{14} J

Chain reaction

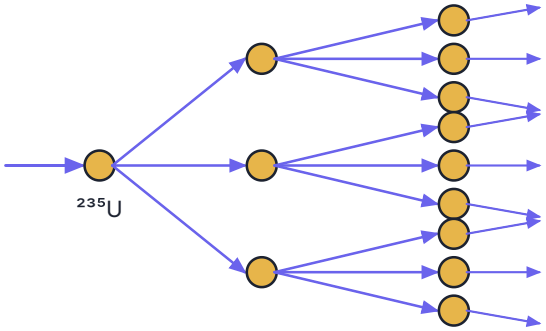


Figure 26.5: a chain reaction.

- Each fission releases 2 or 3 neutrons. If at least one of them causes another fission, the reaction continues by itself: a **chain reaction**.
- If the piece of uranium is too small, too many neutrons escape through the surface and the chain dies out. The **critical mass** is the minimum mass of fissile material needed to sustain a chain reaction.
- Above the critical mass, if every neutron is allowed to cause fission, the number of fissions grows very rapidly (an uncontrolled chain reaction, as in a bomb).

Controlled fission in a nuclear reactor

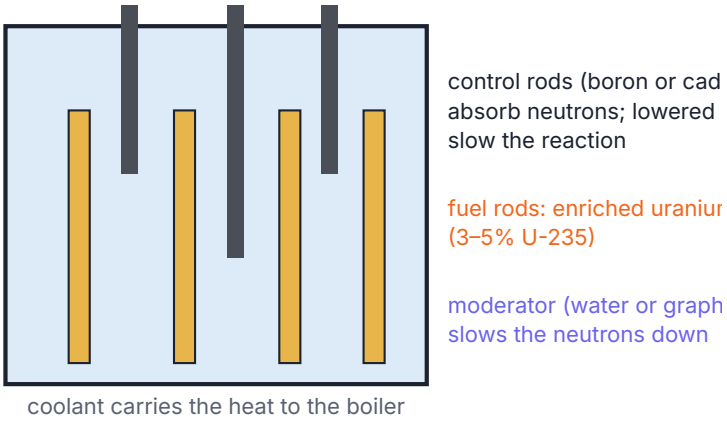


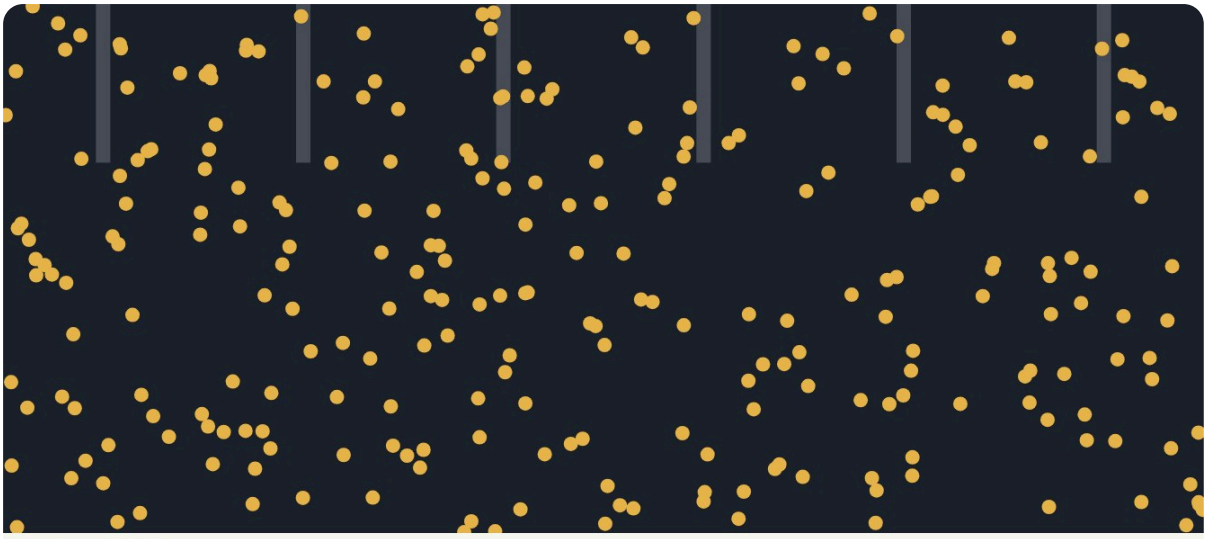
Figure 26.6: the core of a thermal fission reactor.

Part	Purpose
Fuel rods	enriched uranium (3–5% U-235); the U-235 undergoes fission
Moderator (water or graphite)	slows down the fast neutrons from fission by collisions, because slow neutrons are much more likely to cause fission of U-235
Control rods (cadmium or boron)	absorb neutrons. They are raised or lowered so that, on average, exactly **one** neutron from each fission causes another fission: a steady (critical) rate
Coolant	carries the heat away to produce steam, which drives turbines to generate electricity
Concrete shield	absorbs neutrons and γ-radiation to protect the workers

Simulation

Chain reaction and control rods

Fire one neutron into the uranium. Each fission (yellow nucleus) releases 2–3 new neutrons. Push the control rods in to absorb more neutrons: find the setting where the reaction keeps going at a steady rate.



neutrons in flight

fissions so far	neutrons in flight	neutron population
0	0	fire a neutron to start

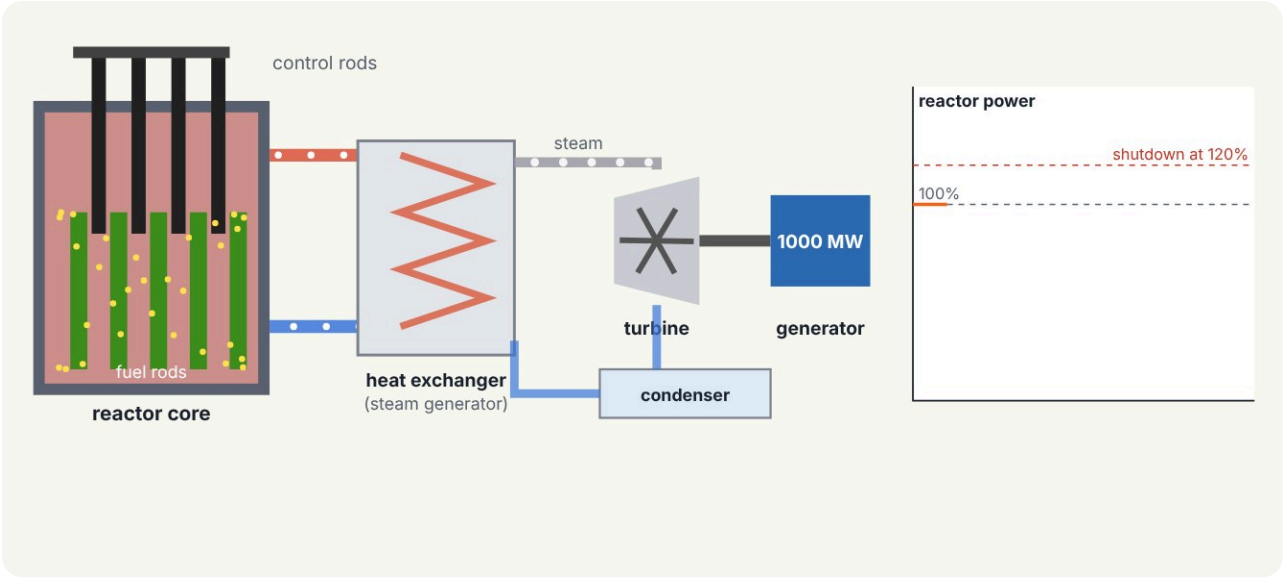
This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 26 to use the simulation.

A nuclear power station

- The fission energy heats the water in the reactor core. In a **pressurised water reactor** the water is kept at high pressure so it does not boil, and the same water acts as moderator and coolant.
- The hot coolant is pumped through a **heat exchanger** (steam generator), where it boils water in a separate loop. The steam turns a **turbine** connected to a **generator**.
- The steam is cooled back to water in a **condenser** and returned. Only about one third of the heat becomes electrical energy; the rest is given out to the cooling water.
- If the power rises too fast, the control rods drop fully into the core (an emergency shutdown) and the chain reaction stops.

Simulation Inside a nuclear power station

Move the control rods. Pulled out, more neutrons survive and the power rises; pushed in, the power falls. Find the position where each fission causes, on average, exactly one more (k = 1) and the power stays steady. If the power goes above 120%, the reactor shuts itself down.



neutrons from each fission causing another fission, k
1.000

state
steady (critical)

thermal power
3000 MW

electrical power (33%)
1000 MW

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 26 to use the simulation.

Example 26.3.5

A nuclear power station produces 1000 MW of electrical power with an efficiency of 33%. Each fission of U-235 releases 200 MeV. Find (a) the thermal power of the reactor, (b) the number of fissions per second, (c) the mass of U-235 used in one day.

Solution

(a) $P_{\text{thermal}} = \dfrac{1000}{0.33} = \mathbf{3.0 \times 10^{3}\ MW}$

(b) energy per fission $= 200 \times 1.60 \times 10^{-13} = 3.2 \times 10^{-11}$ J $\Rightarrow n = \dfrac{3.0 \times 10^{9}}{3.2 \times 10^{-11}} = \mathbf{9.4 \times 10^{19}\ s^{-1}}$

(c) nuclei per day $= 9.4 \times 10^{19} \times 86\,400 = 8.1 \times 10^{24} \Rightarrow$ mass $= \dfrac{8.1 \times 10^{24}}{6.02 \times 10^{23}} \times 0.235$ kg $= \mathbf{3.2\ kg}$

Nuclear fusion

> **Nuclear fusion**
> is **the joining of two light nuclei to form a heavier nucleus**, with the release of energy.

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + ^{1}_{0}n + 17.6\ MeV$$

- Light nuclei have a low binding energy per nucleon. The product (e.g. helium-4) is much more tightly bound, so mass is lost and energy is released. Per nucleon, fusion releases more energy than fission.
- The nuclei are both positive and repel each other. To come within the range of the strong nuclear force (about 10^{-15} m) they need very high kinetic energies, i.e. temperatures of about 10^{7}–10^{8} K. Fusion is therefore called a **thermonuclear** reaction.

Fusion in the Sun

- The Sun's core has a temperature of about 1.5×10^{7} K and a very high density, so hydrogen nuclei collide fast and often enough to fuse.
- The main process is the **proton–proton chain**:

$^{1}_{1}H + ^{1}_{1}H \rightarrow ^{2}_{1}H + ^{0}_{+1}e + \nu$ (positron and neutrino)

$^{2}_{1}H + ^{1}_{1}H \rightarrow ^{3}_{2}He + \gamma$

$^{3}_{2}He + ^{3}_{2}He \rightarrow ^{4}_{2}He + 2^{1}_{1}H$

Overall: $4^{1}_{1}H \rightarrow ^{4}_{2}He + 2^{0}_{+1}e + 2\nu + 2\gamma$ + about **26.7 MeV**

Fission	Fusion
a heavy nucleus splits into lighter ones	light nuclei join into a heavier one
started by neutron bombardment	needs a very high temperature
produces radioactive waste	products mostly not radioactive
controlled in reactors today	very difficult to contain and control

Both release energy, both produce new nuclei, and in both the total mass decreases.

Example 26.3.5

In the fusion ${}_1^3\mathrm{H} + {}_1^2\mathrm{H} \rightarrow {}_2^4\mathrm{He}$, 2 kg of deuterium is used. Find (a) the energy released per helium nucleus, (b) the total mass lost, (c) the total energy released. (${}^2\mathrm{H}$ 2.014 u, ${}^4\mathrm{He}$ 4.002 u)

Solution

(a) $\Delta m = 2(2.014) - 4.002 = 0.026$ u $\Rightarrow Q = 0.026 \times 931.5 =$ **24.2 MeV** (3.88×10^{-12} J)

(b) number of ${}^2\mathrm{H}$ nuclei $= \dfrac{2}{2 \times 10^{-3}} \times 6.02 \times 10^{23} = 6.02 \times 10^{26}$, i.e. 3.01×10^{26} reactions; mass lost $= 3.01 \times 10^{26} \times 0.026$ u $= 7.83 \times 10^{24}$ u = **0.013 kg**

(c) $E = \Delta mc^2 = 0.013 \times (3.00 \times 10^8)^2 =$ **1.17×10^{15} J**

Example 26.3.6

Compare the energy released per nucleon in the fusion ${}_1^3\mathrm{H} + {}_3^6\mathrm{Li} \rightarrow 2\,{}_2^4\mathrm{He}$ ($Q = 22.36$ MeV) with that in the fission ${}_{94}^{239}\mathrm{Pu} + {}_0^1\mathrm{n} \rightarrow {}_{56}^{138}\mathrm{Ba} + {}_{38}^{96}\mathrm{Sr} + 3\,{}_0^1\mathrm{n}$ ($Q = 188.16$ MeV).

Solution

Fusion: 22.36/8 = 2.80 MeV per nucleon. Fission: 188.16/237 = 0.794 MeV per nucleon.

Fusion releases about **3.5 times** more energy per nucleon.

Nuclear waste disposal

Waste	Disposal
Low-level (gloves, clothing, tools)	sealed in drums and buried in shallow engineered landfill sites
Intermediate-level	mixed with concrete or bitumen in steel containers, stored in deep vaults
High-level (used fuel)	cooled for years under water, then sealed in thick steel and concrete casks for deep geological burial, away from groundwater and earthquake zones; some is reprocessed to recover uranium and plutonium

Summary and exercises

Subtopic	Key results
26.1	$\Delta m = Zm_p + Nm_n - M$; $E_B = \Delta mc^2$; 1 u ≙ 931.5 MeV; most stable near $A \approx 56$
26.2	$A = -dN/dt = \lambda N$; $N = N_0 e^{-\lambda t}$; $\lambda = \ln 2 / T_{½}$
26.3	conserve A and Z; $Q = \Delta mc^2$; fission of heavy and fusion of light nuclei both move towards the peak of the E_B/A curve

Exercises

1. A radioactive sample has a half-life of 5.0 days. What fraction remains after 20 days?

   Answer

   4 half-lives: $(½)^4 = 1/16$

2. The activity of a source falls from 800 Bq to 100 Bq in 30 minutes. Find its half-life.

   Answer

   800 → 100 is 3 half-lives: 10 minutes

3. Explain why radioactive decay is described as random.

   Answer

   It is impossible to predict which nucleus will decay next or when; each nucleus has the same probability of decaying per unit time.

4. State the function of the moderator and of the control rods in a fission reactor.

   Answer

   Moderator: slows the neutrons so that they can cause further fission of U-235. Control rods: absorb neutrons to keep the rate of fission steady (one neutron per fission goes on to cause another fission).

5. Use the graph of binding energy per nucleon to explain why both fission and fusion release energy.

   Answer

   Fission (heavy → medium) and fusion (light → heavier) both produce nuclei with greater binding energy per nucleon. The products are more tightly bound, so their total mass is smaller; the lost mass is released as energy, $E = \Delta mc^2$.